# Notes on Quantum Optics

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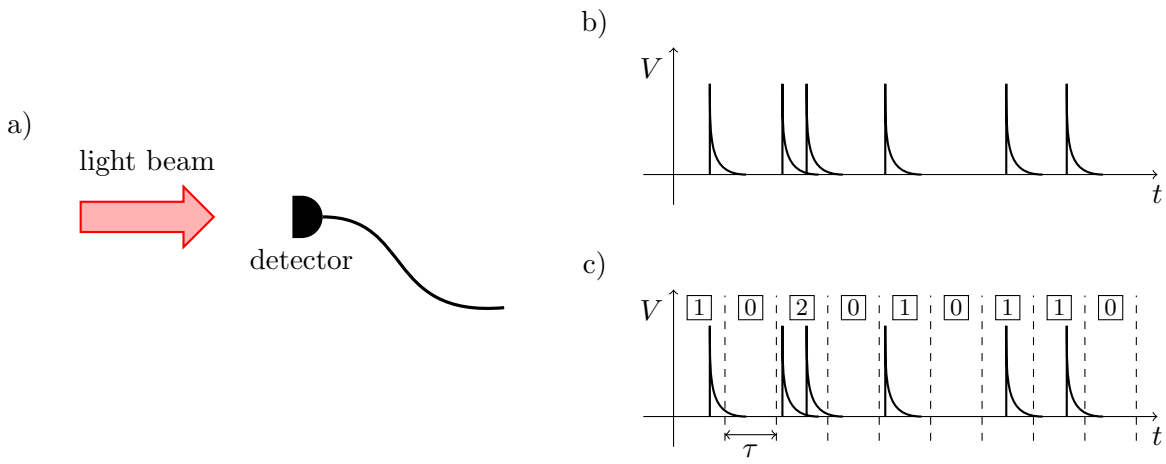


Figure 1: When a light beam illuminates the detector (a), the electrical output records an irregular train of pulses (b) each corresponding to one photo-count event. We divide the arrow of time in bins with duration $\tau$ and count the number of events in each bin. The number of photo-counts in a bin can be modelled as a random variable.

# 1 Photo-counting statistics

## 1.1 Photons and photodetection

Before diving into developing the formal theory of Quantum Optics, we may ask ourselves the question: is it possible to detect or measure a single quantum of light, namely one single **photon**? Is it possible to conceive and build an instrument able to detect and count the single photons that are impinging on it?

The answer is - maybe obviously, at this point - affirmative. Many Quantum Optics experiments, nowadays, are based indeed on *single photon detectors*. We will thus begin our discussion by imagining to build such a sensitive detector, and by imagining to use it to conduct experiments. This will be useful to gain a first feeling of what the word 'non-classical' means, when we talk about light.

Knowledge of the photo-electric effect, even independently of its interpretation in terms of photons, allows us to build a device consisting in a photo-cathode and an anode, enclosed in a vacuum tube. This is the *photo-electric cell*. When a suitable voltage is imposed between the two electrodes, and light is shined onto the photo-cathode, an electric current flow is measured. To increase the sensitivity of our device, we add acceleration grids between the cathode and the anode: in this the seeds of electrons emitted by the cathode are strongly amplified by avalanche mechanisms. We may then increase the amplification to such a level, that a single electron emitted by the cathode gives rise to a self-sustained current. We shall finally add some electronic control mechanism that allows to produce a defined pulse in the output signal, every time that a single electron leaves the photo-cathode. This is the *photo-multiplier tube*.

Here we are. We have got a device with an optical input and an electrical output that works as follows. When it is placed in complete darkness, it gives a flat zero-valued signal.[1] When one shines a (feeble!) light onto its optical input, we observe a sequence of electrical pulses at the output, that we call **photo-counts**. The number of photo-counts per unit time is found to be proportional to the classically measured intensity of the shined light.In general, the train of pulses is irregular and the pulses are not equally spaced in time. The photo-counts appear to follow a statistical process (see Fig. 1).

It would seem natural, once we have such an instrument in our hands, to assume that the sequence of pulses that we see in the output signal is a faithful indication of the sequence of quanta of light

[1] In a more realistic setting, we may find some output signals also when the device is placed in the darkness, due to different noise causes. For simplicity we can assume here that these are negligible.

impinging on the detector. The fact itself that the detector gives at the output isolated 'clicks', might by itself look a direct proof of the granularity of the radiation impinging on it, and thus a proof of the need to adopt a description of the electromagnetic field in terms of photons.
As a matter of fact, the situation is more complicated. On the one hand we have the properties of the light beam incident on the detector, on the other hand the features of the detection process itself. Careful thinking would let us realize that the train of irregularly spaced electric pulses that we measure at the output port of our detector could be explained in completely different ways. In particular we may identify two opposite physical models.

**Quantum model** The light beam is composed of discrete particles (the *photons*), and the detector is responding to the absorption of each particle arriving to its surface, possibly with imperfect efficiency. The train of electric pulses is a somehow faithful reproduction of the train of photons arriving at the detector.

**Semi-classical model** The light beam is not composed of discrete particles, but it is a uniform and indistinct 'magma' of electromagnetic energy flowing towards the detector. However, the quantized properties of the matter that compose the detector constrain the absorption of energy to occur in discrete amounts. In this picture, the photons are just these quanta of allowed optical absorption, and the train of pulses that we see at the output is a manifestation of the statistical feature of this process.

In the next Sections, we will start by modeling the detection process using the second approach, namely the semi-classical one. *Importantly, we will ignore in doing this any quantized notion of light.* It will be instructive to see if and when this model falls short of explaining what is observed in the experiments, as in this way we will learn more precisely why we need indeed a quantum theory of light.

## 1.2 Semi-classical model for photodetection

Starting from the experimental observations, mentioned in the previous Section, we may assume the following reasonable hypotheses to develop a model for photon counting.

**Semi-classical hypotheses for a photon counting model**

1. The probability of getting a photo-count in a small interval of time is proportional to the beam intensity $I$, to the illuminated detector area $A$ and to the duration $\Delta t$ of the small time interval.
2. If $\Delta t \to 0$, namely if we take a sufficiently short time interval, the probability of getting two photo-counts in the same interval is negligible.
3. Detection events in different time intervals are statistically independent.

### Case of constant intensity

We proceed with our modeling by assuming at first that the intensity of the light, incident on the detector, is constant in time.
Let us consider a macroscopic (may be small, however!) time bin $\tau$, and let us divide it into $N$ sub-intervals. We call $\Delta t$ the duration of one these intervals.

$$\tau = N\,\Delta t \tag{1-1}$$

In accordance with our hypotheses, in each interval $\Delta t$ we have a probability $p_1$ of getting *one* click that is written as:

$$p_1 = \xi I A \, \Delta t \tag{1-2}$$

Let $X_1$ be the random variable counting the clicks registered in a given $\Delta t$. This variable can take only the values of 0 or 1, as no more than one click is possible if $\Delta t$ is sufficiently small. The variable $X_1$ can take the two possible values with probabilities:

$$P(0) = 1 - p_1 \qquad\qquad P(1) = p_1 \tag{1-3}$$

The statistics of $X_1$ is thus described by:

$$\mathrm{E}(X_1) = 0 \cdot P(0) + 1 \cdot P(1) = 0 \cdot (1 - p_1) + 1 \cdot p_1 = p_1 \tag{1-4}$$

$$\mathrm{E}(X_1^2) = 0^2 \cdot P(0) + 1^2 \cdot P(1) = 0 \cdot (1 - p_1) + 1 \cdot p_1 = p_1 \tag{1-5}$$

$$\mathrm{Var}(X_1) = \mathrm{E}(X_1^2) - (\mathrm{E}(X_1))^2 = p_1 - p_1^2 = p_1(1 - p_1) \tag{1-6}$$

In fact, the occurrence of a click in the interval is what is known in statistics as a *Bernoulli trial*.
We turn now to studying the probability of getting $n$ clicks in the larger interval $\tau$. In particular, let $X_N$ be the random variable counting these clicks: it is the sum of the random variables counting the clicks in the individual sub-intervals $X_N = \sum_i X_i$. These are all identically distributed as $X_1$, so we write:

$$\bar{n} \equiv \mathrm{E}(X_N) = \sum_i \mathrm{E}(X_i) = N\mathrm{E}(X_1) = Np_1 \tag{1-7}$$

$$\mathrm{Var}(X_N) = \sum_i \mathrm{Var}(X_i) = N\mathrm{Var}(X_1) = Np_1(1 - p_1) \tag{1-8}$$

where we have defined $\bar{n}$ as a short-hand notation for $\mathrm{E}(X_N)$. Actually, being the sum of $N$ identically distributed Bernoulli trials, $X_N$ follows the *binomial distribution* and the probability to get $n$ clicks in $\tau$ is:[2]

$$P(n) = \binom{N}{n} p_1^n (1 - p_1)^{N-n} = \frac{N!}{n!\,(N-n)!} p_1^n (1 - p_1)^{N-n} \tag{1-10}$$

It is practical to rewrite the expressions describing the binomial distribution as functions of $\bar{n}$, exploiting the relation $\bar{n} = Np_1$: We obtain:

$$P(n) = \frac{N!}{n!\,(N-n)!} \left(\frac{\bar{n}}{N}\right)^n \left(1 - \frac{\bar{n}}{N}\right)^{N-n} \tag{1-11}$$

$$\mathrm{E}(X_N) = \bar{n} \tag{1-12}$$

$$\mathrm{Var}(X_N) = \bar{n}\left(1 - \frac{\bar{n}}{N}\right) \tag{1-13}$$

We also consider that:

$$\bar{n} = Np_1 = N\xi I A \, \Delta t = \xi I A \tau \tag{1-14}$$

which means that $\bar{n}$ does not depend on $N$ (or $\Delta t$), but only on the width $\tau$ of the time bin.

[2] The binomial distribution is a well-known result in statistics and Eq. (1-10) is not too difficult to retrieve indeed. For instance, one can start considering the probability to get one click for each of the first $n$ sub-intervals, and zero clicks in the remaining $N - n$, which is:

$$p_1^n p_0^{N-n} = p_1^n (1 - p_1)^{N-n} \tag{1-9}$$

Then, one considers that this is not the only way of getting a total of $n$ clicks in the $N$ available sub-intervals. In fact, there are $N!$ permutations of the $N$ intervals, of which $n$ are identically 'filled' with one click and $N - n$ identically empty. So, the number of possible ways is $\binom{N}{n} = \frac{N!}{n!\,(N-n)!}$. By multiplying the expression (1-9) by the latter factor, one gets Eq. (1-10).

We now take the limits to $N \to +\infty$, while keeping constant the width of the time bin $\tau$. This corresponds to taking $\Delta t \to 0$ and is essential to enforce the second hypothesis of our model. We call $X$ the random variable counting the clicks in $\tau$ in this case. We get:

$$\mathrm{E}(X) = \lim_{N\to+\infty} \mathrm{E}(X_N) \equiv \mathrm{E}(X_N) = \bar{n} \tag{1-15}$$

$$\mathrm{Var}(X) = \lim_{N\to+\infty} \mathrm{Var}(X_N) = \lim_{N\to+\infty} \bar{n}\left(1 - \frac{\bar{n}}{N}\right) = \bar{n} \tag{1-16}$$

In the limit of large $N$, the variance of the distribution equals its average. In fact, the limiting distribution of a binomial, when $N$ grows to infinity and the average is kept fixed, is a *Poissonian distribution*:

$$\boxed{P(X) = \frac{\bar{n}^n}{n!} e^{-\bar{n}}} \tag{1-17}$$

Obtaining this result, which is also well known in statistics, requires some passages that are reported in Box 1-1. An important feature of this distribution, that we have proven here above, is that its variance equals the average:

$$\boxed{\mathrm{Var}(X) \equiv \mathrm{E}(X) \equiv \bar{n}} \tag{1-18}$$

**Case of time-varying intensity**

We study now the case in which the intensity of the beam is varying in time randomly, according to a probability distribution $P(I)$. Rather than on developing the detailed probability distribution of the photo-counts in the time bin $\tau$, which depends on the specific features of $P(I)$, we will focus on deriving expressions for the average $\mathrm{E}(X)$ and for the variance $\mathrm{Var}(X)$ in this case.
First, it is useful to write the expressions of a few statistical figures for the time-varying intensity $I$, that we treat as a random variable itself.

$$\bar{I} = \mathrm{E}(I) = \int_0^{+\infty} I\, P(I)\, dI \tag{1-19}$$

$$\mathrm{E}(I^2) = \int_0^{+\infty} I^2\, P(I)\, dI \tag{1-20}$$

$$\mathrm{Var}(I) = \mathrm{E}(I^2) - (\mathrm{E}(I))^2 \tag{1-21}$$

Second, we note that the Poissonian distribution of the photo-counts that we have worked out in the previous paragraphs is actually a *conditional* probability distribution, namely it describes the probability of getting $n$ clicks in the interval $\tau$, given a *certain* beam intensity $I$. To make the dependence on $I$ explicit, we can add a subscript $I$ where relevant and we transform Eqs. (1-15)-(1-17) by substituting:

$$\bar{n} = \xi I A \tau = cI \tag{1-22}$$

where we have defined also the constant $c = \xi A \tau$, containing the characteristic parameters of the detector ($\xi$ and $A$) and the width of the time bin $\tau$. We obtain:

$$\bar{n}_I = \mathrm{E}(X_I) = \mathrm{Var}(X_I) = cI \tag{1-23}$$

$$P_I(n) = \frac{(cI)^n}{n!} e^{-cI} \tag{1-24}$$

As this will be useful later, we calculate also the average of the square of the number of clicks:

$$\mathrm{E}(X_I^2) = \mathrm{Var}(X_I) + (\mathrm{E}(X_I))^2 = cI + c^2 I^2 \tag{1-25}$$

We can proceed now to the calculation of the average and of the variance of the number of counts in the interval $\tau$. The probability of counting $n$ clicks in the interval $\tau$ is:

$$P(n) = \int_0^{+\infty} P_I(n) \cdot P(I)\, dI \tag{1-26}$$

Box 1-1 - Limit of the binomial distribution for $N \to +\infty$

In order to retrieve the limit for large $N$ of Eq. (1-11), it is useful first to take into account the asymptotic expressions of the factorials (Stirling approximation):

$$N! \sim \sqrt{2\pi N}\left(\frac{N}{e}\right)^N \qquad\qquad (N-n)! \sim \sqrt{2\pi(N-n)}\left(\frac{N-n}{e}\right)^{N-n}$$

In this way we develop the ratio:

$$\frac{N!}{(N-n)!} \sim \frac{\sqrt{2\pi N}\left(\frac{N}{e}\right)^N}{\sqrt{2\pi(N-n)}\left(\frac{N-n}{e}\right)^{N-n}} = \sqrt{\frac{N}{N-n}}\frac{N^n\,(N-n)^n}{(N-n)^N\,e^n}$$

At this point we evaluate further the asymptotic expressions:

$$\sqrt{\frac{N}{N-n}} \sim 1 \qquad\qquad \left(\frac{N}{N-n}\right)^N = \left(1-\frac{n}{N}\right)^{-N} = \left[\left(1-\frac{n}{N}\right)^{-\frac{N}{n}}\right]^n \sim e^n$$

where we have considered $x = -\frac{N}{n}$ and the limit:

$$\lim_{x\to+\infty}\left(1-\frac{1}{x}\right)^{-x} = \lim_{x\to+\infty}\left(\frac{x-1}{x}\right)^{-x} = \lim_{x\to+\infty}\left(\frac{x}{x-1}\right)^{x} = \lim_{y\to+\infty}\left(\frac{y+1}{y}\right)^{y+1} = \lim_{y\to+\infty}\left(1+\frac{1}{y}\right)^{y+1} = e$$

In this way we obtain finally:

$$\frac{N!}{(N-n)!} \sim 1\cdot e^n\frac{(N-n)^n}{e^n} = (N-n)^n$$

With the help of these results we proceed:

$$P(n) \sim \frac{1}{n!}(N-n)^n\left(\frac{\bar{n}}{N}\right)^n\left(1-\frac{\bar{n}}{N}\right)^{N-n}$$

$$P(n) \sim \frac{\bar{n}^n}{n!}\underbrace{\left(\frac{N-n}{N}\right)^n}_{\asymp 1}\left(1-\frac{\bar{n}}{N}\right)^N\underbrace{\left(1-\frac{\bar{n}}{N}\right)^{-n}}_{\asymp 1}$$

$$P(n) \sim \frac{\bar{n}^n}{n!}\left(1-\frac{\bar{n}}{N}\right)^N = \frac{\bar{n}^n}{n!}\underbrace{\left[\left(1-\frac{\bar{n}}{N}\right)^{-\frac{N}{\bar{n}}}\right]^{-\bar{n}}}_{\asymp e^{-\bar{n}}}$$

and finally we can conclude:

$$\lim_{N\to+\infty}P(n) = \lim_{N\to+\infty}\frac{N!}{n!\,(N-n)!}\left(\frac{\bar{n}}{N}\right)^n\left(1-\frac{\bar{n}}{N}\right)^{N-n} = \frac{\bar{n}^n}{n!}e^{-\bar{n}}$$

The average becomes:

$$\mathrm{E}(X) = \sum_{n=0}^{+\infty} n \cdot P(n) = \sum_{n=0}^{+\infty} n \int_0^{+\infty} P_I(n) \cdot P(I)\, dI = \int_0^{+\infty} \left( \sum_{n=0}^{+\infty} n P_I(n) \right) \cdot P(I)\, dI \tag{1-27}$$

Note that we were allowed to exchange the order of the integration and of the summation, as they are conducted on independent variables. We note that, by definition, $\sum_{n=0}^{+\infty} n p_I(n) = \mathrm{E}(X_I)$ so we can substitute Eq. (1-23):

$$\mathrm{E}(X) = \int_0^{+\infty} cI\, P(I)\, dI = c \int_0^{+\infty} I\, P(I)\, dI = c\bar{I} \tag{1-28}$$

With a similar procedure we calculate the average of the square of the number of counts:

$$\begin{aligned}
\mathrm{E}(X^2) &= \sum_{n=0}^{+\infty} n^2 \cdot P(n) = \sum_{n=0}^{+\infty} n^2 \int_0^{+\infty} P_I(n) \cdot P(I)\, dI = \\
&= \int_0^{+\infty} \left( \sum_{n=0}^{+\infty} n^2 P_I(n) \right) \cdot P(I)\, dI = \int_0^{+\infty} \mathrm{E}(X_I^2) \cdot P(I)\, dI = \\
&= \int_0^{+\infty} (cI + c^2 I^2)\, P(I)\, dI = c \int_0^{+\infty} I\, P(I)\, dI + c^2 \int_0^{+\infty} I^2\, P(I)\, dI = \\
&= c\bar{I} + c^2\, \mathrm{E}(I^2)
\end{aligned} \tag{1-29}$$

and finally we retrieve the variance:

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - (\mathrm{E}(X))^2 = c\bar{I} + c^2\, \mathrm{E}(I^2) - c^2 \bar{I}^2 = c\bar{I} + c^2 [\mathrm{E}(I^2) - \bar{I}^2] = c\bar{I} + c^2\, \mathrm{Var}(I) \tag{1-30}$$

For convenience we rewrite together the main results we have obtained:

$$\boxed{\begin{aligned} \mathrm{E}(X) &= c\bar{I} \\ \mathrm{Var}(X) &= c\bar{I} + c^2\, \mathrm{Var}(I) \end{aligned}} \tag{1-31}$$

The case with a constant intensity can be clearly derived from the latter results, just by substituting $\mathrm{Var}(I) = 0$. Note that, since by definition $\mathrm{Var}(I) \geq 0$, we always have $\mathrm{Var}(X) \geq c\bar{I} = \mathrm{E}(X)$.
These results are solidly confirmed by experiments performed with many kind of light sources: candles, fluorescent lamps, incandescent lamps and even lasers. In particular, lasers are the sources that can provide a constant intensity with the greatest accuracy.

## 1.3 Classification of photo-counting statistics

In the previous Section, starting from a semi-classical model for the photo-detection process, we have derived precise rules that the statistics of the photo-counts must follow, summarized in Eqs. (1-31). The fact that the average number of photo-counts per time-bin is maintained proportional to the average intensity $\bar{I}$ is perhaps not too surprising. More interestingly, however, we observe that the variance of the photo-counts cannot take arbitrary values, as it cannot be smaller than the average. The limit case $\mathrm{Var}(X) \equiv \mathrm{E}(X)$ is reached for a light beam with perfectly constant intensity, for which $\mathrm{Var}(I) = 0$. In all other cases, when the intensity fluctuates in time, the variance shall be larger. Indeed, it is customary in Quantum Optics to classify the photo-counting statistics on the basis on the relation between $\mathrm{Var}(X)$ and $\mathrm{E}(X)$.

| Photo-counting statistics | |
|---|---|
| $\mathrm{Var}(X) > \mathrm{E}(X)$ | **super-Poissonian** |
| $\mathrm{Var}(X) = \mathrm{E}(X)$ | **Poissonian** |
| $\mathrm{Var}(X) < \mathrm{E}(X)$ | **sub-Poissonian** |

The first two cases are compatible with the semi-classical picture we have used in our model. The third case instead, would appear impossible on the basis of our previous discussion. As a matter of fact, if we abandon the semi-classical vision and we consider a light beam that is composed of discrete quanta, it is not difficult to conceive a sub-Poissonian photo-counting statistics. It is sufficient to consider a photon stream organized in a suitably ordered fashion. For instance, a stream of single photons equally spaced in time, whose time separation is precisely $\tau$, would give rise to a photo-count statistics with $\mathrm{E}(X) = \bar{n} = 1$ and $\mathrm{Var}(n) = 0 < \bar{n}$. At least, this would happen if we have a perfectly responsive detector that clicks deterministically when each photon arrives. In this case it is the third hypothesis of our model that fails: sub-Poissonian statistics implies indeed the existence of a *temporal correlation* between the photo-count events.
Importantly, an experiment that showed a sub-Poissonian photo-count statistics, would be indeed a strong indication of the necessity of a quantum model for the light field. Namely, its interpretation would require granularity as a fundamental feature of the radiation, and not just of the interaction between radiation and matter. We will report an example of such an experiment in Section 1.5.
By adopting a picture of the radiation in terms of photons, statistical considerations can be extended to the beam itself and not only to the photo-detection process. A photon stream itself may possess a super-Poissonian, Poissonian or sub-Poissonian statistics, and not only the photo-counts. However, it is important to note here that the statistical properties of a light beam can be fragile. As we discuss in the next Section, they can be strongly modified by the presence of optical losses. Inefficiency in the detection process, i.e. the fact that some of the photon impinging on the detector may not give rise to an output 'click' and are not measured, is also equivalent to loss. Important discrepancies can thus exist between the actual statistics of the photon beam and the statistics of the photo-counts. Losses and inefficiencies are particularly detrimental when it comes to reveal experimentally the statistics of sub-Poissonian light.

## 1.4 Effects of losses and inefficiencies

We study here how the statistical properties of a light beam, composed of travelling photons, are altered by introducing losses or detection inefficiencies. Losses can be modeled simply by introducing a beam splitter with transmissivity $\eta$. Namely, each photon impinging on the beam-splitter has a probability $\eta$ of being transmitted and $1 - \eta$ of being reflected away. In this way, as we only look at the transmitted beam, the beam-splitter is giving a controlled loss of photons (Fig. 2).
As already mentioned, inefficiency in the detection process, i.e. the fact that only a fraction $\eta$ of the impinging photons gives rise to photo-counts, are perfectly equivalent to loss. Indeed, if we just look at the photo-counts, it is impossible to distinguish an ideal detector preceded by a loss element with transmission $\eta$, from an imperfect detector with only $\eta$ efficiency and no additional loss at the input. Hence, first of all, let us look in more detail at what can occur at the beam-splitter. We consider a single, isolated photon entering its input port and we call $X_1$ the random variable of the photon counts after the beam splitter. This variable can only take the values 0 or 1 and follows the statistics

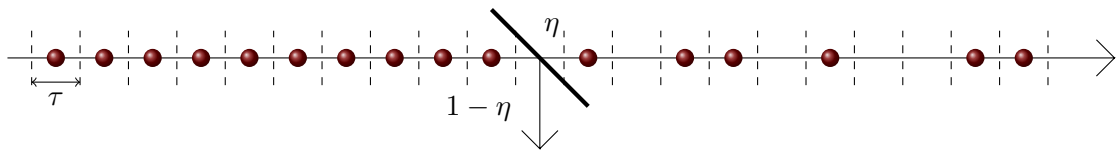


Figure 2: The statistics of a stream of single photons is strongly modified by the interposition of an ideal beam splitter with transmission $\eta$. The beam splitter removes randomly a set of photons from the stream, with each input photon having a probability $1 - \eta$ of being removed. We show here the simple example of a stream of uniformly spaced photons. If we define a proper time bin $\tau$ for photon-counting, we have at the input $\mathrm{E}(X_{in}) = 1$ and $\mathrm{Var}(X_{in}) = 0$. After the beam splitter, instead, $\mathrm{E}(X_{out}) < 1$ and $\mathrm{Var}(X_{out}) > 0$.

of the Bernoulli trial that we have discussed already in Section 1.2. In detail, the probabilities of the two events are:

$$P(0) = 1 - \eta \qquad\qquad P(1) = \eta \tag{1-32}$$

while averages and variances are given by:

$$\mathrm{E}(X_1) = 0 \cdot P_1(0) + 1 \cdot P_1(1) = \eta \tag{1-33}$$
$$\mathrm{E}(X_1^2) = 0 \cdot P_1(0) + 1^2 \cdot P_1(1) = \eta \tag{1-34}$$
$$\mathrm{Var}(X_1) = \mathrm{E}(X_1^2) - (\mathrm{E}(X_1))^2 = \eta - \eta^2 = \eta(1-\eta) \tag{1-35}$$

Let us now consider a set of $m$ photons entering the beam splitter. Each of them has a probability $\eta$ of being transmitted, independently from the other ones. If we call $X_m$ the random variable counting the photons at the output, this will be the sum of random variables distributed identically to $X_1$, so that:

$$\mathrm{E}(X_m) = m\mathrm{E}(X_1) = m\eta \tag{1-36}$$
$$\mathrm{Var}(X_m) = m\mathrm{Var}(X_1) = m\eta(1-\eta) \tag{1-37}$$
$$\mathrm{E}(X_m^2) = \mathrm{Var}(X_m) + (\mathrm{E}(N_m))^2 = m\eta(1-\eta) + m^2\eta^2 \tag{1-38}$$

Its probability distribution will be a binomial distribution:

$$P_m(n) = \binom{m}{n} \eta^n (1-\eta)^{m-n} \tag{1-39}$$

but the latter expression is not too important for what follows.
We assume at this point that an arbitrary light beam, consisting in a stream of photons statistically distributed in time, is incident on the beam splitter. We will call $X_{in}$ the random variable counting the photons in a time bin of duration $\tau$ before the beam splitter, and $X_{out}$ the analogous random variable that refers to the beam splitter output.
Relevant statistical quantities for $X_{in}$ may be the following:

$$\mathrm{E}(X_{in}) = \sum_{m=0}^{+\infty} m P_{in}(m) \tag{1-40}$$
$$\mathrm{E}(X_{in}^2) = \sum_{m=0}^{+\infty} m^2 P_{in}(m) \tag{1-41}$$
$$\mathrm{Var}(X_{in}) = \mathrm{E}(X_{in}^2) - (\mathrm{E}(X_{in}))^2 \tag{1-42}$$

where we have indicated as $P_{in}(m)$ the probability of having $m$ photons in the time bin.
It is not difficult to realize that the probability to detect $n$ photons out of the beam splitter in a given time bin can be written as:

$$P_{out}(n) = \sum_{m=n}^{+\infty} P_m(n) \cdot P_{in}(m) \tag{1-43}$$

Indeed in the above expression we have considered, for each possible $m$-photon group compatible with the detection of $n$ photon at the output (i.e. the ones with $m \geq n$), the probability $P_m(n)$ that $n$ of them are transmitted.
Exploiting this expression, we can begin to calculate the average photon number at the output:

$$\mathrm{E}(X_{out}) = \sum_{n=0}^{+\infty} n \cdot P_{out}(n) = \sum_{n=0}^{+\infty} \sum_{m=n}^{+\infty} n\, P_m(n) \cdot P_{in}(m) \tag{1-44}$$

Here we observe that the double summation $\sum_{n=0}^{+\infty}\sum_{m=n}^{+\infty}$ is running on a subset of all the pairs $(n, m)$, by scanning $n$ from zero to infinity, and for each $n$ by considering the $m$ values for which $m \geq n$. Actually, it would be the same thing to select the pairs $(n, m)$ by running $m$ from zero to infinity and for each $m$ considering the pairs where $n \leq m$. Namely, we can invert the order of the summations on $n$ and on $m$ provided that we adjust the bounds:

$$\sum_{n=0}^{+\infty}\sum_{m=n}^{+\infty} \equiv \sum_{m=0}^{+\infty}\sum_{n=0}^{m} \tag{1-45}$$

Therefore, we can write:

$$\mathrm{E}(X_{out}) = \sum_{m=0}^{+\infty}\sum_{n=0}^{m} nP_m(n)\cdot P_{in}(m) = \sum_{m=0}^{+\infty}\left(\sum_{n=0}^{m} nP_m(n)\right)\cdot P_{in}(m) \tag{1-46}$$

But we note that $\sum_{n=0}^{m} np_m(n) = \mathrm{E}(X_m) = m\eta$, hence:

$$\mathrm{E}(X_{out}) = \sum_{m=0}^{+\infty} m\eta\cdot P_{in}(m) = \eta\sum_{m=0}^{+\infty} m\cdot P_{in}(m) = \eta\,\mathrm{E}(X_{in}) \tag{1-47}$$

In a similar fashion we can compute the average of $X_{out}^2$:

$$\mathrm{E}(X_{out}^2) = \sum_{n=0}^{+\infty} n^2\cdot P_{out}(n) = \sum_{n=0}^{+\infty}\sum_{m=n}^{+\infty} n^2P_m(n)\cdot P_{in}(m) = \sum_{m=0}^{+\infty}\sum_{n=0}^{m} n^2P_m(n)\cdot P_{in}(m) \tag{1-48}$$

Here we note that $\sum_{n=0}^{m} n^2P_m(n) = \mathrm{E}(X_m^2) = m\eta(1-\eta) + m^2\eta^2$, thus it follows:

$$\begin{aligned}\mathrm{E}(X_{out}^2) &= \sum_{m=0}^{+\infty}(m\eta(1-\eta) + m^2\eta^2)\cdot P_{in}(m) = \\ &= \eta(1-\eta)\sum_{m=0}^{+\infty} m\,P_{in}(m) + \eta^2\sum_{m=0}^{+\infty} m^2\,P_{in}(m) = \\ &= \eta(1-\eta)\mathrm{E}(X_{in}) + \eta^2\mathrm{E}(X_{in}^2)\end{aligned} \tag{1-49}$$

We have now all the elements to calculate the variance:

$$\begin{aligned}\mathrm{Var}(X_{out}) &= \mathrm{E}(X_{out}^2) - (\mathrm{E}(X_{out}))^2 = \\ &= \eta(1-\eta)\mathrm{E}(X_{in}) + \eta^2\mathrm{E}(X_{in}^2) - \eta^2(\mathrm{E}(X_{in}))^2 = \eta^2\,\mathrm{Var}(X_{in}) + \eta(1-\eta)\mathrm{E}(X_{in})\end{aligned} \tag{1-50}$$

Before commenting them, let us collect together, for convenience, our main results that summarize how the statistical properties of $X_{out}$ are related to those of $X_{in}$:

$$\boxed{\begin{aligned}\mathrm{E}(X_{out}) &= \eta\,\mathrm{E}(X_{in}) \\ \mathrm{Var}(X_{out}) &= \eta^2\,\mathrm{Var}(X_{in}) + \eta(1-\eta)\mathrm{E}(X_{in})\end{aligned}} \tag{1-51}$$

The first formula just states that the average of the counts, with respect to the input beam, is multiplied by a factor $\eta$. This is simply consistent with the linear optical attenuation operated by the beam splitter.

The second formula describes a non-trivial relation between the variance of the output beam and the average and variance of the input beam. To see more clearly the implications, it may be useful to rewrite it as:

$$\mathrm{Var}(X_{out}) = \eta^2(\mathrm{Var}(X_{in}) - \mathrm{E}(X_{in})) + \eta\mathrm{E}(X_{in}) = \eta^2(\mathrm{Var}(X_{in}) - \mathrm{E}(X_{in})) + \mathrm{E}(X_{out}) \tag{1-52}$$

$$\boxed{\mathrm{Var}(X_{out}) - \mathrm{E}(X_{out}) = \eta^2(\mathrm{Var}(X_{in}) - \mathrm{E}(X_{in}))} \tag{1-53}$$

If the input beam is Poissonian we have $\mathrm{Var}(X_{in}) - \mathrm{E}(X_{in}) = 0$ and we get $\mathrm{Var}(X_{out}) = \mathrm{E}(X_{out})$: the output beam is again perfectly Poissonian irrespectively of the value of $\eta$. This is a peculiar case in which the statistics is not fundamentally altered.
In the case of a super-Poissonian input beam, $\mathrm{Var}(X_{in}) - \mathrm{E}(X_{in}) > 0$ before the beam splitter and $\mathrm{Var}(X_{out}) - \mathrm{E}(X_{out}) > 0$ also afterwards, which means that the beam remains in principle super-Poissonian. Similarly, in the case of a sub-Poissonian input beam $\mathrm{Var}(X_{in}) - \mathrm{E}(X_{in}) < 0$ at the input and $\mathrm{Var}(X_{out}) - \mathrm{E}(X_{out}) < 0$ at the output: the beam remains in principle sub-Poissonian. However, importantly, the difference between variance and average, while keeping its sign, *is scaled down* by a factor $\eta^2$. This means that the output beam has become *more similar* to a Poissonian beam.
In the case of high loss, i.e. $\eta \simeq 0$, after the beam splitter we will have $\mathrm{Var}(X_{out}) - \mathrm{E}(X_{out}) \simeq 0$ for any kind of input beam: the output statistics becomes practically Poissonian irrespective of the input statistics. This shows how critical it is to have low-loss channels and efficient detector, in order to characterize experimentally the photon statistics of an optical beam.

## 1.5 Experimental detection of sub-Poissonian light

Observing experimentally the occurrence of sub-Poissonian photo-count statistics is a compelling evidence for modeling light in terms of photons. However, as proven above, photon statistics is an extremely fragile feature: low-loss optical propagation channels and high-efficiency detectors are indispensable for its direct characterization. It took indeed a few decades, from the first studies on the quantization of the electromagnetic field, to the detection in experiments of **sub-Poissonian light statistics**. We discuss here, as an instructive example, a work reported by P.R. Tapster and collaborators in 1987.[3] The experiment moved along the following lines, which we shall examine in some depth.

1. a carefully designed electric circuit is used to generate a **sub-Poissonian flow of electrons**;
2. a high-efficiency LED, included in such a circuit, generates photons from the flowing electrons;
3. emitted photons are detected with high-efficiency by a photo-diode.

Electrons flowing in a circuit are indeed discrete particles and their flow is characterized by statistics in the same way we have discussed for photo-counts and photons. Precisely as in the latter case, we can imagine to observe the electrons crossing a point in the circuit, and to count how many electrons are passing in time bins of width $\tau$. For uniformity of notation, we will name $X$ the random variable counting the electrons in the time bin. The current intensity $I_e$ is related to the average number of electrons $\mathrm{E}(X)$ by the formula:

$$I_e = \frac{|q_e|\mathrm{E}(X)}{\tau} \tag{1-54}$$

where $|q_e|$ is the modulus of the electron charge. In electric measurements it may be more practical to refer to the bandwidth $\Delta f = \frac{1}{2\tau}$ rather than to the time-bin width $\tau$. So we can write equivalently:

$$I_e = 2\,|q_e|\Delta f\,\bar{n}_e \tag{1-55}$$

The variance of the electron counts $\mathrm{Var}(X)$ is related to the current variance by an analogous proportionality relation (with squared coefficients):

$$\mathrm{Var}(I_e) = 4\,|q_e|^2(\Delta f)^2\,\mathrm{Var}(X) \tag{1-56}$$

This variance is typically referred to in electronics as *noise*. In electronic circuits, noise is due to different causes, often independent one from the other.

[3] P.R. Tapster *et al.* "Generation of Sub-Poissonianm Light by High-Efficiency Light-Emitting Diodes" Europhysics Letters **4**, 293 (1987).

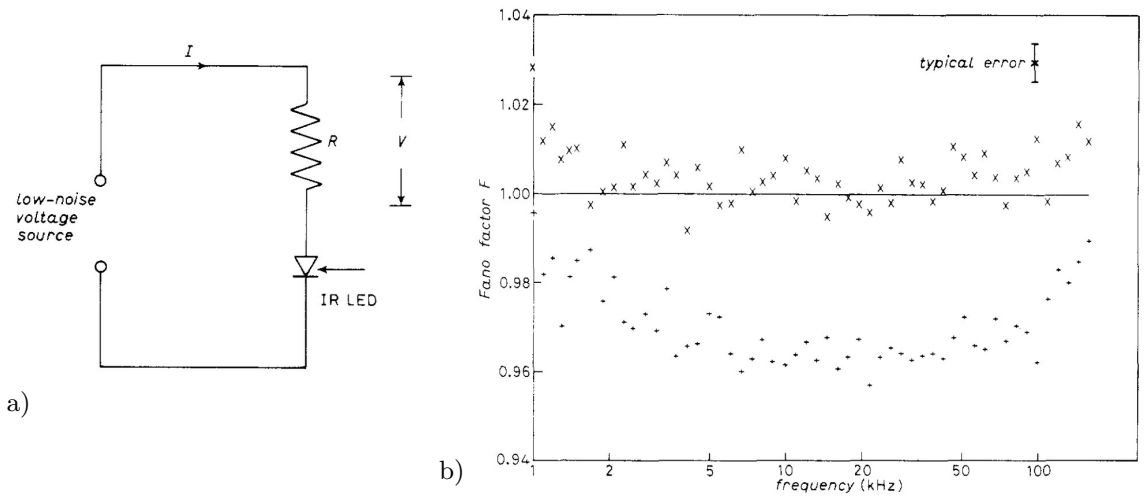


Figure 3: (a) Scheme of the electric circuit including the LED and the power supply, where the current is limited by the resistance $R$. (b) Experimentally measured Fano factors in the case of the LED source ('+' signs) and of a tungsten lamp ('x' sings). Both images are reproduced from the reference in footnote 3.

For instance, if we consider an electric circuit where a vacuum tube is placed in series, and in which the current is effectively *limited* by the random emission mechanisms of the electrons from the cathode, the electron statistics in the circuit follows the statistics of the emission process. The emission events are governed by the same laws and hypotheses that we have used to develop our semi-classical model for photo-detection. Namely, these events follow a Poissonian statistics with $\mathrm{Var}(X) = \mathrm{E}(X)$. The resulting noise in electronics is typically called *shot noise*; if it is the dominant form of noise we have from Eqs. (1-56) and (1-54):

$$[\mathrm{Var}(I_e)]_{\mathrm{shot}} = 4\,|q_e|^2(\Delta f)^2\,\mathrm{E}(I_e) = 2\,|q_e|\Delta f\, I_e \tag{1-57}$$

As a matter of fact, in other types of circuits noise may be mainly governed by other factors, and can be overall higher or lower. If the current in a circuit is limited by a resistor, the main cause of noise may be the thermal noise of the resistor itself. Tapster and collaborators considered indeed this kind of circuit, reported in Fig. 3a, to drive their LED photon source. A stabilized power supply is used to generate a very even and low-noise flow of electrons. The resistor gives the dominant noise contribution, according to the Johnson-Nyquist formula:

$$\mathrm{Var}(I) = \frac{4\kappa_B T}{R}\,\Delta f \tag{1-58}$$

where $\kappa_B$ is the Bolztmann constant, $T$ is the absolute temperature and $R$ the resistance value.
We can compare the actual current variance in the circuit, given by Eq. (1-58), with the variance we would have *in case shot noise was the dominant form of noise.* In detail, if we look for a lower-than-shot noise condition, we can impose:

$$\mathrm{Var}(I) < [\mathrm{Var}(I_e)]_{\mathrm{shot}} \tag{1-59}$$

$$\frac{4\kappa_B T}{R}\,\Delta f < 2\,|q_e|\Delta f\, I_e \tag{1-60}$$

$$\frac{2\kappa_B T}{|q_e|} < V \tag{1-61}$$

where we have called $V = R\,I_e$ the voltage drop across the resistor. If the voltage $V$ is sufficiently high, the current noise in the circuit is lower than the shot-noise limited case. Put in other words, in

this condition the variance in the electron counts $\mathrm{Var}(X)$ is lower than the average $\mathrm{E}(X)$: the electron flow is *sub-Poissonian*. Note that at room temperature this condition is not out of reach, as we have only $\frac{2\kappa_B T}{|q_e|} \sim 50$ mV.
From the experimental point of view, the **Fano factor** $F$ is a useful figure of merit to quantify how much the statistics deviates from the Poissonian one. It is defined as the ratio of the variance observed in the experiment over the variance that would be expected for a process with the same average,[4] if the statistics was Poissonian. The Fano factor naturally equals one, in case of shot-noise limited electronic currents. In case of the mentioned circuit the Fano factor can be written as:

$$F = \frac{\mathrm{Var}(I)}{[\mathrm{Var}(I_e)]_{\mathrm{shot}}} = \frac{2\kappa_B T}{|q_e| V} \tag{1-62}$$

In theory, this could be made arbitrarily small by tuning the voltage and the temperature.
To complete the discussion of the experimental apparatus, we note that the LED transforms the electron stream into a photon stream with some efficiency $\eta_{LED}$. Then, the photon stream is coupled with some optical efficiency[5] $\eta_{opt}$ to a photo-diode which, in turn, converts back photons to flowing electrons with an efficiency $\eta_{PD}$. A suitable low-noise circuitry is used to amplify the photo-diode current and, finally, an electrical noise measurement is performed on the amplified signal.
If all these efficiency terms were equal to one, we would have a one-to-one correspondence between the electrons flowing in the LED and the electrons flowing in the photodiode, with a replication of their statistics. Note that, if the interaction between the LED and the photo-diode wasn't truly mediated by a flux of discrete photons, and light was just a continuous classical wave, the detection process would forcedly follow the semiclassical model discussed in Section 1.2. A sub-shot-noise variance would never be measured in the final signal, in the latter case.
As a matter of fact, non-unit efficiencies bring in a degradation in the statistics, as we have shown in Section 1.4. In particular, tge degradation of the Fano factor is calculated by transforming properly Eq. (1-51). The Fano factor of the initial electron stream can be written in terms of the statistical figures at the particle level as:

$$F = \frac{\mathrm{Var}(X)}{\mathrm{E}(X)} \tag{1-63}$$

We name $X'$ the random variable counting the electrons at the end of the process, and carrying the effect of all inefficiencies, which we collect in a single parameter $\eta = \eta_{LED}\,\eta_{opt}\,\eta_{PD}$. From Eq. (1-51):

$$F' = \frac{\mathrm{Var}(X')}{\mathrm{E}(X')} = \frac{\eta^2\,\mathrm{Var}(X) + \eta(1-\eta)\mathrm{E}(X)}{\eta \mathrm{E}(X)} = \eta F + (1-\eta) \tag{1-64}$$

If the initial Fano factor is very small, we can consider for simplicity $F' \simeq 1 - \eta$.
Figure 3b shows ('+' signs), as a function of the frequency, the Fano factor retrieved experimentally by Tapster and collaborators using as a source the LED, driven by the circuit discussed above and shown in Fig. 3a. The numerator $\mathrm{Var}(I)$ was given by direct electrical measurements, while the denominator was estimated using Eq. (1-57). For comparison, the experimentally measured Fano factor retrieved for a common tungsten lamp, which gives a Poissonian statistics, is also plotted. The measured Fano factor for the LED was only slightly lower than one, due to the effects of the various inefficiencies (it is estimated $\eta \simeq 11.5\%$) and due to other sources of electrical noise in the detector's circuitry. Notwithstanding the small difference of the experimentally measured Fano factor with respect to the 'classical' unit value, the results are solid, because this difference is larger then the experimental uncertainty (also quantified Fig. 3).
This experiment was among the first ones that demonstrated directly the emission of sub-Poissonian light, and as such it was a direct and unequivocal indication of the need to adopt a quantized description of the electromagnetic field.

[4] Note that the variance can be equal to the average only if we consider *particle counts*. If we consider a quantity with some physical dimension, such as a current, equating the variance to the average has basically no meaning, because of dimensional incompatibility.

[5] Namely, not all photons emitted by the LED will reach the photo-diode sensitive surface.

# Notes on Quantum Optics - Section 2

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*Notes for the course 'Quantum Optics and Information' - A.Y. 2023/2024. These notes are provided for personal use only, with no claim of completeness or absence of errors.*

# 2 Classical and quantum harmonic oscillators

## 2.1 Roadmap for the quantization of the electromagnetic field

In order to apply a quantum theory to the electromagnetic field, a first step is to observe that the classical electromagnetic field can be described in terms of **classical harmonic oscillators**. This is an essentially mathematical observation, stemming from the formalism employed to model the electromagnetic radiation. Then, and this is the crucial second step, one **assumes as a postulate** that the formalism of the **quantum harmonic oscillator** can be transferred to that case, with suitable small adaptations.
This is a sort of theoretical 'jump' and one notes that the quantization of the electromagnetic field is not something that is proven or is mathematically demonstrated starting from the principles of the Schroedinger theory, which is valid in the case of massive particles evolving in a potential. In fact, it is a step *beyond* that theory, and further postulates - or principles - are needed.
There are different routes to this passage, which require a more or less involved knowledge and practice with the mathematical formalism of Quantum Mechanics. Here we will adopt one of the simplest approaches, in which, starting from the mentioned observation of the formal analogy between electromagnetic modes and harmonic oscillators, we will take as a postulate the Hamiltonian itself describing the electromagnetic field. Indeed, in building physical theories, the validity of a postulate is not given by how refined is the reasoning that leads to it, but is founded on the fact that the consequences drawn from such a postulate find solid experimental confirmations. In the case of this quantum theory of the electromagnetic field, indeed there are are many of the latter.
In the following, we will recall the fundamental elements of the classical and quantum theories for the harmonic oscillator. General foundations of Quantum Mechanics theory are briefly recalled in Appendix A. In Section 3 we shall elaborate the classical description of the electromagnetic field in a suitable fashion, so that our 'quantum jump' can be done.

## 2.2 Classical harmonic oscillator

The one-dimensional harmonic oscillator consists in a particle with mass $m$ that can move along a linear coordinate $x$, being subject to a potential which is parabolic with $x$. In classical mechanics, the total energy of the system is written as a sum of the kinetic energy $E_{\mathrm{K}}$ and the potential energy $U$ as:

$$\boxed{E_{\mathrm{tot}} = E_{\mathrm{K}} + U = \frac{1}{2m}p^2 + \frac{1}{2}m\omega^2 x^2} \tag{2-1}$$

with $p = m\frac{dx}{dt}$ is the momentum of the particle, and $\omega$ a parameter that describes the curvature of the potential.
The system evolution in time, when energy is conserved, is readily retrieved by imposing $\frac{dE_{\mathrm{tot}}}{dt} = 0$:

$$\frac{dE_{\mathrm{tot}}}{dt} = \frac{1}{2m}\,2p\,\frac{dp}{dt} + \frac{1}{2}m\omega^2\,2x\,\frac{dx}{dt} = 0 \tag{2-2}$$

$$\frac{dx}{dt}\left(m\frac{d^2x}{dt^2}\right) + m\omega^2\,x\,\frac{dx}{dt} = 0 \tag{2-3}$$

$$\frac{dx}{dt}\left(\frac{d^2x}{dt^2} + \omega^2 x\right) = 0 \tag{2-4}$$

The latter equation is solved trivially when the first factor vanishes ($\frac{dx}{dt} = 0$). This solution may be understood as the fact that a particle that is still for an instant, in that instant cannot change its energy. A more interesting condition is found by imposing the second factor as null:

$$\boxed{\frac{d^2x}{dt^2} + \omega^2 x = 0} \tag{2-5}$$

This differential equation has the well known oscillatory solution with angular frequency $\omega$:

$$x = A\cos(\omega t + \phi) \tag{2-6}$$

$$p = m\frac{dx}{dt} = -m\omega A\sin(\omega t + \phi) \tag{2-7}$$

As known, $A$ and $\phi$ are constants that depend on the initial conditions.
Instead of $x$ and $p$, we may choose a different set of coordinates:

$$\boxed{\alpha = m\omega x + \iota p \qquad \alpha^* = m\omega x - \iota p} \tag{2-8}$$

where $\iota$ is the imaginary unit. Basically, $\alpha$ is a complex variable ($\alpha^*$ is its complex conjugate). The real coordinates $x$ and $p$ are proportional respectively to the real and imaginary part of $\alpha$:

$$x = \frac{\alpha + \alpha^*}{2m\omega} = \frac{1}{m\omega}\Re(\alpha) \qquad p = \frac{\alpha - \alpha^*}{2\iota} = \Im(\alpha) \tag{2-9}$$

With these coordinates the motion is described by:

$$\alpha(t) = m\omega A\cos(\omega t + \phi) - \iota m\omega A\sin(\omega t + \phi) = m\omega A\, e^{-\iota(\omega t + \phi)} \tag{2-10}$$

$$\boxed{\alpha(t) = \alpha(0)e^{-\iota\omega t}} \tag{2-11}$$

with $\alpha(0) = m\omega A e^{-\iota\phi}$. Namely, it is a circular motion in the complex plane.
The total mechanical energy takes instead the form:

$$E_{\text{tot}} = \frac{1}{2m}(m^2\omega^2 x^2 + p^2) = \frac{1}{2m}(m\omega x + \iota p)(m\omega x - \iota p) \tag{2-12}$$

$$\boxed{E_{\text{tot}} = \frac{1}{2m}\alpha\alpha^* = \frac{1}{4m}(\alpha\alpha^* + \alpha^*\alpha) = \frac{1}{2m}|\alpha|^2} \tag{2-13}$$

## 2.3 Quantum harmonic oscillator

In the Quantum Mechanics framework, the physical description of a system is enclosed in the Hamiltonian operator:

$$\boxed{\hat{H} = \frac{1}{2m}\hat{p}^2 + \hat{V} = \frac{1}{2m}\hat{p}^2 + \frac{1}{2}m\omega^2\hat{x}^2} \tag{2-14}$$

where the one-dimensional momentum operator is:

$$\hat{p} = -\iota\hbar\frac{d}{dx} \tag{2-15}$$

In the classical model, by imposing the energy conservation, we have retrieved the system evolution in time. Here the time evolution of a system, given an initial state vector $|\psi\rangle$ is governed by the Schroedinger equation. It is convenient to study the energy eigenstates, which solve the equation:

$$\hat{H}\,|\psi\rangle = E\,|\psi\rangle \tag{2-16}$$

where $E$ is the eigenvalue, which is the energy of the state. The evolution in time of these states is simply produced by the multiplication with the phase term $e^{-\iota Et/\hbar}$.
An elegant theory for the harmonic oscillator is developed by defining the pair of operators $\hat{a}$ and $\hat{a}^\dagger$, called **ladder** operator for reasons that will be recalled later.

$$\hat{a} = \frac{1}{\sqrt{2m}}\frac{1}{\sqrt{\hbar\omega}}\,(m\omega\hat{x} + \iota\hat{p}) \tag{2-17}$$

$$\hat{a}^\dagger = \frac{1}{\sqrt{2m}}\frac{1}{\sqrt{\hbar\omega}}\,(m\omega\hat{x} - \iota\hat{p}) \tag{2-18}$$

Note that these operators are not Hermitian and do not correspond by themselves to physical observables. We observe that:

$$\begin{aligned}\hat{a}\hat{a}^\dagger &= \frac{1}{2m\hbar\omega}\left(m\omega\hat{x}+\imath\hat{p}\right)\left(m\omega\hat{x}-\imath\hat{p}\right) = \\ &= \frac{1}{2m\hbar\omega}\left(m^2\omega^2\hat{x}^2+\hat{p}^2-\imath m\omega\left(\hat{x}\hat{p}-\hat{p}\hat{x}\right)\right) = \\ &= \frac{1}{2m\hbar\omega}\left(m^2\omega^2\hat{x}^2+\hat{p}^2-\imath m\omega\left(+\imath\hbar\omega\right)\right) = \\ &= \frac{1}{\hbar\omega}\left(\frac{1}{2}m^2\omega^2\hat{x}^2+\frac{1}{2m}\hat{p}^2\right)+\frac{1}{2} \end{aligned} \tag{2-19}$$

$$\begin{aligned}\hat{a}^\dagger\hat{a} &= \frac{1}{2m\hbar\omega}\left(m\omega\hat{x}-\imath\hat{p}\right)\left(m\omega\hat{x}+\imath\hat{p}\right) = \\ &= \frac{1}{2m\hbar\omega}\left(m^2\omega^2\hat{x}^2+\hat{p}^2+\imath m\omega\left(\hat{x}\hat{p}-\hat{p}\hat{x}\right)\right) = \\ &= \frac{1}{2m\hbar\omega}\left(m^2\omega^2\hat{x}^2+\hat{p}^2+\imath m\omega\left(+\imath\hbar\omega\right)\right) = \\ &= \frac{1}{\hbar\omega}\left(\frac{1}{2}m\omega^2\hat{x}^2+\frac{1}{2m}\hat{p}^2\right)-\frac{1}{2} \end{aligned} \tag{2-20}$$

where we have employed the expression of the commutator[6] $[\hat{x},\hat{p}] = \hat{x}\hat{p}-\hat{p}\hat{x} = \imath\hbar$.
First, Eqs. (2-19) and (2-20) enable to calculate the commutator of $\hat{a}$ and $\hat{a}^\dagger$:

$$\boxed{[\hat{a},\hat{a}^\dagger] = \hat{a}\hat{a}^\dagger-\hat{a}^\dagger\hat{a} = 1} \tag{2-21}$$

Second, those relations allow to rewrite the Hamiltonian (2-14) as:

$$\hat{H} = \frac{1}{2}m\omega^2\hat{x}^2+\frac{1}{2m}\hat{p}^2 = \hbar\omega\hat{a}\hat{a}^\dagger-\frac{1}{2}\hbar\omega\hat{a}\hat{a}^\dagger-\frac{1}{2}\hbar\omega = \hbar\omega\left(\hat{a}\hat{a}^\dagger-\frac{1}{2}\right) \tag{2-22}$$

In fact, three equivalent expressions can be written for $\hat{H}$, by exploiting the commutator relation Eq. (2-21):

$$\boxed{\hat{H} = \hbar\omega\left(\hat{a}\hat{a}^\dagger-\frac{1}{2}\right) = \frac{\hbar\omega}{2}\left(\hat{a}\hat{a}^\dagger+\hat{a}^\dagger\hat{a}\right) = \hbar\omega\left(\hat{a}^\dagger\hat{a}^\dagger+\frac{1}{2}\right)} \tag{2-23}$$

The ladder operators do not commute with the Hamiltonian $\hat{H}$, the commutators being:

$$\begin{aligned}[\hat{H},\hat{a}] &= \hbar\omega[\hat{a}^\dagger\hat{a}+1/2,\hat{a}] = \hbar\omega[\hat{a}^\dagger\hat{a},\hat{a}]+\hbar\omega\underbrace{[1/2,\hat{a}]}_{=0} = \\ &= \hbar\omega(\hat{a}^\dagger\hat{a}\hat{a}-\underbrace{\hat{a}\hat{a}^\dagger}_{=\hat{a}^\dagger\hat{a}+1}\hat{a}) = \hbar\omega(\hat{a}^\dagger\hat{a}\hat{a}-\hat{a}^\dagger\hat{a}\hat{a}-\hat{a}) = -\hbar\omega\hat{a} \end{aligned} \tag{2-24}$$

$$[\hat{H},\hat{a}^\dagger] = ([\hat{a},\hat{H}^\dagger])^\dagger = ([\hat{a},\hat{H}])^\dagger = (-[\hat{H},\hat{a}])^\dagger = (+\hbar\omega\hat{a})^\dagger = \hbar\omega\hat{a}^\dagger \tag{2-25}$$

Finally, in the Heisenberg picture, the evolved operator $\hat{a}$ solve the equation:

$$\frac{d}{dt}\hat{a} = \frac{\imath}{\hbar}[\hat{H},\hat{a}] = -\imath\omega\hat{a} \tag{2-26}$$

[6] In fact, the commutator between $\hat{x} = x$ and $\hat{p} = -\imath\hbar\frac{d}{dx}$ is easily calculated from their definition. Given a generic (wave)function $\psi(x)$:

$$\hat{x}\hat{p}\,\psi(x)-\hat{p}\hat{x}\,\psi(x) = -x\imath\hbar\frac{d}{dx}\psi(x)+\imath\hbar\frac{d}{dx}(x\psi(x)) = -\imath\hbar x\frac{d}{dx}\psi(x)+\imath\hbar\psi(x)+\imath\hbar x\frac{d}{dx}\psi(x) = +\imath\hbar\psi(x)$$

| CLASSICAL | | QUANTUM |
|---|---|---|
| $\alpha = m\omega x + \imath p$ | $\longleftrightarrow$ | $\hat{a} = \frac{1}{\sqrt{2m}}\frac{1}{\sqrt{\hbar\omega}}\left(m\omega\hat{x} + \imath\hat{p}\right)$ |
| $\alpha^* = m\omega x - \imath p$ | $\longleftrightarrow$ | $\hat{a}^\dagger = \frac{1}{\sqrt{2m}}\frac{1}{\sqrt{\hbar\omega}}\left(m\omega\hat{x} - \imath\hat{p}\right)$ |
| $\alpha(t) = \alpha(0)\, e^{-\imath\omega t}$ | $\longleftrightarrow$ | $\hat{a}(t) = \hat{a}(0)\, e^{-\imath\omega t}$ |
| $\alpha^*(t) = \alpha^*(0)\, e^{+\imath\omega t}$ | $\longleftrightarrow$ | $\hat{a}^\dagger(t) = \hat{a}^\dagger(0)\, e^{+\imath\omega t}$ |
| $E_{\text{tot}} = \frac{1}{4m}(\alpha\alpha^* + \alpha^*\alpha)$ | $\longleftrightarrow$ | $\hat{H} = \frac{1}{2}\hbar\omega\left(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}\right)$ |

Table 1: Expressions of scalar quantities that are relevant in the classical model of the harmonic oscillator, compared with the operatorial expressions relevant for the quantum mechanical model.

which implies:

$$\boxed{\hat{a}(t) = \hat{a}(0)\, e^{-\imath\omega t}} \tag{2-27}$$

The relations for $\hat{a}^\dagger$ are retrieved by taking the Hermitian conjugates:

$$\frac{d}{dt}\hat{a}^\dagger = +\imath\omega\hat{a}^\dagger \tag{2-28}$$

$$\boxed{\hat{a}^\dagger(t) = \hat{a}^\dagger(0)\, e^{+\imath\omega t}} \tag{2-29}$$

It is now interesting to observe that the definition of the operators $\hat{a}$ and $\hat{a}^\dagger$ have brought in a cleaner formal similarity between the scalar equations governing the classical harmonic oscillator and the operatorial expressions related to the quantum model. This similarity, made apparent in Table 1, is based on the substitution of the variables $\alpha$ and $\alpha^*$ with the operators $\hat{a}$ and $\hat{a}^\dagger$, with an adjustment of the multiplicative constants (in particular we need to take $\hbar = \frac{1}{2m\omega}$).
Note that the quantum Hamiltonian, written in terms of $\hat{a}$ and $\hat{a}^\dagger$, does not include the mass $m$ as a parameter. This will be crucial to transfer the harmonic oscillator model to another system, the electromagnetic mode, where there is no meaningful reference to any mass.

## 2.4 Properties of the $\hat{a}$ and $\hat{a}^\dagger$ operators

The ladder operators $\hat{a}$ and $\hat{a}^\dagger$ manipulate the eigenstates of the Hamiltonian of the oscillator in a distinctive way, which allows an easy calculation of the eigenenergies.

> **Proposition 2.1** *If $|\psi_n\rangle$ is an eigenstate of $\hat{H}$ with eigenvalue $E_n$, then $\hat{a}\,|\psi_n\rangle$ satisfies the eigenvalue equation, with eigenvalue $E_n - \hbar\omega$.*

We start with the eigenstate $|\psi_n\rangle$, associated to the eigenvalue $E_n$ (which is its energy), namely satisfying $\hat{H}\,|\psi_n\rangle = E_n\,|\psi_n\rangle$. Let us apply $\hat{H}$ to the state $\hat{a}\,|\psi_n\rangle$.

$$\hat{H}\left(\hat{a}\,|\psi_n\rangle\right) = \underbrace{\hat{H}\hat{a}}_{=\hat{a}\hat{H}+[\hat{H},\hat{a}]}\,|\psi_n\rangle = \hat{a}\hat{H}\,|\psi_n\rangle - \hbar\omega\hat{a}\,|\psi_n\rangle = \hat{a}(E_n - \hbar\omega)\,|\psi_n\rangle = (E_n - \hbar\omega)(\hat{a}\,|\psi_n\rangle) \tag{2-30}$$

■

**Proposition 2.2** *If $|\psi_n\rangle$ is an eigenstate of $\hat{H}$ with energy $E_n$, then $\hat{a}^\dagger |\psi_n\rangle$ satisfies the eigenvalue equation, with eigenvalue $E_n + \hbar\omega$.*

We proceed analogously, start the eigenstate $|\psi_n\rangle$ satisfying $\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle$. Let us apply $\hat{H}$ to $\hat{a}^\dagger|\psi_n\rangle$:

$$\hat{H}(\hat{a}^\dagger|\psi_n\rangle) = \underbrace{\hat{H}\hat{a}^\dagger}_{=\hat{a}^\dagger\hat{H}+[\hat{H},\hat{a}^\dagger]}|\psi_n\rangle = \hat{a}^\dagger\hat{H}|\psi_n\rangle + \hbar\omega\hat{a}^\dagger|\psi_n\rangle = \hat{a}^\dagger(E_n+\hbar\omega)|\psi_n\rangle = (E_n+\hbar\omega)(\hat{a}^\dagger|\psi_n\rangle) \tag{2-31}$$

■

We highlight a subtle point here, that is useful to remind. Quantum states should be properly normalized, namely $|\langle\psi_n|\psi_n\rangle|^2 = 1$. However, the eigenvalue equation could be formally satisfied also by a ket multiplied by an arbitrary constant. The propositions above merely tells us that, by applying $\hat{a}$ or $\hat{a}^\dagger$ to the eigenstate $|\psi_n\rangle$, we find another solution to the eigenvalue equation, but the latter solution might not be a normalized state. We may thus add normalization constants:

$$\hat{a}|\psi_n\rangle = C_n|\psi_n'\rangle \tag{2-32}$$

$$\hat{a}^\dagger|\psi_n\rangle = D_n|\psi_n''\rangle \tag{2-33}$$

with $|\psi_n\rangle$, $|\psi_n'\rangle$ and $|\psi_n''\rangle$ are normalized eigenstates of the Hamiltonian, i.e. $|\langle\psi_n|\psi_n\rangle|^2 = |\langle\psi_n'|\psi_n'\rangle|^2 = \langle\psi_n''|\psi_n''\rangle|^2 = 1$. We will find these multiplicative constants later.
The operators $\hat{a}$ and $\hat{a}^\dagger$ are **lowering** and **raising** operators respectively, in the sense that (apart from these possible multiplicative constant) they transform an eigenstate of the energy in another eigenstate with lower or higher energy, by a step of $\hbar\omega$. By applying repeatedly one or the other operator, one might guess that it is always possible to find states with indefinitely low or high energy. This is not true for that regards the lowering part, as there exists indeed a state with minimum energy.

**Proposition 2.3** *There exist a fundamental eigenstate $|\psi_0\rangle$ with minimum energy, such that $\hat{a}|\psi_0\rangle = 0$.*

The condition $\hat{a}|\psi_0\rangle = 0$ means that the application of $\hat{a}$ to this hypothetical $|\psi_0\rangle$ does not produce a ket state, but precisely *zero*, which is no state. The proof of this proposition consists in finding the wavefunction of this $|\psi_0\rangle$, which indeed an evidence that this state exists.

$$\begin{aligned}
\hat{a}\psi(x) &= 0 \\
\frac{1}{\sqrt{2m}}\frac{1}{\sqrt{\hbar\omega}}(m\omega\hat{x} + \imath\hat{p})\psi(x) &= 0 \\
(m\omega x + \hbar\frac{d}{dx})\psi(x) &= 0 \\
m\omega x\psi(x) + \hbar\frac{d\psi(x)}{dx} &= 0 \\
\frac{d\psi(x)}{\psi(x)} &= -\frac{m\omega}{\hbar}x\,dx \\
\int_{\psi(0)}^{\psi(x)}\frac{d\psi(x)}{\psi(x)} &= -\int_0^x\frac{m\omega}{\hbar}x\,dx \\
\ln\frac{\psi(x)}{\psi(0)} &= -\frac{m\omega}{2\hbar}x^2 \\
\psi(x) &= \psi(0)\,e^{-\frac{m\omega}{2\hbar}x^2}
\end{aligned}$$

We have found the well known gaussian wavefunction of the ground state of the quantum harmonic oscillator. The factor $\psi(0)$ can be found by imposing the normalization condition $\int_{-\infty}^{+\infty}|\psi(x)|^2\,dx = 1$. In particular, one finds $\psi(0) = \sqrt[4]{\frac{m\omega}{\hbar\pi}}$. ■
Note that a corresponding limitation on the energy from the top does not exist. If we write the equation $\hat{a}^\dagger\psi(x) = 0$, trying to find the state with maximum energy, we end up with no solution.

In fact we would substitute $\hat{a}^\dagger = \frac{1}{\sqrt{2m}}\frac{1}{\sqrt{\hbar\omega}}(m\omega x - \hbar\frac{d}{dx})$ and get in the end $\psi(x) = \psi(0)\, e^{+\frac{m\omega}{2\hbar}x^2}$; but due to the presence of the '+' sign at the exponent, this function is not normalizable and does not represent a wavefunction.
With the properties proven up to now we can have a clear representation of the structure of the energy eigenstates. Starting from the bottom rung $|\psi_0\rangle$, by repeated application of $\hat{a}^\dagger$ one can go through all the eigenstates $|\psi_n\rangle$. We need just to allow for a possible multiplicative constant, that here we write as $A_n$:

$$|\psi_n\rangle = A_n(\hat{a}^\dagger)^n\,|\psi_0\rangle \tag{2-34}$$

In particular, having called $E_n$ the energy of the eigenstate $|\psi_n\rangle$:

$$\hat{H}\,|\psi_n\rangle = E_n\,|\psi_n\rangle \tag{2-35}$$

one proves that the eigenstates build a ladder with energies equally spaced by $\hbar\omega$.

**Proposition 2.4** *The eigenvalue associated to the state $|\psi_n\rangle$, namely its energy, is given by:*

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega \tag{2-36}$$

We can start from $\psi_0$:

$$\hat{H}\,|\psi_0\rangle = \hbar\omega(\hat{a}^\dagger\hat{a} + 1/2)\,|\psi_0\rangle = \hbar\omega\hat{a}^\dagger\underbrace{\hat{a}\,|\psi_0\rangle}_{=0} + \frac{1}{2}\hbar\omega\,|\psi_0\rangle = \frac{1}{2}\hbar\omega\,|\psi_0\rangle \tag{2-37}$$

which means $E_0 = \frac{1}{2}\hbar\omega$.
Considering that from Eq. (2-34), $|\psi_n\rangle = \frac{A_n}{A_{n-1}}\hat{a}^\dagger\,|\psi_{n-1}\rangle$ we can retrieve the following recursive relation:

$$\begin{aligned} E_n\,|\psi_n\rangle = \hat{H}\,|\psi_n\rangle = \hat{H}\left(\frac{A_n}{A_{n-1}}\hat{a}^\dagger\,|\psi_{n-1}\rangle\right) &= \frac{A_n}{A_{n-1}}\hat{H}\left(\hat{a}^\dagger\,|\psi_{n-1}\rangle\right) = \\ &= \frac{A_n}{A_{n-1}}(E_{n-1} + \hbar\omega)\left(\hat{a}^\dagger\,|\psi_{n-1}\rangle\right) = (E_{n-1} + \hbar\omega)\,|\psi_n\rangle \end{aligned} \tag{2-38}$$

$$E_n = E_{n-1} + \hbar\omega \tag{2-39}$$

and this proves Eq. (2-36). ■
It is further useful to define the **number operator**:

$$\boxed{\hat{n} = \hat{a}^\dagger\hat{a}} \tag{2-40}$$

The eigenstates of $\hat{H}$ are also eigenstates of $\hat{n}$ with eigenvalue $n$, namely:

$$\hat{n}\,|\psi_n\rangle = (\hat{a}^\dagger\hat{a})\,|\psi_n\rangle = \left(\frac{\hat{H}}{\hbar\omega} - \frac{1}{2}\right)|\psi_n\rangle = \left(\frac{E_n}{\hbar\omega} - \frac{1}{2}\right)|\psi_n\rangle = n\,|\psi_n\rangle \tag{2-41}$$

Indeed typically called **number states** and are indicated with the ket $|n\rangle$.

$$\hat{n}\,|n\rangle = n\,|n\rangle \tag{2-42}$$

$$\hat{H}\,|n\rangle = E_n\,|n\rangle = \hbar\omega(n + 1/2)\,|n\rangle \tag{2-43}$$

To complete our survey on the operators $\hat{a}$ and $\hat{a}^\dagger$, and fully characterize their action on the normalized states, we can work out the multiplicative constants $C_n$ and $D_n$ in Eqs. (2-32) and (2-33).

**Proposition 2.5** *The operators $\hat{a}$ and $\hat{a}^\dagger$ act on the (normalized) number states as follows:*

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle \tag{2-44}$$

$$\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle \tag{2-45}$$

First we consider $C_n$. We have understood that $\hat{a}$ lowers the state $|n\rangle$ by one step:

$$\hat{a}\,|n\rangle = C_n\,|n-1\rangle \tag{2-46}$$

The Hermitian conjugate of this relation reads, on the other hand:

$$\langle n|\,\hat{a}^\dagger = \langle n-1|\,C_n^* \tag{2-47}$$

Therefore we can write:

$$1 = \langle n-1|n-1\rangle = \frac{1}{|C_n|^2}\,\langle n|\,\hat{a}^\dagger\hat{a}\,|n\rangle = \frac{1}{|C_n|^2}\,\langle n|\,\hat{n}\,|n\rangle = \frac{n}{|C_n|^2}\langle n|n\rangle = \frac{n}{|C_n|^2} \tag{2-48}$$

We proceed similarly for $D_n$:

$$\hat{a}^\dagger\,|n\rangle = D_n\,|n+1\rangle \qquad \langle n|\,\hat{a} = \langle n+1|\,D_n^* \tag{2-49}$$

$$1 = \langle n+1|n+1\rangle = \frac{1}{|D_n|^2}\,\langle n|\underbrace{\hat{a}\hat{a}^\dagger}_{=\hat{a}^\dagger\hat{a}+1=\hat{n}+1}|n\rangle = \frac{1}{|D_n|^2}\,\langle n|\,\hat{n}+1\,|n\rangle = \frac{n+1}{|D_n|^2}\langle n|n\rangle = \frac{n+1}{|D_n|^2} \tag{2-50}$$

Equations (2-48) and (2-50) only fix the moduli of $C_n$ and $D_n$, and would allow in principle for arbitrary phase terms. As global phase terms are not relevant, we keep $C_n$ and $D_n$ as real and we take:

$$C_n = \sqrt{n} \qquad\qquad D_n = \sqrt{n+1} \tag{2-51}$$

■

As a consequence of Prop. 2.5, the number states $|n\rangle$ can be written as:

$$\boxed{|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}\,|0\rangle} \tag{2-52}$$

the ket $|0\rangle$ indicating the ground state of the oscillator.
It is not difficult to prove that the number states are mutually orthogonal:

$$\langle m|n\rangle = \delta_{mn} \tag{2-53}$$

They also form a complete basis of the Hilbert space; this is expressed through the closure relation:

$$\sum_{n=0}^{+\infty}|n\rangle\,\langle n| = \hat{\mathbb{I}} \tag{2-54}$$

We can also express operators in this basis; in detail, an operator $\hat{Q}$ can be written as:

$$\hat{Q} = \hat{\mathbb{I}}\,\hat{Q}\,\hat{\mathbb{I}} = \left(\sum_{m=0}^{+\infty}|m\rangle\,\langle m|\right)\hat{Q}\left(\sum_{n=0}^{+\infty}|n\rangle\,\langle n|\right) = \sum_{m,n}\langle m|\,\hat{Q}\,|n\rangle\;|m\rangle\,\langle n| \tag{2-55}$$

Therefore, our ladder and number operators reads:

$$\hat{a} = \sum_{n=0}^{+\infty}\sqrt{n+1}\,|n\rangle\,\langle n+1| \tag{2-56}$$

$$\hat{a}^\dagger = \sum_{n=0}^{+\infty}\sqrt{n+1}\,|n+1\rangle\,\langle n| \tag{2-57}$$

$$\hat{n} = \hat{a}^\dagger\hat{a} = \sum_{n=0}^{+\infty} n\,|n\rangle\,\langle n| \tag{2-58}$$

# Notes on Quantum Optics - Electromagnetic fields: from classical to quantum

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# 3 Electromagnetic fields: from classical to quantum

## 3.1 Classical electromagnetic potentials in the vacuum

The **Maxwell equations** for the electric field $\vec{E}$ and magnetic field $\vec{B}$ in the absence of charges and currents read:

$$\begin{aligned} \nabla \cdot \vec{E} &= 0 & \nabla \cdot \vec{B} &= 0 \\ \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} & \nabla \times \vec{B} &= \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \end{aligned} \tag{3-1}$$

where we give the usual meaning to the constants $\mu_0$ and $\varepsilon_0$.
For what follows it is convenient to write the fields in terms of the **vector potential** $\vec{A}$ and of the **scalar potential** $\varphi$, defined by the relations:

$$\nabla \times \vec{A} = \vec{B} \tag{3-2}$$

$$\nabla \varphi = -\vec{E} - \frac{\partial \vec{A}}{\partial t} \tag{3-3}$$

Furthermore, we assume the Coulomb gauge:

$$\nabla \cdot \vec{A} = 0 \tag{3-4}$$

Relations (3-2) and (3-3) satisfy identically the second and third Maxwell equations ($\nabla \cdot \vec{B} = 0$ and $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$) which become somehow redundant in this framework. We remain with the first equation, that we can transform using Eq. (3-3):

$$\nabla \cdot \vec{E} = 0 \qquad \Rightarrow \qquad \nabla \cdot \left( \nabla \varphi + \frac{\partial \vec{A}}{\partial t} \right) = \nabla^2 \varphi + \frac{\partial (\nabla \cdot \vec{A})}{\partial t} = 0 \qquad \Rightarrow \qquad \nabla^2 \varphi = 0 \tag{3-5}$$

This allows us to licitly assume that the scalar potential is vanishing everywhere in the space:

$$\varphi = 0 \tag{3-6}$$

In addition, with the aid of Eqs. (3-2) and the fourth Maxwell equation becomes a wave equation for the vector potential:

$$\nabla \times \nabla \times \vec{A} = \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left( -\frac{\partial \vec{A}}{\partial t} \right) \qquad \Rightarrow \qquad \nabla^2 \vec{A} - \mu_0 \varepsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2} = 0 \tag{3-7}$$

In summary, the above discussion shows that in the absence of charges and currents the electromagnetic field is uniquely determined by the vector potential $\vec{A}$ in accordance to Eq. (3-7), the scalar potential $\varphi$ being null. Overall, the fields are determined by the following set of equations:

$$\boxed{\begin{aligned} \nabla \cdot \vec{A} &= 0 & \nabla^2 \vec{A} &= \mu_0 \varepsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2} \\ \vec{E} &= -\frac{\partial \vec{A}}{\partial t} & \vec{B} &= \nabla \times \vec{A} \end{aligned}} \tag{3-8}$$

### 3.2 Solution of the classical wave equation

Equation (3-7) is a vectorial wave equation, which will have different solutions depending on the details boundary conditions. In general, we can look for **solutions with separated variables** of the form:

$$\boxed{\vec{A}(\vec{r},t) = A_0\,\vec{f}(\vec{r})\,\alpha(t)} \tag{3-9}$$

Substituting Eq. (3-9) in Eq. (3-7) we obtain:

$$\nabla^2\left(A_0\,\vec{f}(\vec{r})\,\alpha(t)\right) - \mu_0\varepsilon_0\frac{\partial^2}{\partial t^2}\left(A_0\,\vec{f}(\vec{r})\,\alpha(t)\right) = 0 \tag{3-10}$$

$$\alpha(t)\nabla^2\vec{f}(\vec{r}) - \vec{f}(\vec{r})\,\mu_0\varepsilon_0\frac{\partial^2\alpha(t)}{\partial t^2} = 0 \tag{3-11}$$

which means that for each component $f_i(\vec{r})$ with $i = x, y, z$, we have:

$$\frac{1}{\mu_0\varepsilon_0}\frac{1}{f_i(\vec{r})}\nabla^2 f_i(\vec{r}) = \frac{1}{\alpha(t)}\frac{\partial^2\alpha(t)}{\partial t^2} = C \tag{3-12}$$

In the latter expression $C$ is an arbitrary constant that is convenient to rewrite as:

$$C = -\omega^2 \tag{3-13}$$

For the moment, $\omega$ is just another arbitrary constant: its physical meaning will be clear later. Finally, from Eq. (3-12) we can retrieve independent differential equations respectively in the spatial and time coordinates:

$$\nabla^2\vec{f}(\vec{r}) + \mu_0\varepsilon_0\,\omega^2\,\vec{f}(\vec{r}) = 0 \tag{3-14}$$

$$\frac{\partial^2\alpha(t)}{\partial t^2} + \omega^2\alpha(t) = 0 \tag{3-15}$$

We recognize that Eq. (3-14) is the vectorial Helmholtz equation: its solutions $\vec{f}(\vec{r})$ depend on the boundary conditions of the specific problem, and we will discuss them in some more detail in the following. Equation (3-15), on the other hand, is the familiar equation of the harmonic oscillator. The solutions $\alpha(t)$ as a function of the parameter $\omega$ thus come with the form:

$$\boxed{\alpha(t) = \alpha_\omega(t) = \alpha_\omega(0)\,e^{-\iota\omega t}} \tag{3-16}$$

which is a complex oscillation with angular frequency $\omega$.
We note that each of the factors in (3-9) might be scaled by a constant, and still it would be a valid solution of the wave equation. In fact, multiplicative constants and normalization criteria can be chosen with some arbitrariness. Here we shall adopt the following ones:

- $A_0$ keeps the physical constants and dimensions;
- $\vec{f}(\vec{r})$ is an adimensional vector function; it carries the direction and orientation of the $\vec{A}$ field, which determines in the end the polarization direction of the electromagnetic wave; depending on the boundary conditions of the specific problem, a normalization constraint can be further chosen;
- $\alpha(t)$ is adimensional and carries the amplitude of the oscillating field.

Before proceeding, we can also note that if we find a solution $\vec{A}(\vec{r},t)$ in the complex field, also its complex conjugate will be a solution. This holds also for the individual factors: if $\vec{f}(\vec{r})$ is a solution of Eq. (3-14), also $\vec{f}^*(\vec{r})$ is a solution; if $\alpha(t)$ is a solution of Eq. (3-15), also $\alpha^*(t)$ is a solution. In addition, as it is always the case for linear equations, sums of solutions are again solutions.

On the basis of these observations, in order to keep the fields as real quantities, we can improve the form of our solution and take, instead of Eq. (3-9):

$$\boxed{\vec{A} = A_0 \left[\vec{f}(\vec{r})\alpha(t) + \vec{f}^*(\vec{r})\alpha^*(t)\right]} \tag{3-17}$$

Each of these solutions corresponds to a **normal mode** of the electromagnetic field in the considered setting, determined by the boundary conditions. $\alpha(t)$ may be called the **normal variable** of that mode. From Eq. (3-17) we can derive an expression for the electric field $\vec{E}$ of the electromagnetic mode:

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} = -A_0 \left[\vec{f}(\vec{r})\frac{\partial \alpha(t)}{dt} + \vec{f}^*(\vec{r})\frac{\partial \alpha^*(t)}{dt}\right] \tag{3-18}$$

and taking into account Eq. (3-16):

$$\boxed{\vec{E} = \iota\omega A_0 \left[\vec{f}(\vec{r})\alpha(t) - \vec{f}^*(\vec{r})\alpha^*(t)\right]} \tag{3-19}$$

The expression of the magnetic field $\vec{B}$ is a bit more complex, due to the rotor operation, whose result depends on the specific form of $\vec{f}(\vec{r})$:

$$\boxed{\vec{B} = A_0 \left[\left(\nabla \times \vec{f}(\vec{r})\right)\alpha(t) + \left(\nabla \times \vec{f}^*(\vec{r})\right)\alpha^*(t)\right]} \tag{3-20}$$

## 3.3 Spatial part and discretization of the electromagnetic modes

As mentioned, the form of the spatial part $\vec{f}(\vec{r})$ depends on the specific problem and in particular on the boundary conditions imposed to the Helmholtz equation. For what follows, we will consider linearly polarized **plane wave solutions** with the form:[7]

$$\boxed{\vec{f}(\vec{r}) = f(\vec{r})\,\vec{\epsilon} = e^{\iota\vec{k}\cdot\vec{r}}\,\vec{\epsilon}} \tag{3-21}$$

where $\vec{k} = k_x\vec{u}_x + k_y\vec{u}_y + k_z\vec{u}_z$ is a vector such that $|k|^2 = k_x^2 + k_y^2 + k_z^2 = \mu_0\varepsilon_0\omega^2 = \frac{\omega^2}{c^2}$, called the *wavevector*, and $\vec{\epsilon}$ is a unit vector indicating the polarization of the wave.
One can show[8] that the condition $\nabla \cdot \vec{A} = 0$ forces the polarization vector $\vec{\epsilon}$ to be orthogonal with respecto to $\vec{k}$. It is thus meaningful to define for each mode the right-handed triplet of vectors $\vec{\epsilon}$, $\vec{\epsilon}'$ and $\vec{u}_k = \vec{k}/|\vec{k}|$ such that:

$$\vec{\epsilon} \times \vec{\epsilon}' = \vec{u}_k \qquad \vec{\epsilon}' \times \vec{u}_k = \vec{\epsilon} \qquad \vec{u}_k \times \vec{\epsilon} = \vec{\epsilon}' \tag{3-22}$$

[7] It is easy to check that Eq. (3-21) is indeed a solution of the Helmholtz equation:

$$f(\vec{r}) = \exp(\iota\vec{k}\cdot\vec{r}) = \exp(\iota k_x x + \iota k_y y + \iota k_z z) = \exp(\iota k_x x)\exp(\iota k_y y)\exp(\iota k_z z)$$

thus we have:

$$\nabla^2\vec{f} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\exp(\iota k_x x)\exp(\iota k_y y)\exp(\iota k_z z)\,\vec{\epsilon} = -k_x^2\vec{f} - k_y^2\vec{f} - k_z^2\vec{f} = -k^2\vec{f}$$

and provided that $k^2 = \mu_0\varepsilon_0\omega^2$, Eq. (3-14) is satisfied.

[8] Let us take for a moment $\vec{\epsilon} \equiv \vec{u}_x$, then:

$$\nabla \cdot \vec{A} = 0 \qquad \Rightarrow \qquad \nabla \cdot (f(\vec{r})\,\vec{\epsilon}) = \nabla \cdot (e^{\iota(k_x x + k_y y + k_z z)}\,\vec{u}_x) = \iota k_x e^{\iota(k_x x + k_y y + k_z z)} = 0$$

and this means:

$$k_x = 0 \qquad \Rightarrow \qquad \vec{k}\cdot\vec{\epsilon} = 0$$

Given these premises, we can work out the expressions for the vector potential and for the electric field directly from Eqs. (3-17) and (3-18):

$$\vec{A} = A_0 \left[ e^{\iota \vec{k}\cdot\vec{r}} \alpha(t) + e^{-\iota \vec{k}\cdot\vec{r}} \alpha^*(t) \right] \vec{\epsilon} \tag{3-23}$$

$$\vec{E} = \iota\omega A_0 \left[ e^{\iota \vec{k}\cdot\vec{r}} \alpha(t) - e^{-\iota \vec{k}\cdot\vec{r}} \alpha^*(t) \right] \vec{\epsilon} \tag{3-24}$$

To obtain the expression for the magnetic field we need to calculate $\nabla \times (f(\vec{r})\vec{\epsilon})$. For the sake of clarity and without loosing generality we may provisionally take a coordinate system $x, y, z$ where the three axes are respectively parallel to the unit vectors $\vec{\epsilon}$, $\vec{\epsilon}'$ and $\vec{u}_k$. In this frame:

$$\nabla \times (f(\vec{r})\vec{\epsilon}) \equiv \nabla \times (e^{\iota kz}\vec{u}_x) = \frac{\partial e^{\iota kz}}{\partial z}\vec{u}_y = \iota k f(\vec{r})\vec{u}_y \tag{3-25}$$

Coming back to the system of vectors $\vec{\epsilon}$, $\vec{\epsilon}'$ and $\vec{u}_k$, and being $k = \omega/c$, we rewrite Eq. (3-25) as:

$$\nabla \times (f(\vec{r})\vec{\epsilon}) = \iota k e^{\iota \vec{k}\cdot\vec{r}}\, \vec{\epsilon}' = \iota \frac{\omega}{c} e^{\iota \vec{k}\cdot\vec{r}}\, \vec{\epsilon}' \tag{3-26}$$

Hence, we obtain the expression for the magnetic field:

$$\vec{B} = \iota \frac{\omega}{c} A_0 \left[ e^{\iota \vec{k}\cdot\vec{r}} \alpha(t) - e^{-\iota \vec{k}\cdot\vec{r}} \alpha^*(t) \right] \vec{\epsilon}' \tag{3-27}$$

Note that, as it is well known from elementary electromagnetism, plane electromagnetic waves are found to be transverse ($\vec{E} \perp \vec{k}$, $\vec{B} \perp \vec{k}$), with $\vec{B} \perp \vec{E}$ and $|\vec{B}| = |\vec{E}|/c$.
We shall also consider **periodic boundary conditions** on a cubic volume with side $L$:

$$\begin{cases} f(\vec{r}) = f(\vec{r} + L\vec{u}_x) \\ f(\vec{r}) = f(\vec{r} + L\vec{u}_y) \\ f(\vec{r}) = f(\vec{r} + L\vec{u}_z) \end{cases} \tag{3-28}$$

These correspond to the following constraints:

$$\begin{cases} e^{\iota \vec{k}\cdot\vec{r}} = e^{\iota \vec{k}\cdot\vec{r} + \iota L k_x} \\ e^{\iota \vec{k}\cdot\vec{r}} = e^{\iota \vec{k}\cdot\vec{r} + \iota L k_y} \\ e^{\iota \vec{k}\cdot\vec{r}} = e^{\iota \vec{k}\cdot\vec{r} + \iota L k_z} \end{cases} \quad \longrightarrow \quad \begin{cases} L k_x = 2 n_x \pi \\ L k_y = 2 n_y \pi \\ L k_z = 2 n_z \pi \end{cases} \tag{3-29}$$

where $n_x, n_y, n_z \in \mathbb{Z}$. Namely, only discrete values are possible for the components $k_x$, $k_y$ and $k_z$:

$$\boxed{k_x = \frac{2\pi}{L} n_x \qquad k_y = \frac{2\pi}{L} n_x \qquad k_z = \frac{2\pi}{L} n_z} \tag{3-30}$$

Importantly, this discretization condition implies that only a limited set of values for $|\vec{k}|$ is valid. This further constrains the allowed values for $\omega$, which must satisfy $|\omega| = \frac{|\vec{k}|}{\sqrt{\mu_0 \varepsilon_0}}$. Anyway, by letting $L \to +\infty$ (to get the possible solutions in the infinite empty space) the allowed $\kappa$ values become infinitely closely spaced, which means that practically all frequencies $\omega$ are possible.
Now, for each $\vec{k}$ vector, **two polarization directions** are possible, mutually orthogonal, which we can indicate as $\vec{\epsilon}_1$ and $\vec{\epsilon}_2$. Therefore, each normal mode can be identified by the 4-tuple of values $(k_x, k_y, k_z, \epsilon_i)$.
The set of all possible modes results to be infinite, but discrete and countable. Hence, we may define an ordering and a numbering of all modes with a single index $l$, such that there is a one-to-one correspondence between the values of $l$ and the possible 4-tuples of parameters.

$$l \quad \longleftrightarrow \quad (k_x, k_y, k_z, \epsilon_i) \tag{3-31}$$

For convenience, we choose the ordering in such a way that the mode $-l$ is associated to the mode with the same polarization of $l$, but opposite $\vec{k}$ vector.

$$-l \quad \longleftrightarrow \quad (-k_x, -k_y, -k_z, \epsilon_i) \tag{3-32}$$

In the general case, the electromagnetic field can have active contributions from all the possible modes, and using the index $l$ we can take all of them into account using a single sum:

$$\vec{A}(\vec{r},t) = \sum_l \frac{\mathcal{E}_l}{\omega_l}\,\vec{\epsilon}_l\,\left[e^{\iota\vec{k}_l\cdot\vec{r}}\alpha_l(t) + e^{-\iota\vec{k}_l\cdot\vec{r}}\alpha_l^*(t)\right] \tag{3-33}$$

$$\vec{E}(\vec{r},t) = \sum_l \iota\mathcal{E}_l\,\vec{\epsilon}_l\,\left[e^{\iota\vec{k}_l\cdot\vec{r}}\alpha_l(t) - e^{-\iota\vec{k}_l\cdot\vec{r}}\alpha_l^*(t)\right] \tag{3-34}$$

$$\vec{B}(\vec{r},t) = \sum_l \iota\frac{\mathcal{E}_l}{c}\,\vec{\epsilon}_l'\,\left[e^{\iota\vec{k}_l\cdot\vec{r}}\alpha_l(t) - e^{-\iota\vec{k}_l\cdot\vec{r}}\alpha_l^*(t)\right] \tag{3-35}$$

In the expressions above we have adopted for convenience a new constant:

$$\mathcal{E}_l = \omega_l A_{0,l} \tag{3-36}$$

We observe that the fields are completely determined by the values of the normal variables $\alpha_l(t)$.
As a final note, it is useful to keep in mind that this is not the only way of solving the wave equation Eq. (3-7). For instance, in an optical cavity comprised between plane mirrors the field should be imposed as vanishing at the boundaries, and this condition results in $f(\vec{r})$ with the form of stationary waves instead of traveling ones.

### 3.4 Energy of the classical electromagnetic field

To calculate the energy of the electromagnetic field we write the integral of the energy density on the whole cubic volume $V$:

$$\mathcal{U} = \int_V u(\vec{r})\,dV = \int_V \frac{1}{2}\varepsilon_0\left(|\vec{E}|^2 + c^2|\vec{B}|^2\right)\,dV = \frac{1}{2}\varepsilon_0\int_V |\vec{E}|^2\,dV + \frac{1}{2}\varepsilon_0 c^2\int_V |\vec{B}|^2\,dV \tag{3-37}$$

We begin with the evaluation of:

$$\begin{aligned}
&\int_V |\vec{E}|^2\,dV = \int_V (\vec{E}\cdot\vec{E})dV = \\
&= -\sum_l\sum_{l'}\mathcal{E}_l\mathcal{E}_{l'}\,(\vec{\epsilon}_l\cdot\vec{\epsilon}_{l'})\int_V\left(e^{\iota\vec{k}_l\cdot\vec{r}}\alpha_l - e^{-\iota\vec{k}_l\cdot\vec{r}}\alpha_l^*\right)\left(e^{\iota\vec{k}_{l'}\cdot\vec{r}}\alpha_{l'} - e^{-\iota\vec{k}_{l'}\cdot\vec{r}}\alpha_{l'}^*\right)\,dxdydz = \\
&= -\sum_l\sum_{l'}\mathcal{E}_l\mathcal{E}_{l'}\,(\vec{\epsilon}_l\cdot\vec{\epsilon}_{l'})\int_V\Big(e^{\iota(\vec{k}_l+\vec{k}_{l'})\cdot\vec{r}}\alpha_l\alpha_{l'} - e^{\iota(\vec{k}_l-\vec{k}_{l'})\cdot\vec{r}}\alpha_l\alpha_{l'}^* + \\
&\qquad\qquad -e^{-\iota(\vec{k}_l-\vec{k}_{l'})\cdot\vec{r}}\alpha_l^*\alpha_{l'} + e^{-\iota(\vec{k}_l+\vec{k}_{l'})\cdot\vec{r}}\alpha_l^*\alpha_{l'}^*\Big)\,dxdydz
\end{aligned} \tag{3-38}$$

where we have used the decomposition (3-34) with indices $l$ and $l'$, and we have dropped the indication of the time dependency of the $\alpha$ variables for simplicity of notation. We observe, interestingly, that:

- The integral contains products of the kind:

  $$e^{\pm\iota\vec{k}_l\cdot\vec{r}}\cdot e^{\pm\iota\vec{k}_l\cdot\vec{r}} = \exp\left(\pm\iota(\vec{k}_l \mp \vec{k}_{l'})\cdot\vec{r}\right) \tag{3-39}$$

  These oscillating functions have periods in $x$, $y$ and $z$ that are integer sub-multiples of the side length $L$, due to the periodic boundary conditions. Hence, the integral on the volume $V = L^3$ vanishes, unless:

  $$\vec{k}_l = \pm\vec{k}_{l'} \tag{3-40}$$

  If this condition is met, we have either $e^{\pm\iota(\vec{k}_l-\vec{k}_{l'})\cdot\vec{r}} = e^0 = 1$ or $e^{\pm\iota(\vec{k}_l+\vec{k}_{l'})\cdot\vec{r}} = e^0 = 1$ and some of the integrals do not vanish.

- $\vec{\epsilon}_l \cdot \vec{\epsilon}_{l'} = 0$ if $l$ and $l'$ correspond to modes with orthogonal polarization.

As a consequence, the terms in the double sum $\sum_l \sum_{l'}$ are vanishing except for two cases:

- The case $\boxed{l = l'}$, where $\vec{\kappa}_l = \vec{\kappa}_{l'}$ and the integral takes the value

$$\int_V \left(e^{\iota \vec{k}_l \cdot \vec{r}} \alpha_l - e^{-\iota \vec{k}_l \cdot \vec{r}} \alpha_l^*\right) \left(e^{\iota \vec{k}_l \cdot \vec{r}} \alpha_l - e^{-\iota \vec{k}_l \cdot \vec{r}} \alpha_l^*\right) dxdydz = -V \alpha_l \alpha_l^* - V \alpha_l^* \alpha_l = -2V\, \alpha_l \alpha_l^* \quad (3\text{-}41)$$

- The case $\boxed{l = -l'}$, where $\vec{\kappa}_l = -\vec{\kappa}_{l'}$ and the integral takes the value

$$\int_V \left(e^{\iota \vec{k}_l \cdot \vec{r}} \alpha_l - e^{-\iota \vec{k}_l \cdot \vec{r}} \alpha_l^*\right) \left(e^{-\iota \vec{k}_l \cdot \vec{r}} \alpha_{-l} - e^{\iota \vec{k}_l \cdot \vec{r}} \alpha_{-l}^*\right) dxdydz = +V \alpha_l \alpha_{-l} + V \alpha_l^* \alpha_{-l}^* \quad (3\text{-}42)$$

We can conclude:

$$\int_V |\vec{E}|^2\, dV = \sum_l |\mathcal{E}_l|^2 V \left(2\alpha_l \alpha_l^* - \alpha_l \alpha_{-l} - \alpha_l^* \alpha_{-l}^*\right) \quad (3\text{-}43)$$

The integral containing $|\vec{B}|^2$ is very similar to the one for $|\vec{E}|^2$ :

$$\begin{aligned} \int_V |\vec{B}|^2\, dV &= \int_V (\vec{B} \cdot \vec{B}) dV = \\ &= -\frac{1}{c^2} \sum_l \sum_{l'} \mathcal{E}_l \mathcal{E}_{l'} \left(\vec{\epsilon}_l' \cdot \vec{\epsilon}_{l'}'\right) \int_V \left(e^{\iota \vec{k}_l \cdot \vec{r}} \alpha_l - e^{-\iota \vec{k}_l \cdot \vec{r}} \alpha_l^*\right) \left(e^{\iota \vec{k}_{l'} \cdot \vec{r}} \alpha_{l'} - e^{-\iota \vec{k}_{l'} \cdot \vec{r}} \alpha_{l'}^*\right) dxdydz \end{aligned} \quad (3\text{-}44)$$

The only differences with respect to Eq. (3-38) are the presence of a multiplicative factor $1/c^2$ and of the scalar product $\vec{\epsilon}_l' \cdot \vec{\epsilon}_{l'}'$ instead of $\vec{\epsilon}_l \cdot \vec{\epsilon}_{l'}$. Again, the only terms that matter in the double sum are those with $l = l'$ and $l = -l'$. However, we should note that in the case $l = -l'$ here we have:

$$\vec{\epsilon}_l' \cdot \vec{\epsilon}_{-l}' = (\vec{u}_{k_l} \times \vec{\epsilon}_l) \cdot (-\vec{u}_{k_l} \times \vec{\epsilon}_l) = -1 \quad (3\text{-}45)$$

and this changes the sign of the terms with the products $\alpha_l \alpha_{-l}$ and $\alpha_l^* \alpha_{-l}^*$. Namely, we get in the end:

$$\int_V |\vec{B}|^2\, dV = \frac{1}{c^2} \sum_l |\mathcal{E}_l|^2 V \left(2\alpha_l \alpha_l^* + \alpha_l \alpha_{-l} + \alpha_l^* \alpha_{-l}^*\right) \quad (3\text{-}46)$$

In conclusion, the total energy of the electromagnetic field is:

$$\begin{aligned} \mathcal{U} &= \frac{1}{2}\varepsilon_0 \int_V |\vec{E}|^2\, dV + \frac{1}{2}\varepsilon_0 c^2 \int_V |\vec{B}|^2\, dV = \\ &= \frac{1}{2}\varepsilon_0 \sum_l |\mathcal{E}_l|^2 V \left(2\alpha_l \alpha_l^* - \alpha_l \alpha_{-l} - \alpha_l^* \alpha_{-l}^*\right) + \frac{1}{2}\varepsilon_0 c^2 \frac{1}{c^2} \sum_l |\mathcal{E}_l|^2 V \left(2\alpha_l \alpha_l^* + \alpha_l \alpha_{-l} + \alpha_l^* \alpha_{-l}^*\right) \end{aligned} \quad (3\text{-}47)$$

which becomes a very compact expression:

$$\boxed{\mathcal{U} = \frac{1}{2}\varepsilon_0 V \sum_l |\mathcal{E}_l|^2 (4\alpha_l^* \alpha_l) = 2\varepsilon_0 V \sum_l |\mathcal{E}_l|^2 |\alpha_l|^2} \quad (3\text{-}48)$$

This shows that the energy of the electromagnetic field is the **sum of the energies** associated with each normal mode, **without cross-terms**.

| ELECTROMAGNETIC MODE | | HARMONIC OSCILLATOR | |
|---|---|---|---|
| Normal variable | $\alpha_l(t) = \alpha_l(0)\, e^{-\iota\omega_l t}$ | Normal variable | $\alpha(t) = \alpha(0)\, e^{-\iota\omega_l t}$ |
| Tot. energy | $U = \varepsilon_0 V \lvert\mathcal{E}_l\rvert^2(\alpha_l^*\alpha_l + \alpha_l\alpha_l^*)$ | Tot. energy | $E = \frac{1}{4m}(\alpha_l^*\alpha_l + \alpha_l\alpha_l^*)$ |
| Potential vector | $A(t) = \frac{\mathcal{E}_l}{\omega_l}(\alpha(t) + \alpha^*(t))$ | Position | $x(t) = \frac{1}{2m\omega}(\alpha(t) + \alpha^*(t))$ |
| Electric field | $E(t) = -\frac{\mathcal{E}_l}{\iota}(\alpha(t) - \alpha^*(t))$ | Momentum | $p(t) = +\frac{1}{2\iota}(\alpha(t) - \alpha^*(t))$ |

Table 2: Comparison of the expressions of physical quantities that characterize, on the one side, a plane-wave electromagnetic mode (in the position $\vec{r} = 0$) and, on the other side, a one-dimensional harmonic oscillator.

### 3.5 Quantization of the electromagnetic field

We can now proceed to a comparison between a classical electromagnetic field and a classical harmonic oscillator, from a formal point of view. In Table 2 we consider a few characteristic physical quantities that describe one electromagnetic mode (in detail a plane-wave mode) and compare them with quantities that describe a one-dimensional harmonic oscillator. For simplicity, we evaluate the quantities of the electromagnetic mode in the position $\vec{r} = 0$.
We observe that in both cases the relevant physical quantities are functions of a suitably defined complex variable $\alpha(t)$. Notably, this variable has the same time evolution in both cases, describing a uniform circular motion in the complex plane with angular frequency $\omega$. The energy of the mode, as well as the energy of the oscillator, is proportional to $|\alpha|^2$. Finally, the potential vector and the electric field in one case, and the position and momentum in the other case, are functions of $\alpha(t) \pm \alpha^*(t)$. In other words they are functions of the real and imaginary part of $\alpha(t)$ respectively. Note that the expressions of the electromagnetic quantities in a different position $\vec{r} \neq 0$ correspond to taking an initial phase $\phi \neq 0$ in the solution of the harmonic oscillator.
We conclude that **each classical electromagnetic mode is formally analogous to an harmonic oscillator**. More in general, as the classical electromagnetic field consists in the combination of many independent modes, it can be assumed formally analogous to a set of independent harmonic oscillators, each one providing its additive energy contribution.
The crucial passage to the application of a quantum theory to the electromagnetic field is to postulate that the quantum theory of the harmonic oscillator can be adapted to the field-mode oscillator. Intuitively[9], this may be done by replacing $\alpha$ with a suitable $\hat{a}$ operator and $\alpha^*$ with $\hat{a}^\dagger$. Starting from the expression of the total energy of a single mode, we would write a Hamiltonian:

$$\hat{H}_l = \varepsilon_0 V |\mathcal{E}_l|^2 (\hat{a}^\dagger \hat{a} + \hat{a}\hat{a}^\dagger) \tag{3-49}$$

This Hamiltonian should be made compatible with the results of Planck's work. Namely, we know that each electromagnetic mode can be excited with discrete quanta of energy of magnitude $\hbar\omega$. The expression in Eq. (3-49) is compatible with this result if take:

$$\mathcal{E}_l = \sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}} \tag{3-50}$$

We should then assume a form, at least, for the operator of the potential vector. In this way we can link the quantum oscillator to the macroscopic observables of the electromagnetic field. We may formalize our postulate as follows.

[9] Recall the observations conducted at the end of Section 2.3

**Electromagnetic-field quantization postulate**

To each electromagnetic mode is associated an independent quantum harmonic oscillator, with **Hamiltonian**:

$$\hat{H}_l = \hbar\omega \left( \hat{a}_l^\dagger \hat{a}_l + \frac{1}{2} \right) \tag{3-51}$$

where the index $l$ represents the label of the electromagnetic mode. The **potential vector** operator is:

$$\hat{\vec{A}}_l = \frac{\mathcal{E}_l}{\omega_l} \vec{\epsilon}_l \left[ e^{\iota \vec{k}_l \cdot \vec{r}} \hat{a} + e^{-\iota \vec{k}_l \cdot \vec{r}} \hat{a}^\dagger \right] \tag{3-52}$$

where $\mathcal{E}_l = \sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}$.

**Electric- and magnetic-field operators** are derived from Eq. (3-52):

$$\boxed{\hat{\vec{E}}_l = \iota \mathcal{E}_l \vec{\epsilon}_l \left[ e^{\iota \vec{k}_l \cdot \vec{r}} \hat{a} - e^{-\iota \vec{k}_l \cdot \vec{r}} \hat{a}^\dagger \right]} \tag{3-53}$$

$$\boxed{\hat{\vec{B}}_l = \iota \frac{\mathcal{E}_l}{c} \vec{\epsilon}_l' \left[ e^{\iota \vec{k}_l \cdot \vec{r}} \hat{a} - e^{-\iota \vec{k}_l \cdot \vec{r}} \hat{a}^\dagger \right]} \tag{3-54}$$

where the constant $\mathcal{E}_l$ again takes the expression given by Eq. (3-50).
If many electromagnetic modes are considered, these correspond to independent oscillators. The overall Hamiltonian is the simple sum of the Hamiltonians of the modes:

$$\hat{H}_R = \sum_l \hat{H}_l \tag{3-55}$$

The potential vector and the fields, in turn, are given by:

$$\hat{\vec{A}} = \sum_l \hat{\vec{A}}_l \qquad \hat{\vec{E}} = \sum_l \hat{\vec{E}}_l \qquad \hat{\vec{B}} = \sum_l \hat{\vec{B}}_l \tag{3-56}$$

The eigenstates of the Hamiltonian are formally the same number states $|n\rangle$ studied for the harmonic oscillator.

$$\hat{H}_l |n\rangle = E_n |n\rangle \qquad E_n = \hbar\omega \left( n + \frac{1}{2} \right) \tag{3-57}$$

In the context of quantum optics, these states often take the name of **Fock states**. They have highly non-classical features, as we will see.
The operators $\hat{a}_l$ and $\hat{a}_l^\dagger$ possess all the properties discussed in Section 2.4 and in particular:

$$\boxed{\begin{aligned} &\hat{a}_l |n\rangle = \sqrt{n} |n-1\rangle \\ &\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle \\ &[\hat{a}, \hat{a}^\dagger] = 1 \end{aligned}} \tag{3-58}$$

To correctly account for the independence of the different modes, operators of modes with different indices $l \neq l'$ commute:

$$[\hat{a}_l, \hat{a}_{l'}^\dagger] = \delta_{l\,l'} \tag{3-59}$$

$$[\hat{a}_l, \hat{a}_{l'}] = [\hat{a}_l^\dagger, \hat{a}_{l'}^\dagger] = 0 \tag{3-60}$$

In the following, when discussing a single mode of the field and if there is no risk of ambiguity, we will often drop the subscript $l$.

The quanta of excitation of the radiation mode can be interpreted as the photons.

**Photons**

A **photon** is defined as a quantum of excitation of one electromagnetic mode. To each photon is associated an energy contribution:

$$E_{ph} = \hbar\omega \tag{3-61}$$

The operators $\hat{a}$ and $\hat{a}^\dagger$ respectively subtract and add a photon to the electromagnetic mode; in fact, they are called the **annihilation** (or destruction) and **creation** operators. Fock states are characterized by a determined fixed of photons in the mode; the $|0\rangle$ state is also named the **vacuum** state, as it contains no photons.

# Notes on Quantum Optics - Section 4

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# 4 Quasi-classical states

## 4.1 Definition of the quasi-classical states

We may ask ourselves whether it is possible to represent classical light waves with the developed quantum formalism. Or, somehow equivalently, we may ask in which conditions quantum states of light reach the classical limit.
Starting from Eq. (3-48) and taking $\mathcal{E}_l = \sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}}$, we can write the total energy of a single radiation mode as:

$$H_{\text{Cl}} = \hbar\omega|\alpha|^2 \tag{4-1}$$

where $\alpha = \alpha(t)$ is the complex-valued normal variable associated to the mode. The electric field is instead given by:

$$E_{\text{Cl}} = \iota\mathcal{E}\left[\alpha(t)\, e^{\iota\vec{k}\cdot\vec{r}} - \alpha^*(t)\, e^{-\iota\vec{k}\cdot\vec{r}}\right] \tag{4-2}$$

On the other hand, in the quantum model we have:

$$\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right) \tag{4-3}$$

$$\hat{E} = \iota\mathcal{E}\left[\hat{a}\, e^{\iota\vec{k}\cdot\vec{r}} - \hat{a}^\dagger\, e^{-\iota\vec{k}\cdot\vec{r}}\right] \tag{4-4}$$

Note that for simplicity we have considered the scalar part of the vectorial electric field operator $\hat{\vec{E}}$, thus dropping the polarization unit vector $\vec{\epsilon}$.
We now look for quantum states that are **as similar as possible to classical waves**, which means:

$$\langle\hat{H}\rangle \simeq H_{\text{Cl}} \tag{4-5}$$

$$\langle\hat{E}\rangle = E_{\text{Cl}} \tag{4-6}$$

This can be accomplished if we find a quantum state $|\psi\rangle$ of the electromagnetic field such that:

$$\langle\hat{a}\rangle = \langle\psi|\,\hat{a}\,|\psi\rangle = \alpha \qquad\qquad \langle\hat{a}^\dagger\hat{a}\rangle = \langle\psi|\,\hat{a}^\dagger\hat{a}\,|\psi\rangle = \alpha^*\alpha \tag{4-7}$$

The first relation implies $\langle\hat{a}^\dagger\rangle = \langle\psi|\,\hat{a}^\dagger\,|\psi\rangle = \alpha^*$, which means that for such a state $|\psi\rangle$ one reaches precisely Eq. (4-6). With regard to the Hamiltonian, instead one can only write an approximate equality:

$$\langle\hat{H}\rangle = \hbar\omega\left(\alpha^*\alpha + \frac{1}{2}\right) \simeq \hbar\omega\alpha^*\alpha = H_{\text{Cl}} \tag{4-8}$$

The similarity between the expectation value of the Hamiltonian operator and the classical expression of the energy can get better as $|\alpha| = \alpha^*\alpha \gg 1$, so that the term $1/2$ becomes more and more negligible.
We develop now another implication of the conditions (4-7) that allows for a better definition of these states of radiation. Let us define the operator $\hat{b}$ as:

$$\hat{b} = \hat{a} - \alpha\hat{\mathbb{I}} \tag{4-9}$$

where $\hat{a}$ is the usual annihilation operator, $\hat{\mathbb{I}}$ is the identity operator and $\alpha$ is now a generic complex number. It is easy to show that the expectation value of $\hat{b}^\dagger\hat{b}$ is zero if conditions (4-7) hold, indeed:

$$\begin{aligned}\langle\hat{b}^\dagger\hat{b}\rangle &= \langle(\hat{a}^\dagger - \alpha^*\hat{\mathbb{I}})(\hat{a} - \alpha\hat{\mathbb{I}})\rangle = \langle\hat{a}^\dagger\hat{a}\rangle - \alpha^*\langle\hat{a}\rangle - \alpha\langle\hat{a}^\dagger\rangle + \alpha^*\alpha = \\ &= \alpha^*\alpha - \alpha^*\alpha - \alpha\alpha^* + \alpha^*\alpha = 0\end{aligned} \tag{4-10}$$

Namely this means that, if there is a state $|\psi\rangle$ that satisfies (4-7) and we define $|\psi'\rangle = \hat{b}\,|\psi\rangle$, then:

$$0 = \langle\hat{b}^\dagger\hat{b}\rangle = \langle\psi|\,\hat{b}^\dagger\hat{b}\,|\psi\rangle = \langle\psi'|\psi'\rangle \tag{4-11}$$

We note that due to the normalization condition there is no quantum state $|\psi'\rangle \neq 0$ that entails $\langle\psi'|\psi'\rangle = 0$. Put in other words:

$$\langle\psi'|\psi'\rangle = 0 \qquad \Leftrightarrow \qquad \left|\psi'\right\rangle = 0 \tag{4-12}$$

By developing the latter relation we get:

$$0 = \left|\psi'\right\rangle = \hat{b}\,|\psi\rangle = (\hat{a} - \alpha\hat{\mathbb{I}})\,|\psi\rangle = 0 \tag{4-13}$$

which implies:

$$\hat{a}\,|\psi\rangle - \alpha\,|\psi\rangle = 0 \tag{4-14}$$

$$\hat{a}\,|\psi\rangle = \alpha\,|\psi\rangle \tag{4-15}$$

Interestingly, we can conclude that those states $|\psi\rangle$ that we are looking for, which satisfy Eqs. (4-7) and thus satisfy also Eq. (4-11), are nothing else that **the eigenstates of the annihilation operator** $\hat{a}$. In the following we will indicate the eigenstate associated to the eigenvalue $\alpha$ simply by writing the ket $|\alpha\rangle$.
In the light of this discussion, we can conveniently define the **quasi-classical states** $|\alpha\rangle$ as the states that satisfy:

$$\boxed{\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle} \tag{4-16}$$

From the definitory relation (4-16) one has directly:

$$\langle\alpha|\,\hat{a}^\dagger = \langle\alpha|\,\alpha \tag{4-17}$$

It is thus easy to show that the states $|\alpha\rangle$ not only give:

$$\langle\hat{a}\rangle = \alpha \qquad \langle\hat{a}^\dagger\rangle = \alpha^* \qquad \langle\hat{a}^\dagger\hat{a}\rangle = \alpha^*\alpha$$

but more in general:

$$\langle(\hat{a}^\dagger)^m\hat{a}^n\rangle = (\alpha^*)^m\alpha^n \tag{4-18}$$

## 4.2 Quasi-classical states in the Fock-states basis

To visualize a geometric vector in the space, we typically specify its Cartesian components, namely we write it as a combination of the vectors of the Cartesian basis. Analogously, to understand whether the states $|\alpha\rangle$ are meaningful and to characterize them better, a good way is to write them as combinations of states of a known basis. The Fock states form a complete basis of the Hilbert space of a single mode of radiation, and are indeed a convenient basis to use in this case. In mathematical terms, we now want to write a state $|\alpha\rangle$ as combination of states $|n\rangle$:

$$\boxed{|\alpha\rangle = \sum_{n=0}^{+\infty} c_n\,|n\rangle} \tag{4-19}$$

where the complex coefficients $c_n$ are defined as:

$$c_n = \langle n|\alpha\rangle \tag{4-20}$$

From the known relation that defines the action of the creation operator on a Fock state, $\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$, we write the action on the state $|n-1\rangle$ as:

$$\hat{a}^\dagger\,|n-1\rangle = \sqrt{n}\,|n\rangle \tag{4-21}$$

Then, we take the Hermitian conjugate of this expression and divide by $\sqrt{n}$, obtaining:

$$\frac{1}{\sqrt{n}} \langle n-1| \hat{a} = \langle n| \tag{4-22}$$

By using Eq. (4-22) and the definition of quasi-classical states in Eq. (4-16), we can develop Eq. (4-20) as follows:

$$\langle n|\alpha\rangle = \frac{1}{\sqrt{n}} \langle n-1| \hat{a} |\alpha\rangle = \frac{1}{\sqrt{n}} \langle n-1| \alpha |\alpha\rangle = \frac{\alpha}{\sqrt{n}} \langle n-1|\alpha\rangle \tag{4-23}$$

which means:

$$\boxed{c_n = \frac{\alpha}{\sqrt{n}} c_{n-1}} \tag{4-24}$$

This is a recursion relation that allows to find all $c_n$ starting from $c_0$:

$$c_1 = \frac{\alpha}{\sqrt{1}} c_0 \tag{4-25}$$

$$c_2 = \frac{\alpha}{\sqrt{2}} c_1 = \frac{\alpha^2}{\sqrt{2 \cdot 1}} c_0 \tag{4-26}$$

$$c_3 = \frac{\alpha}{\sqrt{3}} c_2 = \frac{\alpha^3}{\sqrt{3 \cdot 2 \cdot 1}} c_0 \tag{4-27}$$

$$\vdots$$

$$c_n = \frac{\alpha}{\sqrt{n}} c_{n-1} = \frac{\alpha^n}{\sqrt{n!}} c_0 \tag{4-28}$$

The state $|\alpha\rangle$ is thus written as:

$$|\alpha\rangle = c_0 \sum_{n=0}^{+\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle \tag{4-29}$$

The coefficient $c_0$ can be found by imposing the normalization condition $\langle\alpha|\alpha\rangle = 1$. First, we can write:

$$\langle\alpha|\alpha\rangle = \left( c_0^* \sum_{m=0}^{+\infty} \frac{(\alpha^*)^n}{\sqrt{m!}} \langle m| \right) \left( c_0 \sum_{n=0}^{+\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle \right) = c_0^* c_0 \sum_{m=0}^{+\infty} \sum_{n=0}^{+\infty} \frac{(\alpha^*)^m \alpha^n}{\sqrt{m!n!}} \langle m|n\rangle \tag{4-30}$$

Then, by recalling that $\langle m|n\rangle = \delta_{mn}$, we write:

$$1 = \langle\alpha|\alpha\rangle = |c_0|^2 \sum_{n=0}^{+\infty} \frac{(\alpha^*)^n \alpha^n}{\sqrt{n!n!}} \underbrace{\langle n|n\rangle}_{=1} = |c_0|^2 \sum_{n=0}^{+\infty} \frac{\left(|\alpha|^2\right)^n}{n!} = |c_0|^2 e^{+|\alpha|^2} \tag{4-31}$$

$$|c_0|^2 = e^{-|\alpha|^2} \tag{4-32}$$

Since a global phase term in (4-29) is physically insignificant, we can take $c_0$ as real, and we conclude:

$$c_0 = e^{-\frac{|\alpha|^2}{2}} \tag{4-33}$$

The resulting expression for a quasi-classical state $|\alpha\rangle$ written in the Fock state basis is:

$$\boxed{|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{+\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle} \tag{4-34}$$

The quasi-classical states are also called **coherent** states. As a matter of fact, these states and their number-state expansion (4-34) were discussed already by Schroedinger himself as the classical limit for

a quantum harmonic oscillator.[10] Their adaptation to the oscillator of a single-mode of electromagnetic field is due to Glauber.[11]
From the decomposition (4-34) we see that the quasi-classical state with $\alpha = 0$ corresponds to the Fock state with $n = 0$, namely the unique ground state of the field oscillator $|0\rangle$.
Quasi-classical states associated to different eigenvalues **are not orthogonal**. Namely, given $|\alpha\rangle$ and $|\beta\rangle$ with $\alpha \neq \beta$, then $|\langle\beta|\alpha\rangle|^2 \neq 0$, indeed:

$$\langle\beta|\alpha\rangle = \left(e^{-\frac{|\beta|^2}{2}} \sum_{m=0}^{+\infty} \frac{(\beta^*)^m}{\sqrt{m!}} \langle m|\right) \left(e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{+\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle\right) = \tag{4-35}$$

$$= e^{-\frac{|\alpha|^2}{2}-\frac{|\beta|^2}{2}} \sum_{m=0}^{+\infty}\sum_{n=0}^{+\infty} \frac{(\beta^*)^m\alpha^n}{\sqrt{m!n!}} \langle m|n\rangle = \tag{4-36}$$

$$= e^{-\frac{|\alpha|^2}{2}-\frac{|\beta|^2}{2}} \sum_{n=0}^{+\infty} \frac{(\beta^*\alpha)^n}{\sqrt{n!n!}} \langle n|n\rangle = \tag{4-37}$$

$$= e^{-\frac{|\alpha|^2}{2}-\frac{|\beta|^2}{2}} \sum_{n=0}^{+\infty} \frac{(\beta^*\alpha)^n}{n!} = \tag{4-38}$$

$$= e^{-\frac{|\alpha|^2}{2}-\frac{|\beta|^2}{2}+\beta^*\alpha} \neq 0 \tag{4-39}$$

The modulus squared of the latter expression is:

$$|\langle\beta|\alpha\rangle|^2 = e^{-|\alpha|^2-|\beta|^2} \left|e^{\beta^*\alpha}\right|^2 \tag{4-40}$$

Now we recall that, for any complex number $z$ written as $z = x + \imath y$, we have $|e^z|^2 = |e^x e^{\imath y}|^2 = e^{2x} = e^{2\Re\{z\}} = e^{z+z^*}$. Therefore:

$$e^{-|\alpha|^2-|\beta|^2} \left|e^{\beta^*\alpha}\right|^2 = e^{-|\alpha|^2-|\beta|^2} e^{\beta^*\alpha+\beta\alpha^*} =$$
$$= \exp\left[-(\alpha^*\alpha + \beta^*\beta - \beta^*\alpha - \beta\alpha^*)\right] = \exp\left[-(\alpha^* - \beta^*)(\alpha - \beta)\right] \tag{4-41}$$

$$\boxed{|\langle\beta|\alpha\rangle|^2 = e^{-|\alpha-\beta|^2}} \tag{4-42}$$

Importantly, however, this shows that the states $|\alpha\rangle$ and $|\beta\rangle$ may become *almost* orthogonal if $|\alpha - \beta|$ is very large, as in that case $|\langle\beta|\alpha\rangle|^2 \to 0$.
Even though they do not form an orthonormal set, it can be shown that the quasi-classical states $|\alpha\rangle$ form an (over)complete basis of the Hilbert space of the single mode of radiation. In particular, the following completeness relation could be proven:

$$\frac{1}{\pi} \int d^2\alpha \; |\alpha\rangle \langle\alpha| = \hat{\mathbb{I}} \tag{4-43}$$

## 4.3 Photon statistics

Equation (4-34) allows for a simple evaluation of the photon statistics expected for the state $|\alpha\rangle$. The probability to find in the optical mode precisely $n$ photons is:

$$p_n = |\langle n|\alpha\rangle|^2 = |c_n|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!} \tag{4-44}$$

which is a **Poisson distribution** with mean $|\alpha|^2$ (see Fig. 4).

[10] E. Schroedinger, "Der stetige Übergang von der Mikro-zur Makromechanik" *Naturwissenschaften* **14**, 664-666 (1926).
[11] R.J. Glauber, "Coherent and incoherent states of the radiation field" *Physical Review* **131**, 2766 (1963).

In fact, we can also evaluate the average photon number and its variance:

$$\langle \hat{n} \rangle = \langle \alpha | \hat{a}^\dagger \hat{a} | \alpha \rangle = (\langle \alpha | \hat{a}^\dagger)(\hat{a} | \alpha \rangle) = \alpha^* \underbrace{\langle \alpha | \alpha \rangle}_{=1} \alpha = \alpha^* \alpha = |\alpha|^2 \tag{4-45}$$

$$\langle (\Delta n)^2 \rangle = \langle \hat{n}^2 \rangle - (\langle \hat{n} \rangle)^2 = \langle \alpha | \hat{a}^\dagger \underbrace{\hat{a}\hat{a}^\dagger}_{=\hat{a}^\dagger \hat{a}+1} \hat{a} | \alpha \rangle - \Big( \langle \alpha | \hat{a}^\dagger \hat{a} | \alpha \rangle \Big)^2 = \tag{4-46}$$

$$= \langle \alpha | \hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{a} | \alpha \rangle + \langle \alpha | \hat{a}^\dagger \hat{a} | \alpha \rangle - \Big( \langle \alpha | \hat{a}^\dagger \hat{a} | \alpha \rangle \Big)^2 = \tag{4-47}$$

$$= |\alpha|^4 + |\alpha|^2 - \big(|\alpha|^2\big)^2 = |\alpha|^2 \tag{4-48}$$

These results indeed agree with the known results for Poisson distributions, for which $\langle (\Delta n)^2 \rangle = \langle n \rangle$. For quasi-classical states, the number of photons populating the mode has an intrinsic uncertainty $\Delta n = |\alpha| = \sqrt{\langle n \rangle}$, which in absolute terms increases with $|\alpha|$. However, if we look at the *relative* uncertainty we have $\Delta n / n = 1/\sqrt{n}$, which decreases with $|\alpha|$.

## 4.4 Time evolution of quasi-classical states

We now investigate the time evolution of quasi-classical states, under the action of the free-radiation Hamiltonian $\hat{H} = \hbar\omega(\hat{a}^\dagger \hat{a} + 1/2)$. This amounts to answering the question: given a certain quasi-classical state $|\alpha_0\rangle$, identified as such at time $t = 0$, how can I write the state of the field in an arbitrary $t > 0$?

To perform this calculation it may be useful to make a step back and consider the evolution of the Fock states, which are the eigenstates of the Hamiltonian. In general, the time evolution of an eigenstate $|\psi\rangle$ with energy $E$ is given by $|\psi(t)\rangle = \exp(-\iota E t/\hbar) |\psi(0)\rangle$. The Fock state $|n\rangle$, associated to energy $E_n = \hbar\omega(n + \frac{1}{2})$ evolves in time as:

$$|n(t)\rangle = e^{-\iota \frac{\omega}{2} t} e^{-\iota \omega n t} |n(0)\rangle \tag{4-49}$$

Because of the linearity of the Schroedinger equation, a generic state written at time $t = 0$ as $|\psi\rangle = \sum_{n=0}^{+\infty} c_n |n\rangle$, at a time $t > 0$ becomes:

$$|\psi(t)\rangle = e^{-\iota \frac{\omega}{2} t} \sum_{n=0}^{+\infty} e^{-\iota \omega n t} c_n |n\rangle \tag{4-50}$$

We consider here a quasi-classical state that is associated to the complex eigenvalue $\alpha$ at $t = 0$.

$$|\psi(0)\rangle \equiv |\alpha\rangle \tag{4-51}$$

By substituting the Fock-state decomposition of the quasi-classical state (4-34) into Eq. (4-50), we obtain:

$$|\psi(t)\rangle = e^{-\iota \frac{\omega}{2} t} e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{+\infty} e^{-\iota \omega n t} \frac{\alpha^n}{\sqrt{n!}} |n\rangle = e^{-\iota \frac{\omega}{2} t} e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{+\infty} \frac{(e^{-\iota \omega t} \alpha)^n}{\sqrt{n!}} |n\rangle \tag{4-52}$$

$$\boxed{|\psi(t)\rangle = e^{-\iota \frac{\omega}{2} t} \left| \alpha e^{-\iota \omega t} \right\rangle = e^{-\iota \frac{\omega}{2} t} |\alpha(t)\rangle} \tag{4-53}$$

Basically, apart from an irrelevant global phase factor $e^{-\iota \frac{\omega}{2} t}$, a quasi-classical state with eigenvalue $\alpha(0)$ evolved in time is again a quasi-classical state, but with a different eigenvalue $\alpha(t) = \alpha(0) e^{-\iota \omega t}$. Note that the eigenvalue evolves in time precisely as the classical normal variable, by acquiring a complex phase term $e^{-\iota \omega t}$.

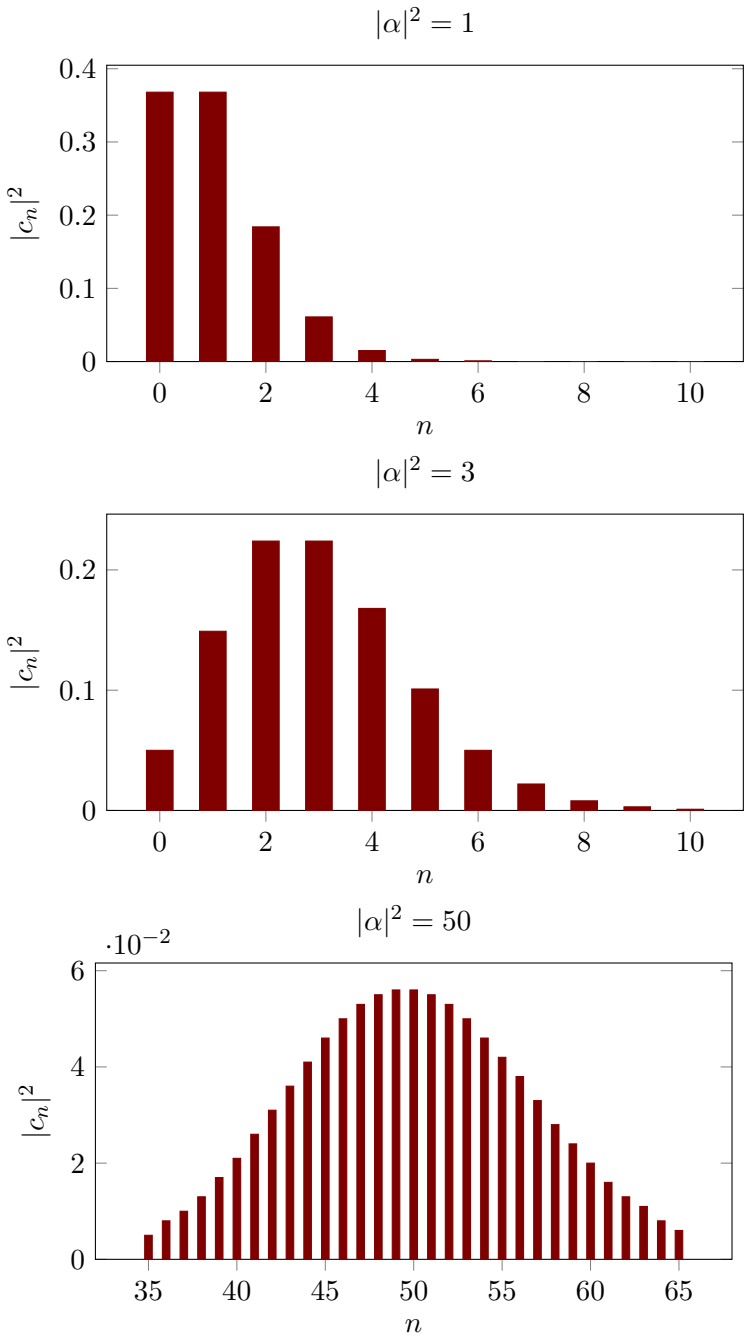


Figure 4: Photon statistics of coherent states. Examples of probability distributions of measuring $n$ photons, for quasi-classical states with different values of $\bar{n} = |\alpha|^2$. These graphs allow to observe a few peculiar features of the Poissonian distribution. As $p_{n+1} - p_n = \frac{\bar{n}^n}{n!} e^{-\bar{n}} \left( \frac{\bar{n}}{n+1} - 1 \right)$ the probability $p_n$ is increasing with $n$ until $n + 1 > \bar{n}$, when it begins to decrease. The most likely photon number is thus around $\bar{n}$. In addition, for large $\bar{n}$ (see e.g. the case $|\alpha|^2 = 50$), the Poissonian resembles a Gaussian.

### 4.5 The displacement operator

It can be useful to define an operator $\hat{D} = \hat{D}(\alpha)$ that acts on a quasi-classical state $|\beta\rangle$ in this way:

$$\hat{D}(\alpha)\,|\beta\rangle = e^{\iota\phi}\,|\alpha+\beta\rangle \tag{4-54}$$

where $\phi$ is a phase term. Namely, the operator $\hat{D}(\alpha)$ (function of the complex parameter $\alpha$) transforms the quasi-classical state $|\beta\rangle$ into the quasi-classical state $|\alpha+\beta\rangle$ apart from a possible phase term $e^{\iota\phi}$, which is negligible as it multiplies the whole state. The operator $\hat{D}$ is called **displacement operator**. Another way of expressing this concept is that the state given by $\hat{D}(\alpha)\,|\beta\rangle$ is again an eigenstate of the annihilation operator $\hat{a}$, with eigenvalue $(\alpha+\beta)$.

$$\boxed{\hat{a}\,\hat{D}(\alpha)\,|\beta\rangle = (\alpha+\beta)\,\hat{D}(\alpha)\,|\beta\rangle} \tag{4-55}$$

These relations imply that $\hat{D}(\alpha)$ is unitary, as it does not change the norm of the state. Its inverse coincides with its Hermitian conjugate and, in particular:

$$\hat{D}^{-1}(\alpha) = \hat{D}^{\dagger}(\alpha) = \hat{D}(-\alpha) \tag{4-56}$$

It also follows, for $\alpha = 0$:

$$\hat{D}(0) = \hat{D}^{-1}(0) \qquad \Rightarrow \qquad \hat{D}(0) \equiv \hat{\mathbb{I}} \tag{4-57}$$

Note that:

$$\hat{a}\,\hat{D}(-\alpha)\,|\alpha\rangle = (-\alpha+\alpha)\,\hat{D}(-\alpha)\,|\alpha\rangle = 0 \tag{4-58}$$

but $\hat{a}\,|\psi\rangle = 0$ if and only if $|\psi\rangle = |0\rangle$, namely it is the vacuum state. Therefore:

$$\hat{D}(-\alpha)\,|\alpha\rangle = |0\rangle \tag{4-59}$$

and thus:

$$\boxed{|\alpha\rangle = \hat{D}(\alpha)\,|0\rangle} \tag{4-60}$$

The quasi-classical states $|\alpha\rangle$ are obtained by applying the operator $\hat{D}(\alpha)$ to the vacuum states.
It is found[12] that a suitable form for the displacement operator $\hat{D}(\alpha)$ is:

$$\boxed{\hat{D}(\alpha) = e^{\alpha\hat{a}^{\dagger}-\alpha^{*}\hat{a}}} \tag{4-61}$$

To prove that an operator $\hat{D}$ defined as in Eq. (4-61) works as it should, we can start by calculating

$$\hat{D}(\alpha)\,|0\rangle = e^{\alpha\hat{a}^{\dagger}-\alpha^{*}\hat{a}}\,|0\rangle \tag{4-62}$$

We exploit to this purpose the following theorem, which is true for operators $\hat{A}$ and $\hat{B}$ that commute with their commutator (i.e. when $[\hat{A},[\hat{A},\hat{B}]] = [\hat{B},[\hat{A},\hat{B}]] = 0$):

$$e^{\hat{A}+\hat{B}} = e^{\hat{A}}e^{\hat{B}}e^{-\frac{1}{2}[\hat{A},\hat{B}]} \tag{4-63}$$

In our case $\hat{A} = \alpha\hat{a}^{\dagger}$, $\hat{B} = -\alpha^{*}\hat{a}$ and $[\hat{A},\hat{B}] = -\alpha\alpha^{*}[\hat{a}^{\dagger},\hat{a}] = \alpha\alpha^{*} = |\alpha|^2$. As the commutator here is a scalar, it obviously commutes with any operator and the theorem holds true. Hence we write:

$$\hat{D}(\alpha)\,|0\rangle = e^{\alpha\hat{a}^{\dagger}-\alpha^{*}\hat{a}}\,|0\rangle = e^{\alpha\hat{a}^{\dagger}}e^{-\alpha\hat{a}}e^{-\frac{1}{2}|\alpha|^2}\,|0\rangle = e^{-\frac{1}{2}|\alpha|^2}e^{\alpha\hat{a}^{\dagger}}e^{-\alpha^{*}\hat{a}}\,|0\rangle \tag{4-64}$$

We evaluate $e^{-\alpha^{*}\hat{a}}\,|0\rangle$ by decomposing the exponential in a sum of powers as follows:

$$e^{-\alpha^{*}\hat{a}}\,|0\rangle = \sum_{n=0}^{+\infty}\frac{(-\alpha^{*})^n}{n!}(\hat{a})^n\,|0\rangle = \frac{(-\alpha^{*})^0}{0!}\underbrace{\hat{a}^0}_{=\hat{\mathbb{I}}}\,|0\rangle + \sum_{n=1}^{+\infty}\frac{(-\alpha^{*})^n}{n!}\underbrace{(\hat{a})^n\,|0\rangle}_{=0} = |0\rangle + 0 = |0\rangle \tag{4-65}$$

[12] R.J. Glauber, "Coherent and incoherent states of the radiation field" *Physical Review* **131**, 2766 (1963).

And proceeding by the same technique with $e^{\alpha\hat{a}^\dagger}$ we can finally verify Eq. (4-60):

$$\hat{D}(\alpha)\left|0\right\rangle = e^{-\frac{1}{2}|\alpha|^2}e^{\alpha\hat{a}^\dagger}\left|0\right\rangle = e^{-\frac{1}{2}|\alpha|^2}\sum_{n=0}^{+\infty}\frac{(\alpha)^n}{n!}\underbrace{(\hat{a}^\dagger)^n\left|0\right\rangle}_{=\sqrt{n!}|n\rangle} = e^{-\frac{1}{2}|\alpha|^2}\sum_{n=0}^{+\infty}\frac{(\alpha)^n}{\sqrt{n!}}\left|n\right\rangle \equiv \left|\alpha\right\rangle \tag{4-66}$$

We go now for the verification of Eq. (4-54) or Eq. (4-55). We calculate $\hat{D}(\alpha)\left|\beta\right\rangle$.

$$\hat{D}(\alpha)\left|\beta\right\rangle = \hat{D}(\alpha)\hat{D}(\beta)\left|0\right\rangle = e^{\alpha\hat{a}^\dagger-\alpha^*\hat{a}}e^{\beta\hat{a}^\dagger-\beta^*\hat{a}}\left|0\right\rangle \tag{4-67}$$

To compute the product of two exponentials, when the arguments are two operators, we cannot write the exponential of the sum of the arguments. Instead, we may try to apply Eq. (4-63) in the form:

$$e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}}e^{+\frac{1}{2}[\hat{A},\hat{B}]} \tag{4-68}$$

which again is valid in the hypothesis that $\hat{A}$ and $\hat{B}$ commute with their commutator. In this case $\hat{A} = \alpha\hat{a}^\dagger - \alpha^*\hat{a}$ and $\hat{B} = \beta\hat{a}^\dagger - \beta^*\hat{a}$, so that:

$$\begin{aligned}[\hat{A},\hat{B}] = [\alpha\hat{a}^\dagger - \alpha^*\hat{a}, \beta\hat{a}^\dagger - \beta^*\hat{a}] = \\ = \alpha\beta[\hat{a}^\dagger,\hat{a}^\dagger] - \alpha\beta^*[\hat{a}^\dagger,\hat{a}] - \alpha^*\beta[\hat{a},\hat{a}^\dagger] + \alpha^*\beta^*[\hat{a},\hat{a}] = \alpha\beta^* - \alpha^*\beta\end{aligned} \tag{4-69}$$

which is a scalar, so it commutes with any operator and the hypothesis is verified. By applying the Theorem we obtain:

$$\begin{aligned}\hat{D}(\alpha)\left|\beta\right\rangle = e^{\alpha\hat{a}^\dagger-\alpha^*\hat{a}}e^{\beta\hat{a}^\dagger-\beta^*\hat{a}}\left|0\right\rangle = e^{(\alpha+\beta)\hat{a}^\dagger-(\alpha+\beta)^*\hat{a}}e^{(\alpha\beta^*-\alpha^*\beta)/2}\left|0\right\rangle = \\ = \hat{D}(\alpha+\beta)e^{(\alpha\beta^*-\alpha^*\beta)/2}\left|0\right\rangle = e^{(\alpha\beta^*-\alpha^*\beta)/2}\hat{D}(\alpha+\beta)\left|0\right\rangle = e^{(\alpha\beta^*-\alpha^*\beta)/2}\left|\alpha+\beta\right\rangle\end{aligned} \tag{4-70}$$

We finally observe that:

$$(\alpha\beta^* - \alpha^*\beta)/2 = \iota\,\Im\{\alpha\beta^*\} \tag{4-71}$$

namely it is a purely imaginary number with null real part. This means that the exponential $e^{(\alpha\beta^*-\alpha^*\beta)/2}$ only adds a phase term to $\left|\alpha+\beta\right\rangle$ and Eq. (4-54) is verified. We have thus shown that the operator $\hat{D}$ defined by Eq. (4-61) is a possible form of the displacement operator we need.

# Notes on Quantum Optics - Section 5

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*Notes for the course 'Quantum Optics and Information' - A.Y. 2023/2024. These notes are provided for personal use only, with no claim of completeness or absence of errors.*

# 5 Field quadratures and phase space representation

## 5.1 Classical quadrature amplitudes

Let us consider the scalar expression of the electric field for a **classical electromagnetic wave**, as a function of the time $t$ and of the spatial position $\vec{r}$. According to the former analysis, this can be written as:

$$E(\vec{r},t) = \iota\mathcal{E}\left(e^{\iota\vec{k}\cdot\vec{r}}\alpha(t) - e^{-\iota\vec{k}\cdot\vec{r}}\alpha^*(t)\right) = E_0 \sin\left(\omega t - \vec{k}\cdot\vec{r} + \phi\right) \tag{5-1}$$

for a properly defined real constant $E_0$, having substituted $\alpha(t) = \alpha(0)e^{-\iota\omega t} = |\alpha(0)|e^{-\iota\phi}e^{-\iota\omega t}$. The constant $E_0$ thus represents the field amplitude. We have also included an initial phase term $\phi$.
By exploiting simple trigonometric formulas, we can write Eq. (5-1) also as:

$$E(\vec{r},t) = E_0\cos\phi\sin(\omega t - \vec{k}\cdot\vec{r}) + E_0\sin\phi\cos(\omega t - \vec{k}\cdot\vec{r}) \tag{5-2}$$

This form is significant if we define the two **quadrature amplitudes** $E_Q$ and $E_P$:

$$E_Q = E_0\cos\phi \qquad\qquad E_P = E_0\sin\phi \tag{5-3}$$

Indeed, by substituting these expression the classical electric field in Eq. (5-2) becomes:

$$E(\vec{r},t) = E_Q\sin(\omega t - \vec{k}\cdot\vec{r}) + E_P\cos(\omega t - \vec{k}\cdot\vec{r}) \tag{5-4}$$

The two quadrature amplitudes $E_Q$ and $E_P$ multiply sinusoidal terms shifted in phase by $\pi/2$, which is the motivation for their name. We note that they encode in another way the same information contained in the total field amplitude $E_0$ and in the initial phase $\phi$, or equivalently in the initial value $\alpha(0)$ of the complex normal variable.

## 5.2 Quadrature operators

The concept of quadrature is extended quite straightforwardly to the quantum realm. Here, they are often defined as adimensional quantities. In particular, for a **single mode** of a **quantized field** we define the **quadrature operators**:

$$\boxed{\hat{a}_q = \frac{\hat{a} + \hat{a}^\dagger}{2}} \tag{5-5}$$

$$\boxed{\hat{a}_p = \frac{\hat{a} - \hat{a}^\dagger}{2\iota}} \tag{5-6}$$

We will show in the following how these can be related to the classical quantities and why they can be significant in the quantum description of the field.
By inverting the above relations one can also express $\hat{a}$ and $\hat{a}^\dagger$ as a function of the quadratures:

$$\hat{a} = \hat{a}_q + \iota\hat{a}_p \qquad\qquad \hat{a}^\dagger = \hat{a}_q - \iota\hat{a}_p \tag{5-7}$$

The two quadrature operators do not commute:

$$\begin{aligned}
[\hat{a}_q, \hat{a}_p] &= \frac{1}{4\iota}[\hat{a} + \hat{a}^\dagger, \hat{a} - \hat{a}^\dagger] = \\
&= \frac{1}{4\iota}\left([\hat{a},\hat{a}] - [\hat{a},\hat{a}^\dagger] + [\hat{a}^\dagger,\hat{a}] - [\hat{a}^\dagger,\hat{a}^\dagger]\right) = \\
&= \frac{1}{4\iota}(0 - 1 - 1 + 0) = \frac{\iota}{2}
\end{aligned} \tag{5-8}$$

This constrains the product of the uncertainties, according to the generalized Heisenberg relation:

$$\Delta a_q \cdot \Delta a_p \geq \left| \frac{\langle [\hat{a}_q, \hat{a}_p] \rangle}{2\iota} \right| \tag{5-9}$$

$$\boxed{\Delta a_q \cdot \Delta a_p \geq \frac{1}{4}} \tag{5-10}$$

Note that the number operator $\hat{n} = \hat{a}^\dagger \hat{a}$ can be rewritten using $\hat{a}_q$ and $\hat{a}_p$ as:

$$\begin{aligned} \hat{n} = \hat{a}^\dagger \hat{a} &= (\hat{a}_q - \iota \hat{a}_p)(\hat{a}_q + \iota \hat{a}_p) = \\ &= \hat{a}_q^2 + \hat{a}_p^2 + \iota [\hat{a}_q, \hat{a}_p] = \hat{a}_q^2 + \hat{a}_p^2 - \frac{1}{2} \end{aligned} \tag{5-11}$$

Thus the field Hamiltonian for a single mode becomes:

$$\hat{H} = \hbar\omega \left( \hat{a}^\dagger \hat{a} + \frac{1}{2} \right) = \hbar\omega \left( \hat{a}_q^2 + \hat{a}_p^2 \right) \tag{5-12}$$

Basically, the two quadratures can play the role of the position and momentum of the harmonic oscillator (in adimensional units). This analogy can be made even more explicit by defining suitable $\hat{x}$ and $\hat{p}$ operators such that:

$$\hat{a}_q = \sqrt{\frac{\omega}{2\hbar}} \hat{x} \qquad \hat{a}_p = \frac{1}{\sqrt{2\hbar\omega}} \hat{p} \tag{5-13}$$

In this way:

$$\hat{H} = \frac{1}{2} \hat{p}^2 + \omega^2 \hat{x}^2 \tag{5-14}$$

which is the Hamiltonian of a quantum harmonic oscillator with unit mass.
To understand the physical meaning of the quadrature operators, it is useful to consider the electric field operator in the Heisenberg picture (we take only the scalar part for simplicity):

$$\hat{E}(t) = \iota \mathcal{E} \left[ \hat{a}(t)\, e^{\iota \vec{k} \cdot \vec{r}} - \hat{a}^\dagger(t)\, e^{-\iota \vec{k} \cdot \vec{r}} \right] \tag{5-15}$$

As $\hat{E}$ is a linear combination of $\hat{a}$ and $\hat{a}^\dagger$, the evolved $\hat{E}$ operator in the Heisenberg picture can be retrieved by substituting these operators with $\hat{a}(t) = \hat{a} e^{-\iota\omega t}$ and $\hat{a}^\dagger(t) = \hat{a}^\dagger e^{+\iota\omega t}$, where we have identified the operators at time $t = 0$ with the operators $\hat{a}$ and $\hat{a}^\dagger$ of the Schroedinger picture. In detail, one has:

$$\hat{E}(t) = \iota \mathcal{E} \left[ \hat{a} e^{\iota \left( \vec{k} \cdot \vec{r} - \omega t \right)} - \hat{a}^\dagger e^{-\iota \left( \vec{k} \cdot \vec{r} - \omega t \right)} \right] \tag{5-16}$$

Upon substitution of the relations (5-7) we get $\hat{E}(t)$ as a function of the quadrature operators:

$$\begin{aligned} \hat{E}(t) &= \iota \mathcal{E} \left[ (\hat{a}_q + \iota \hat{a}_p) e^{\iota \left( \vec{k} \cdot \vec{r} - \omega t \right)} - (\hat{a}_q - \iota \hat{a}_p) e^{-\iota \left( \vec{k} \cdot \vec{r} - \omega t \right)} \right] = \\ &= \iota \mathcal{E} \hat{a}_q \left( e^{\iota \left( \vec{k} \cdot \vec{r} - \omega t \right)} - e^{-\iota \left( \vec{k} \cdot \vec{r} - \omega t \right)} \right) - \mathcal{E} \hat{a}_p \left( e^{\iota \left( \vec{k} \cdot \vec{r} - \omega t \right)} + e^{-\iota \left( \vec{k} \cdot \vec{r} - \omega t \right)} \right) = \\ &= -2\mathcal{E} \hat{a}_q \sin \left( \vec{k} \cdot \vec{r} - \omega t \right) - 2\mathcal{E} \hat{a}_p \cos \left( \vec{k} \cdot \vec{r} - \omega t \right) = \\ &= 2\mathcal{E} \hat{a}_q \sin \left( \omega t - \vec{k} \cdot \vec{r} \right) - 2\mathcal{E} \hat{a}_p \cos \left( \omega t - \vec{k} \cdot \vec{r} \right) \end{aligned} \tag{5-17}$$

By inspecting Eq. (5-17) one readily notes that:

$$\hat{a}_q \equiv \frac{\hat{E}(t)}{2\mathcal{E}} \qquad \text{for } \omega t - \vec{k} \cdot \vec{r} = \frac{\pi}{2} + 2m\pi \tag{5-18}$$

$$\hat{a}_p \equiv \frac{\hat{E}(t)}{2\mathcal{E}} \qquad \text{for } \omega t - \vec{k} \cdot \vec{r} = \pi + 2m\pi \tag{5-19}$$

In fact, the two quadrature observables amount to snapshots of the electric field (apart from normalization constants) taken at different points or instants of the propagation.
Analogously, it is possible to define a generic quadrature:

$$\boxed{\hat{a}_\theta = \frac{\hat{a}e^{-\iota\theta} + \hat{a}^\dagger e^{+\iota\theta}}{2}} \tag{5-20}$$

that corresponds to taking a snapshot of the (normalized) electric field at $t$ such that $\omega t - \vec{k}\cdot\vec{r} = \theta + \frac{\pi}{2} + 2m\pi$. In particular $\hat{a}_q$ and $\hat{a}_p$ corresponds to $\hat{a}_\theta$ with $\theta = 0$ and $\theta = \pi/2$ respectively. Quadratures corresponding to $\theta$ and to $\theta + \pi/2$ are conjugated observables, and the commutator is again:

$$[\hat{a}_\theta, \hat{a}_{\theta+\pi/2}] = \iota/2 \tag{5-21}$$

It is understood that the definition of the quadratures, in a real experimental setting, implies the choice of an instant $t = 0$ or, equivalently, of a phase reference, which contains always some degree of arbitrariness. Conversely, measuring a given quadrature while shifting the origin of the temporal axis is equivalent to measure all quadratures with different $\theta$.

## 5.3 Quadrature values and uncertainties for quasi-classical states

It is interesting to calculate the expectation values of the quadratures in the case of a coherent or quasi-classical state. By definition, for a state $|\alpha\rangle$ we have:

$$\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle \qquad\qquad \langle\alpha|\,\hat{a}^\dagger = \langle\alpha|\,\alpha^*$$

Therefore:

$$\langle a_q\rangle = \frac{\langle\alpha|\,(\hat{a}+\hat{a}^\dagger)\,|\alpha\rangle}{2} = \frac{\langle\alpha|\,\hat{a}\,|\alpha\rangle + \langle\alpha|\,\hat{a}^\dagger\,|\alpha\rangle}{2} = \frac{\alpha+\alpha^*}{2} = \Re(\alpha) \tag{5-22}$$

$$\langle a_q\rangle = \frac{\langle\alpha|\,(\hat{a}-\hat{a}^\dagger)\,|\alpha\rangle}{2\iota} = \frac{\langle\alpha|\,\hat{a}\,|\alpha\rangle - \langle\alpha|\,\hat{a}^\dagger\,|\alpha\rangle}{2\iota} = \frac{\alpha-\alpha^*}{2\iota} = \Im(\alpha) \tag{5-23}$$

If we consider a coherent state characterized by the eigenvalue $\alpha = |\alpha|e^{-\iota\phi}$ we have $\langle a_q\rangle = |\alpha|\cos(-\phi)$ and $\langle a_p\rangle = |\alpha|\sin(-\phi)$. The expectation value of the electric field at time $t$ is:

$$\langle E(t)\rangle = 2\mathcal{E}\left[\langle a_q\rangle \sin\left(\omega t - \vec{k}\cdot\vec{r}\right) - \langle a_p\rangle\cos\left(\omega t - \vec{k}\cdot\vec{r}\right)\right] = \tag{5-24}$$

$$= 2\mathcal{E}|\alpha|\cos\phi\sin\left(\omega t - \vec{k}\cdot\vec{r}\right) + 2\mathcal{E}|\alpha|\sin\phi\cos\left(\omega t - \vec{k}\cdot\vec{r}\right) \tag{5-25}$$

This fully agrees with the classical case of Eq. (5-2) if one takes:

$$E_0 = 2\mathcal{E}|\alpha| \tag{5-26}$$

In fact, as it is evident by comparing Eq. (5-24) with Eq. (5-4), the classical quadratures appear to be linked to the expectation values of the quantum operators as follows:

$$E_Q = 2\mathcal{E}\langle a_q\rangle \qquad\qquad E_P = -2\mathcal{E}\langle a_p\rangle \tag{5-27}$$

The expectation values of the quadratures further confirm the similarity of the state $|\alpha\rangle$ with the intuition of a classical wave. As a matter of fact, whereas classical quadratures have well defined values, the quantum observables may have an intrinsic **uncertainty**. This may be quantified by the quadrature variance. For the coherent state $|\alpha\rangle$ we have:

$$\begin{aligned}
\langle a_q^2\rangle &= \langle\alpha|\,\frac{(\hat{a}+\hat{a}^\dagger)(\hat{a}+\hat{a}^\dagger)}{4}\,|\alpha\rangle = \\
&= \frac{1}{4}\left[\langle\alpha|\,\hat{a}^2\,|\alpha\rangle + \langle\alpha|\,\hat{a}^\dagger\hat{a}\,|\alpha\rangle + \underbrace{\langle\alpha|\,\hat{a}\hat{a}^\dagger\,|\alpha\rangle}_{=\langle\alpha|\hat{a}^\dagger\hat{a}|\alpha\rangle+1} + \langle\alpha|\,\hat{a}^{\dagger\,2}\,|\alpha\rangle\right] = \\
&= \frac{1}{4}\left[\langle\alpha|\,\hat{a}^2\,|\alpha\rangle + 2\,\langle\alpha|\,\hat{a}^\dagger\hat{a}\,|\alpha\rangle + \langle\alpha|\,\hat{a}^{\dagger\,2}\,|\alpha\rangle + 1\right] = \\
&= \frac{1}{4}\left[\alpha^2 + 2\alpha^*\alpha + \alpha^{*\,2} + 1\right] = \\
&= \left(\frac{\alpha+\alpha^*}{2}\right)^2 + \frac{1}{4} = (\Re(\alpha))^2 + \frac{1}{4}
\end{aligned} \tag{5-28}$$

$$\begin{aligned}
\langle a_p^2\rangle &= -\,\langle\alpha|\,\frac{(\hat{a}-\hat{a}^\dagger)(\hat{a}-\hat{a}^\dagger)}{4}\,|\alpha\rangle = \\
&= -\frac{1}{4}\left[\langle\alpha|\,\hat{a}^2\,|\alpha\rangle - \langle\alpha|\,\hat{a}^\dagger\hat{a}\,|\alpha\rangle - \underbrace{\langle\alpha|\,\hat{a}\hat{a}^\dagger\,|\alpha\rangle}_{=\langle\alpha|\hat{a}^\dagger\hat{a}|\alpha\rangle+1} + \langle\alpha|\,\hat{a}^{\dagger\,2}\,|\alpha\rangle\right] = \\
&= -\frac{1}{4}\left[\langle\alpha|\,\hat{a}^2\,|\alpha\rangle - 2\,\langle\alpha|\,\hat{a}^\dagger\hat{a}\,|\alpha\rangle + \langle\alpha|\,\hat{a}^{\dagger\,2}\,|\alpha\rangle - 1\right] = \\
&= -\frac{1}{4}\left[\alpha^2 - 2\alpha^*\alpha + \alpha^{*\,2} - 1\right] = \\
&= \left(\frac{\alpha-\alpha^*}{2\iota}\right)^2 + \frac{1}{4} = (\Im(\alpha))^2 + \frac{1}{4}
\end{aligned} \tag{5-29}$$

thus the variances are:

$$\langle\Delta a_q^2\rangle = \langle a_q^2\rangle - (\langle a_q\rangle)^2 = (\Re(\alpha))^2 + \frac{1}{4} - (\Re(\alpha))^2 = \frac{1}{4} \tag{5-30}$$

$$\langle\Delta a_p^2\rangle = \langle a_p^2\rangle - (\langle a_p\rangle)^2 = (\Im(\alpha))^2 + \frac{1}{4} - (\Im(\alpha))^2 = \frac{1}{4} \tag{5-31}$$

We note that variance is identically equal to $\frac{1}{4}$ both for the $a_q$ and $a_p$ quadratures, irrespective of the parameter $\alpha$. Since no dependence is given on the initial phase, this implies that actually **all quadratures have the same variance.** Namely, for all values of $\theta$ it holds:

$$\langle\Delta a_\theta^2\rangle = \frac{1}{4} \tag{5-32}$$

In addition, the variance does not depend on the amplitude $|\alpha|$ of the coherent state. Thus, the uncertainty may be particularly relevant for weak coherent states with low values of $|\alpha|$; on the other hand, it becomes negligible for more intense beams, recovering also in this aspect the classical behaviour.
We finally point out that the product of the uncertainties on $a_q$ and $a_p$ reaches indeed the minimum value allowed by the Heisenberg relations:

$$\langle\Delta a_q\rangle\cdot\langle\Delta a_p\rangle = \sqrt{\frac{1}{4}}\cdot\sqrt{\frac{1}{4}} = \frac{1}{4} \tag{5-33}$$

Coherent states belong to a class of states called **minimum-uncertainty states**.

## 5.4 Non-classical features of Fock states

The expectation values of the $\hat{a}$ and $\hat{a}^\dagger$ operators are readily calculated for a generic Fock state $|n\rangle$:

$$\langle \hat{a} \rangle = \langle n| \hat{a} |n\rangle = \underbrace{\langle n|n-1\rangle}_{=0} \sqrt{n} = 0 \tag{5-34}$$

$$\langle \hat{a}^\dagger \rangle = \langle n| \hat{a}^\dagger |n\rangle = \underbrace{\langle n|n+1\rangle}_{=0} \sqrt{n+1} = 0 \tag{5-35}$$

and similarly for their powers one proves:

$$\langle \hat{a}^m \rangle = \langle n| \hat{a}^m |n\rangle = 0 \qquad \langle (\hat{a}^\dagger)^m \rangle = \langle n| (\hat{a}^\dagger)^m |n\rangle = 0 \tag{5-36}$$

However, for $\hat{n} = \hat{a}^\dagger \hat{a}$ we know that:

$$\langle \hat{a}^\dagger \hat{a} \rangle = \langle \hat{n} \rangle = \langle n| \hat{n} |n\rangle = n \tag{5-37}$$

These results immediately imply that **the expectation value of all quadratures is zero.**

$$\langle \hat{a}_q \rangle = \frac{1}{2}(\langle \hat{a} \rangle + \langle \hat{a}^\dagger \rangle) = 0 \tag{5-38}$$

$$\langle \hat{a}_p \rangle = \frac{1}{2\iota}(\langle \hat{a} \rangle - \langle \hat{a}^\dagger \rangle) = 0 \tag{5-39}$$

$$\langle \hat{a}_\theta \rangle = \frac{1}{2}\left(\langle \hat{a} \rangle e^{-\iota\theta} + \langle \hat{a}^\dagger \rangle e^{+\iota\theta}\right) = 0 \tag{5-40}$$

In fact, for Fock states, the expectation value of the electric field is zero, in any point of the space and in any instant of time:

$$\langle \hat{E} \rangle = \iota\mathcal{E}\left[\langle \hat{a} \rangle e^{\iota\vec{k}\cdot\vec{r}} - \langle \hat{a}^\dagger \rangle e^{-\iota\vec{k}\cdot\vec{r}}\right] = 0 \tag{5-41}$$

**The uncertainty in the quadratures however is not zero.** Let us calculate first $\langle \hat{a}_p^2 \rangle$ and $\langle \hat{a}_p^2 \rangle$:

$$\begin{aligned}\langle \hat{a}_q^2 \rangle &= \frac{1}{4}\langle \hat{a}^2 + (\hat{a}^\dagger)^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} \rangle = \frac{1}{4}\langle \hat{a}^2 \rangle + \frac{1}{4}\langle (\hat{a}^\dagger)^2 \rangle + \frac{1}{4}\langle \underbrace{\hat{a}\hat{a}^\dagger}_{=\hat{a}^\dagger\hat{a}+1} + \hat{a}^\dagger\hat{a} \rangle = \\ &= 0 + 0 + \frac{1}{4}\langle 2\hat{a}^\dagger\hat{a} + 1 \rangle = \frac{2n+1}{4}\end{aligned} \tag{5-42}$$

$$\begin{aligned}\langle \hat{a}_p^2 \rangle &= \frac{1}{(2\iota)^2}\langle \hat{a}^2 + (\hat{a}^\dagger)^2 - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} \rangle = -\frac{1}{4}\langle \hat{a}^2 \rangle - \frac{1}{4}\langle (\hat{a}^\dagger)^2 \rangle + \frac{1}{4}\langle \underbrace{\hat{a}\hat{a}^\dagger}_{=\hat{a}^\dagger\hat{a}+1} + \hat{a}^\dagger\hat{a} \rangle = \\ &= 0 + 0 + \frac{1}{4}\langle 2\hat{a}^\dagger\hat{a} + 1 \rangle = \frac{2n+1}{4}\end{aligned} \tag{5-43}$$

Thus the variances are:

$$\langle \Delta a_q^2 \rangle = \langle \hat{a}_q^2 \rangle - (\langle \hat{a}_q \rangle)^2 = \frac{2n+1}{4} - 0 = \frac{2n+1}{4} \tag{5-44}$$

$$\langle \Delta a_p^2 \rangle = \langle \hat{a}_p^2 \rangle - (\langle \hat{a}_p \rangle)^2 = \frac{2n+1}{4} - 0 = \frac{2n+1}{4} \tag{5-45}$$

As well, the uncertainty in the electric field value is nonvanishing. We calculate $\langle \hat{E}^2 \rangle$ as follows:

$$\begin{aligned}\langle \hat{E}^2 \rangle = -\mathcal{E}^2\langle \left[\hat{a}e^{\iota\vec{k}\cdot\vec{r}} - \hat{a}^\dagger e^{-\iota\vec{k}\cdot\vec{r}}\right]^2 \rangle = -\mathcal{E}^2 \left[\underbrace{\langle \hat{a}^2 \rangle}_{=0} e^{\iota 2\vec{k}\cdot\vec{r}} + \underbrace{\langle (\hat{a}^\dagger)^2 \rangle}_{=0} e^{-\iota 2\vec{k}\cdot\vec{r}} - \langle \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger \rangle\right] = \\ = \mathcal{E}^2\langle 2\hat{a}^\dagger\hat{a} + 1 \rangle = \mathcal{E}^2(2n+1) = \frac{\hbar\omega}{\varepsilon_0 V}\left(n + \frac{1}{2}\right)\end{aligned} \tag{5-46}$$

However, being $\langle \hat{E} \rangle = 0$, the variance of the electric field is simply:

$$\langle \Delta E^2 \rangle = \langle \hat{E}^2 \rangle - (\langle \hat{E} \rangle)^2 = \frac{\hbar\omega}{\varepsilon_0 V}\left(n + \frac{1}{2}\right) \tag{5-47}$$

The above results have profound implications and allows to understand better how Fock states are **deeply non-classical**. The Fock states have expectation values for the quadratures, as well as expectation values for the electric field, that are *identically zero*. However, the uncertainty in these values is not zero, which reveals the existence of quantum fluctuations and the presence of a finite non-vanishing energy in the electromagnetic mode.
It is important to understand in which sense the expectation values are zero, in order not to be confused with averaged values over a region of space or an interval of time. A classical wave, for instance, has an *average* electric field that can be zero if evaluated on a certain region of the space which contains an integer number of wavelengths, or in an interval of time that is multiple of the period. However, if we were able to measure the electric field of a classical wave in a given position $\vec{r}$ and in a given instant of time $t$, we would retrieve a definite and generally non-zero value for the electric field. The uncertainty could be made arbitrarily small by improving in the measurement procedure. In the case of Fock states, on the contrary, in a given position and instant of time the *expectation value* (which correspond to a hypothetical ensemble average on a set of identically prepared system) of the field gives zero with some uncertainty. Thus, a hypothetic measurement at position $\vec{r}$ and time $t$ would provide generally a non-zero value, picked from a random distribution with average zero and variance given by (5-47).
These findings are particularly striking in the case of the vacuum state $|0\rangle$, which is the Fock state with $n = 0$. The electromagnetic mode is excited with precisely $n = 0$ photons. However, the energy is non-zero and equal to the point-zero energy $E_0 = \frac{1}{2}\hbar\omega$. The expectation value of the electric field is zero, but its variance is not, and is equal to:

$$\langle \Delta E^2 \rangle_{vacuum} = \frac{\hbar\omega}{2\varepsilon_0 V} \tag{5-48}$$

These are the vacuum fluctuations of the electric field, predicted by quantum theory.

## 5.5 Phase space representation

The state of an electromagnetic mode, in Classical Physics, is completely determined the values of a pair of orthogonal quadratures, such as $E_Q$ and $E_P$. These two quantities can be considered as generalized coordinates that describes the state of the physical system, namely define its **phase space**. It is thus possible to represent this status in a cartesian graph with axes $E_Q$ and $E_P$: one point in the graph determines the phase and amplitude of the wave. In a more complex situation, in which there is some statistical uncertainty on the coordinates, instead of a single point we can indicate a function $p(E_P, E_Q)$, which is a probability distribution and represents in each point the probability of measuring simultaneously the two quadratures with values in the intervals $[E_Q, E_Q + dE_Q]$ and $[E_P, E_P + dE_P]$.
We may ask whether it is possible to extend to the quantum realm this phase space representation of the status of an electromagnetic mode. To this purpose, we might think of using the adimensional quadratures $\hat{a}_q$ and $\hat{a}_p$. However, we realize immediately that we cannot represent a point on the graph, or a probability distribution, with the same meaning as in the classical case: the two quadratures are incompatible observables as their operators do not commute. The probability of getting two specific values, say $\alpha_q$ and $\alpha_p$, in the simultaneous measurement of the two quadratures, does not have a relevant physical meaning and cannot describe the state.
To extend the phase space representation to the quantum regime we have indeed to change perspective, and there are several different way to do this. One of these is the following. We no longer think of measuring the quadratures independently, but we think of quantifying somehow **the overlap of the quantum state of the mode with a coherent state** with complex eigenvalue $\alpha = \alpha_q + \iota\alpha_p$. Here

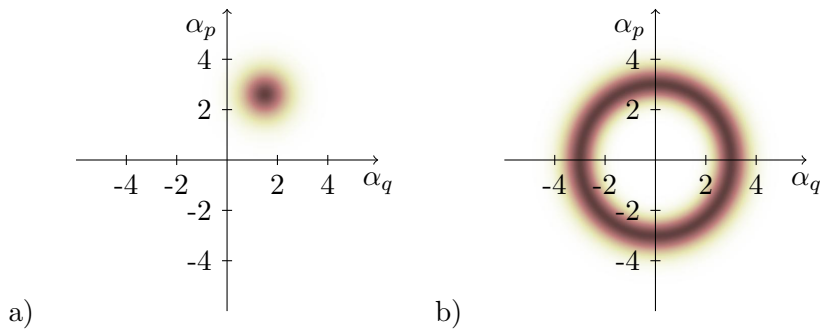


Figure 5: Color-scale representation of the Husimi $Q$ function (a) for a coherent state $|\beta\rangle$ with $\beta = 3e^{\iota\pi/3} = \frac{3}{2} + \iota\frac{3\sqrt{3}}{2}$ and (b) for a Fock state $|n\rangle$ with $n = 9$.

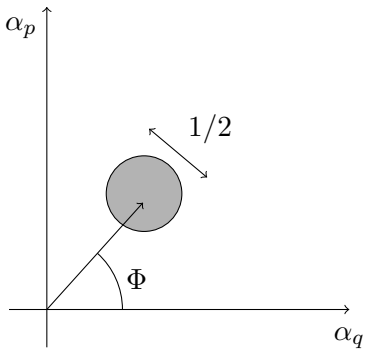


Figure 6: A coherent state $|\beta\rangle$ is represented by a circular 'cloud' in the phase space, centered on the point $(\Re\{\beta\}, \Im\{\beta\})$. Having traced a vector joining the origin to one point inside the cloud, the angle between its direction and the $\alpha_q$ axis is indicative of the optical phase $\Phi$ (though this association cannot be rigorous).

$\alpha_q$ and $\alpha_p$ are the real and imaginary parts of the complex eigenvalue. This overlap is quantified by the so-called **Husimi-Q function**, that we write for an unknown state $|\psi\rangle$ to be characterized as:

$$Q(\alpha) = \frac{1}{\pi}|\langle\alpha|\psi\rangle|^2 \tag{5-49}$$

The $1/\pi$ factor is a normalization factor, in order to obtain:

$$\int Q(\alpha)d^2\alpha = 1 \tag{5-50}$$

Note that, since coherent states with different eigenvalues are not orthogonal, the $Q$ function characterizing a coherent state does not identify a single point on the plane, but is actually a larger bell-shaped function. In detail, the $Q$ function for a coherent state with eigenvalue $\beta$ would be written as:

$$Q(\alpha) = \frac{1}{\pi}e^{-|\alpha-\beta|^2} = \frac{1}{\pi}e^{-(\alpha_q-\Re(\beta))^2-(\alpha_p-\Im(\beta))^2} \tag{5-51}$$

This is a Gaussian centered on the complex point $\beta$, namely, in the two dimensional graph, on the point $(\Re\{\beta\}, \Im\{\beta\})$ (see Figure 5a). It describes indeed a circular 'cloud' around this central point that reflects the intrinsic uncertainty in the quadrature values, that we have just proven in Section 5.3. Note that as the time passes, this cloud shows a circular motion around the origin with angular frequency $\omega$, because the coherent state evolves as $\beta = \beta_0 e^{-\iota\omega t}$.
If we now trace a vector from the origin of the phase space to a point inside this cloud as in Fig. 6, the allowed dispersion in the angle $\Phi$ determined with the horizontal axis represents pictorially the

uncertainty in the optical phase.[13] If we quantify the diameter of the cloud with the standard deviation in the quadrature values, which is 1/2, we may write:

$$\Delta\Phi \simeq \frac{1/2}{|\beta|} = \frac{1}{2|\beta|} \tag{5-52}$$

as the uncertainty in the optical phase.
For the coherent state, we have $|\beta| = \sqrt{\langle n \rangle} = \Delta n$. We could then obtain from Eq. (5-52):

$$\Delta\Phi\,\Delta n = \frac{1}{2} \tag{5-53}$$

which is a sort of Heisenberg-type uncertainty relation that constrains together the number of photons and the optical phase. The larger the modulus of $\beta$, the larger is the absolute uncertainty on the number of photons, but the optical phase is determined with less uncertainty. In fact, this corresponds to the classical limit where we have a coherent state with large amplitude, and it becomes more meaningful to define an optical phase of the beam.
In the case of a Fock state $|n\rangle$, the $Q$ function would be:

$$Q(\alpha) = \frac{1}{\pi}\frac{|\alpha|^{2n}}{n!}e^{-|\alpha|^2} \tag{5-54}$$

this is a ring-shaped function centered on the origin (see Figure 5b). The radius of the ring depends on the number of photons $n$, and its width interprets the uncertainty in the quadrature amplitude. The fact that it is a circularly symmetric function around the origin represents the complete indetermination of the optical phase for this quantum state of the field.

[13] It is actually tricky to define rigorously the optical phase in quantum optics. Here we give this sort of heuristic treatment, which shouldn't be pushed too much forward beyond the results that we give here.

# Notes on Quantum Optics - Section 6

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*Notes for the course 'Quantum Optics and Information' - A.Y. 2023/2024. These notes are provided for personal use only, with no claim of completeness or absence of errors.*

# 6 Beam splitters and interferometers

## 6.1 Classical and quantum description of a beam splitter

A beam splitter is an optical device that couples linearly two optical modes. While different physical systems may be coinceived to implement this function, the simplest idealized representation of a beam splitter may be a semi-reflecting mirror onto which two optical beams are impinging (Fig. 7). The beams are incident on opposite sides and are symmetrical with respect to the beam splitter plane; in this way, the partly reflected waves of one beam are perfectly overlapped to the partly transmitted waves of the other. Here we will consider beams consisting each of a single mode of the electromagnetic field, with the same frequency $\omega$.
The **classical description** of the beam splitter is conveniently carried out referring to phasors of the input and output monochromatic electromagnetic waves. We write the electric field of the wave as:

$$E = \imath\mathcal{E}\left(\alpha_\omega(t)e^{\imath\vec{k}\cdot\vec{r}} - \alpha_\omega^*(t)e^{-\imath\vec{k}\cdot\vec{r}}\right) = \imath\mathcal{E}\left(\alpha e^{\imath(\vec{k}\cdot\vec{r}-\omega t)} - \alpha^* e^{-\imath(\vec{k}\cdot\vec{r}-\omega t)}\right) \tag{6-1}$$

The normal variable associated to the optical mode evolves indeed in time as:

$$\alpha_\omega(t) = \alpha e^{-\imath\omega t}$$

where the complex number $\alpha$ may include a phase argument (namely, it may be written as $\alpha = |\alpha|e^{-\imath\phi}$) and acts as a phasor for this wave.
The beam splitter operation can be written as a linear relation among the phasors of the input and output modes. We may name $\alpha$ and $\beta$ the phasors of the input modes, and $\alpha'$ and $\beta'$ the phasors of the output modes. If suitable phase references are considered for the input and output modes, a generic lossless beam splitter can be described by the relations:

$$\alpha' = t\alpha + \imath r\beta \tag{6-2}$$
$$\beta' = \imath r^*\alpha + t\beta \tag{6-3}$$

where $r$ is a complex number while $t$ can be kept as real, and $t^2 + |r|^2 = 1$. Without loosing generality one can write:

$$t = \cos(\theta/2) \qquad\qquad r = e^{\imath\varphi}\sin(\theta/2) \tag{6-4}$$

The constraints for the coefficients in Eqs. (6-2) and (6-3) are basically a consequence of energy conservation (see Box 6-1). The beam splitter relations are written more compactly in matrix form:

$$\boxed{\begin{bmatrix}\alpha'\\ \beta'\end{bmatrix} = \begin{bmatrix} t & \imath r\\ \imath r^* & t\end{bmatrix}\cdot\begin{bmatrix}\alpha\\ \beta\end{bmatrix} = M\cdot\begin{bmatrix}\alpha\\ \beta\end{bmatrix}} \tag{6-5}$$

The condition of energy conservation is expressed here as the requirement for $M$ to be unitary, namely:

$$M^{-1} = M^\dagger \tag{6-6}$$

Following the approach used for the field quantization, it would look natural to describe the beam splitter in **quantum mechanical terms** by replacing the normal variables with the annihilation operators of the optical modes. We write:

$$\hat{a}' = t\hat{a} + \imath r\hat{b} \tag{6-7}$$
$$\hat{b}' = \imath r^*\hat{a} + t\hat{b} \tag{6-8}$$

where we have indicated with $\hat{a}$ and $\hat{b}$ the annihilation operators of the input modes and with $\hat{a}'$ and $\hat{b}'$ the annihilation operators of the output modes.

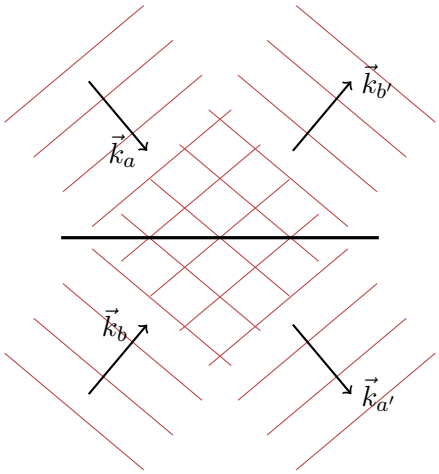


Figure 7: Scheme of a beam-splitter, represented as a planar semi-reflecting mirror, with two single-mode optical beams incident from symmetric directions, indicated by the wavevectors $\vec{k}_a$ and $\vec{k}_b$. The partly reflected or transmitted beams have wavevectors $\vec{k}_{a'}$ and $\vec{k}_{b'}$.

These relations are also written in the matrix form:

$$\boxed{\begin{bmatrix}\hat{a}' \\ \hat{b}'\end{bmatrix} = \begin{bmatrix} t & \imath r \\ \imath r^* & t \end{bmatrix} \cdot \begin{bmatrix}\hat{a} \\ \hat{b}\end{bmatrix} = M \cdot \begin{bmatrix}\hat{a} \\ \hat{b}\end{bmatrix}} \tag{6-9}$$

where indeed $M$ is the same of the classical case. **We will now assume these linear relations to hold** and we will later show that they give the classical limit in the case of coherent input states.
Before proceeding, it is very important to state correctly what we are referring to with the operators $\hat{a}'$ and $\hat{b}'$. In quantum mechanics we can describe the beam splitter as a component that implements a certain Hamiltonian $\hat{H}_{\text{BS}}$, that makes the two modes interact together. For the moment we may ignore the precise form of this Hamiltonian (which is discussed in Box 6-2). Anyway, this Hamiltonian would make an input state evolve according to an evolution operator $\hat{U}(t) = e^{-\imath \frac{\hat{H}_{\text{BS}}}{\hbar} t}$. In the beam splitter the interaction occurs for a fixed limited time $\tau$, thus if we consider only the relation between input and output states we can limit ourself to use the operator $\hat{U} \equiv \hat{U}(\tau)$.
In this framework, the output-mode operators $\hat{a}'$ and $\hat{b}'$ are correctly defined as the **evolved operators** in the Heisenberg picture:

$$\hat{a}' = \hat{U}^\dagger \, \hat{a} \, \hat{U} \qquad\qquad \hat{b}' = \hat{U}^\dagger \, \hat{b} \, \hat{U} \tag{6-10}$$

The matricial beam splitter relations indeed mean:

$$\boxed{\begin{bmatrix}\hat{U}^\dagger \, \hat{a} \, \hat{U} \\ \hat{U}^\dagger \, \hat{b} \, \hat{U}\end{bmatrix} = M \cdot \begin{bmatrix}\hat{a} \\ \hat{b}\end{bmatrix} = \begin{bmatrix} t & \imath r \\ \imath r^* & t \end{bmatrix} \cdot \begin{bmatrix}\hat{a} \\ \hat{b}\end{bmatrix}} \tag{6-11}$$

By taking the Hermitian conjugate we can also derive the relations for the evolved creation operators:

$$\boxed{\begin{bmatrix}\hat{U}^\dagger \, \hat{a}^\dagger \, \hat{U} \\ \hat{U}^\dagger \, \hat{b}^\dagger \, \hat{U}\end{bmatrix} = M^* \cdot \begin{bmatrix}\hat{a}^\dagger \\ \hat{b}^\dagger\end{bmatrix} = \begin{bmatrix} t & -\imath r^* \\ -\imath r & t \end{bmatrix} \cdot \begin{bmatrix}\hat{a}^\dagger \\ \hat{b}^\dagger\end{bmatrix}} \tag{6-12}$$

Note that $(\hat{U}^\dagger \, \hat{a} \, \hat{U})^\dagger = (\hat{U}^\dagger)^\dagger \, \hat{a}^\dagger \hat{U}^\dagger = \hat{U}^\dagger \, \hat{a}^\dagger \, \hat{U}$. Also note that in Eq. (6-12) we don't find the Hermitian conjugate of $M$ but only its complex conjugate $M^*$, because what we do is actually to take the Hermitian conjugate of the Eqs. (6-7) and (6-8): the operators become their Hermitian conjugates but the multiplication coefficients (which are the matrix entries) just become complex conjugates.

Box 6-1 - Form for the beam splitter relations

Let's assume a very generic linear relation:

$$\boxed{\begin{aligned}\alpha' &= t_a\alpha + r_a\beta \\ \beta' &= r_b\alpha + t_b\beta\end{aligned}}$$

The energy carried by the optical modes is proportional to the squared modulus of the normal variables, namely to the squared modulus of the phasors. **Energy conservation** corresponds to imposing:

$$|\alpha'|^2 + |\beta'|^2 = \alpha'^*\alpha' + \beta'^*\beta' = \alpha^*\alpha + \beta^*\beta = |\alpha|^2 + |\beta|^2$$

By substituting the relations for $\alpha'$ and $\beta'$ one obtains:

$$|\alpha'|^2 + |\beta'|^2 = (|t_a|^2 + |r_b|^2)\,|\alpha|^2 + (|r_a|^2 + |t_b|^2)\,|\beta|^2 + (t_a^* r_a + t_b r_b^*)\alpha^*\beta + (t_a r_a^* + t_b^* r_b)\alpha\beta^*$$

in order for this to be equal to $|\alpha|^2 + |\beta|^2$ for any $\alpha$ and $\beta$, one needs the following three conditions to be satisfied together:

$$\begin{cases} |t_a|^2 + |r_b|^2 = 1 \\ |r_a|^2 + |t_b|^2 = 1 \\ t_a^* r_a = -t_b r_b^* \end{cases} \qquad (\star)$$

We take the squared modulus of the third one and substitute from the other two:

$$|t_a|^2|r_a|^2 = |t_b|^2|r_b|^2 \quad \rightarrow \quad |t_a|^2(1-|t_b|^2) = |t_b|^2(1-|t_a|^2) \quad \rightarrow \quad |t_a|^2 = |t_b|^2$$

$$|t_a|^2|r_a|^2 = |t_b|^2|r_b|^2 \quad \rightarrow \quad (1-|r_b|^2)|r_a|^2 = (1-|r_a|^2)|r_b|^2 \quad \rightarrow \quad |r_a|^2 = |r_b|^2$$

We consider first the transmission coefficients $t_a$ and $t_b$. We have just shown that they have the same modulus. If we fix **convenient phase references** for the output beams we can take them both as real and write:

$$\boxed{t_a = t_b = t}$$

Then we consider the reflection coefficients $r_a$ and $r_b$. If we take $\boxed{r_a = \imath r}$ the third of the three conditions $(\star)$ reads:

$$t\imath r = -t r_b^* \quad \rightarrow \quad r_b = (-\imath r)^* \quad \rightarrow \quad \boxed{r_b = \imath r^*}$$

This justifies the form of Eqs. (6-2) and (6-3). To justify also Eq. (6-4) we observe that the first two conditions in $(\star)$ now both translate into $t^2 + |r|^2 = 1$. This means that the coefficient $t$ is a real number comprised between -1 and 1, which can always be written as:

$$\boxed{t = \cos(\theta/2)}$$

for a given $\theta$. For the coefficient $r$ we have as a consequence:

$$|r|^2 = 1 - t^2 = 1 - \cos^2(\theta/2) = \sin^2(\theta/2)$$

To be general, we provide $r$ with an arbitrary complex phase and write it as:

$$\boxed{r = e^{\imath\varphi}\sin(\theta/2)}$$

Box 6-2 - The beam splitter Hamiltonian

We prove here that the evolution operator $\hat{U}$ can be obtained as $\hat{U} = e^{-\iota \hat{H}_{BS}\tau/\hbar}$ considering the following Hamiltonian, which describes the interaction between two optical modes:

$$\boxed{\hat{H}_{BS} = -\hbar\frac{g}{2}(e^{-\iota\varphi}\hat{a}\hat{b}^\dagger + e^{+\iota\varphi}\hat{a}^\dagger\hat{b})}$$

where $g$ determines the strength of the interaction.
We start by writing the evolution operator in a more compact form as:

$$\hat{U} = e^{-\iota \hat{H}_{BS}\tau/\hbar} = e^{-\iota\hat{G}\theta/2}$$

with $\hat{G} = -(e^{-\iota\varphi}\hat{a}\hat{b}^\dagger + e^{+\iota\varphi}\hat{a}^\dagger\hat{b})$ and $\theta = g\tau$. The evolution of the operators $\hat{a}$ and $\hat{b}$ in the Heisenberg picture is given by:

$$\hat{a}' = \hat{U}^\dagger\hat{a}\hat{U} = e^{+\iota\hat{G}\theta/2}\,\hat{a}\,e^{-\iota\hat{G}\theta/2} \qquad \hat{b}' = \hat{U}^\dagger\hat{b}\hat{U} = e^{+\iota\hat{G}\theta/2}\,\hat{b}\,e^{-\iota\hat{G}\theta/2}$$

It is useful at this point to exploit the Baker-Hausdorff lemma, which holds for two operators $\hat{A}$ and $\hat{B}$:

$$e^{\hat{B}}\,\hat{A}\,e^{-\hat{B}} = \hat{A} + [\hat{B},\hat{A}] + \frac{1}{2!}[\hat{B},[\hat{B},\hat{A}]] + \ldots = \hat{A} + \sum_{n=1}^{+\infty}\frac{\overbrace{[\hat{B},[\hat{B},\ldots[\hat{B},}^{n\text{ times}}\hat{A}]]]}{n!}$$

In this case, considering for instance the evolution of the operator $\hat{a}$, we have:

$$e^{+\iota\hat{G}\theta/2}\,\hat{a}\,e^{-\iota\hat{G}\theta/2} = \hat{a} + \textstyle\sum_{n=1}^{+\infty}\left(\iota\frac{\theta}{2}\right)^n\frac{\overbrace{[\hat{G},[\hat{G},\ldots[\hat{G},}^{n\text{ times}}\hat{a}]]]}{n!}$$

The nested commutators $\overbrace{[\hat{G},[\hat{G},\ldots[\hat{G},}^{n\text{ times}}\hat{a}]]]$ are not difficult to evaluate. For $n = 1$ one has:

$$[\hat{G},\hat{a}] = -[e^{-\iota\varphi}\hat{a}\hat{b}^\dagger,\hat{a}] - [e^{+\iota\varphi}\hat{a}^\dagger\hat{b},\hat{a}] = -e^{-\iota\varphi}[\hat{a},\hat{a}]\,\hat{b}^\dagger - e^{+\iota\varphi}[\hat{a}^\dagger,\hat{a}]\hat{b} = e^{+\iota\varphi}\hat{b}$$

while for $n = 2$:

$$[\hat{G},[\hat{G},\hat{a}]] = [\hat{G},e^{+\iota\varphi}\hat{b}] = -[e^{-\iota\varphi}\hat{a}\hat{b}^\dagger,e^{+\iota\varphi}\hat{b}] - [e^{+\iota\varphi}\hat{a}^\dagger\hat{b},e^{+\iota\varphi}\hat{b}] = -\hat{a}[\hat{b}^\dagger,\hat{b}] - e^{+2\iota\varphi}\hat{a}^\dagger[\hat{b},\hat{b}] = \hat{a}$$

Going on, depending on $n$ being even or odd, one has simply:

$$\overbrace{[\hat{G},[\hat{G},\ldots[\hat{G},}^{n\text{ (even) times}}\hat{a}]]] = \hat{a} \qquad \overbrace{[\hat{G},[\hat{G},\ldots[\hat{G},}^{n\text{ (odd) times}}\hat{a}]]] = \hat{e}^{+\iota\varphi}\hat{b}$$

We thus obtain:

$$e^{+\iota\hat{G}\theta/2}\,\hat{a}\,e^{-\iota\hat{G}\theta/2} = \sum_{n\text{ odd}}\frac{\left(\iota\frac{\theta}{2}\right)^n}{n!}e^{+\iota\varphi}\hat{b} + \sum_{n\text{ even}}\frac{\left(\iota\frac{\theta}{2}\right)^n}{n!}\hat{a}$$

Note that, starting the index of the summation for $n$ even from 0, we include also the isolated term $\hat{a}$ that was present in the former expression. We recognize in the two sums for odd and even $n$ respectively the series expansion of sine and cosine functions. We conclude:

$$\hat{a}' = \hat{U}^\dagger\hat{a}\hat{U} = e^{+\iota\hat{G}\theta/2}\,\hat{a}\,e^{-\iota\hat{G}\theta/2} = \cos(\theta/2)\hat{a} + e^{\iota\varphi}\sin(\theta/2)\hat{b}$$

The above equation is precisely Eq. (6-7). A similar procedure can be carried out for $\hat{b}' = \hat{U}^\dagger\hat{b}\hat{U}$, to retrieve Eq. (6-8).

## 6.2 Action of the beam splitter on quantum states of light

According to the formalism discussed above, the action of the beam splitter on a **generic input state** $|\psi_{in}\rangle$, to generate the output state $|\psi_{out}\rangle$, can be written as:

$$|\psi_{out}\rangle = \hat{U}\,|\psi_{in}\rangle \tag{6-13}$$

In quantum optics, light states are often described with the action of an operator on the vacuum state. For instance, the single photon state of one mode is $|1\rangle = \hat{a}^\dagger\,|0\rangle$ while a coherent state is written using the Glauber displacement operator $|\alpha\rangle = \hat{D}(\alpha)\,|0\rangle$. If we write:

$$\boxed{|\psi_{in}\rangle = \hat{A}\,|0\rangle} \tag{6-14}$$

we have:

$$|\psi_{out}\rangle = \hat{U}\hat{A}\,|0\rangle \tag{6-15}$$

We may now observe that the action of the beam splitter on a vacuum state, physically, can only produce the vacuum state at the output. Thus it is licit to write:

$$\hat{U}\,|0\rangle = |0\rangle \qquad\qquad |0\rangle = \hat{U}^\dagger\,|0\rangle \tag{6-16}$$

which applied to Eq. (6-15) gives:

$$\boxed{|\psi_{out}\rangle = \hat{U}\,\hat{A}\,\hat{U}^\dagger\,|0\rangle} \tag{6-17}$$

This results to be a more convenient form, for some very practical reasons we now illustrate. The operator $\hat{A}$ is generally a function of creation and annihilation operators of the optical input modes. We know from Eqs. (6-11) and (6-12) how to describe the time evolution of these operators. Hence, we can derive immediately, for the annihilation operators:

$$\hat{U}\begin{bmatrix}\hat{U}^\dagger\,\hat{a}\,\hat{U}\\ \hat{U}^\dagger\,\hat{b}\,\hat{U}\end{bmatrix}\hat{U}^\dagger = \hat{U}\cdot M\begin{bmatrix}\hat{a}\\ \hat{b}\end{bmatrix}\hat{U}^\dagger = M\cdot\hat{U}\begin{bmatrix}\hat{a}\\ \hat{b}\end{bmatrix}\hat{U}^\dagger \tag{6-18}$$

which gives:

$$\begin{bmatrix}\hat{a}\\ \hat{b}\end{bmatrix} = M\cdot\begin{bmatrix}\hat{U}\,\hat{a}\,\hat{U}^\dagger\\ \hat{U}\,\hat{b}\,\hat{U}^\dagger\end{bmatrix} \tag{6-19}$$

and by exploiting the fact that $M^{-1} = M^\dagger$:

$$\boxed{\begin{bmatrix}\hat{U}\,\hat{a}\,\hat{U}^\dagger\\ \hat{U}\,\hat{b}\,\hat{U}^\dagger\end{bmatrix} = M^\dagger\cdot\begin{bmatrix}\hat{a}\\ \hat{b}\end{bmatrix} = \begin{bmatrix}t & -\iota r\\ -\iota r^* & t\end{bmatrix}\cdot\begin{bmatrix}\hat{a}\\ \hat{b}\end{bmatrix}} \tag{6-20}$$

Analogously, for the creation operators one writes:

$$\boxed{\begin{bmatrix}\hat{U}\,\hat{a}^\dagger\,\hat{U}^\dagger\\ \hat{U}\,\hat{b}^\dagger\,\hat{U}^\dagger\end{bmatrix} = M^T\cdot\begin{bmatrix}\hat{a}^\dagger\\ \hat{b}^\dagger\end{bmatrix} = \begin{bmatrix}t & \iota r^*\\ \iota r & t\end{bmatrix}\cdot\begin{bmatrix}\hat{a}^\dagger\\ \hat{b}^\dagger\end{bmatrix}} \tag{6-21}$$

As a first example, we derive the evolution of a **single-photon state** injected in mode $a$, together with the vacuum state in mode $b$.

$$|\psi_{in}\rangle = |1\rangle_a\,|0\rangle_b = \hat{a}^\dagger\,|0\rangle \tag{6-22}$$

We have, according to the discussion above:

$$|\psi_{out}\rangle = \hat{U}\,|\psi_{in}\rangle = \hat{U}\hat{a}^\dagger\,|0\rangle = \hat{U}\hat{a}^\dagger\hat{U}^\dagger\,|0\rangle \tag{6-23}$$

and then:

$$|\psi_{out}\rangle = (t\hat{a}^\dagger + \iota r^*\hat{b}^\dagger)\,|0\rangle = \cos(\theta/2)\,|1\rangle_a\,|0\rangle_b + \iota\sin(\theta/2)e^{-\iota\varphi}\,|0\rangle_a\,|1\rangle_b \tag{6-24}$$

This is a quantum superposition of the single photon remaining on the mode $a$ (with probability $\cos^2\frac{\theta}{2}$) and of the single photon being transferred to mode $b$ (with probability $\sin^2\frac{\theta}{2}$). Obviously a single photon cannot be split, and at the output can be measured in *either* of the two modes. The state $|\psi_{out}\rangle$ in Eq. (6-24) involves two optical modes and is **entangled**, because it cannot be written as a product of independent states of the modes.
Proceeding with a similar method one could also study the case of an input state with $n$ photons on one mode, namely an $n$-photon Fock state. As discussed in Box 6-3, one would find that the photons are probabilistically split in the two outputs according to a binomial distribution.
It is worth studying in detail the case of a **coherent input state** on one mode (e.g. mode $a$):

$$|\psi_{in}\rangle = |\alpha\rangle_a = \hat{D}_a(\alpha)\,|0\rangle \tag{6-25}$$

with $\hat{D}_a(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}$. This requires mathematical passages that are a bit more involved than those employed for simple Fock states, as we need to transform the displacement operator:

$$|\psi_{out}\rangle = \hat{U}\,\hat{D}_a(\alpha)\,\hat{U}^\dagger\,|0\rangle \tag{6-26}$$

We observe that the operator $\hat{D}_a(\alpha)$ can be written as a power series of the operator $\hat{B} = \alpha\hat{a}^\dagger - \alpha^*\hat{a}$:

$$\hat{D}_a(\alpha) = e^{\hat{B}} = \sum_{n=0}^{+\infty}\frac{\hat{B}^n}{n!} \tag{6-27}$$

Thus, by developing the transformation $\hat{U}\,e^{\hat{B}}\,\hat{U}^\dagger$ we get:

$$\begin{aligned}\hat{U}\,\hat{e}^{\hat{B}}\,\hat{U}^\dagger &= \sum_{n=0}^{+\infty}\hat{U}\frac{\hat{B}^n}{n!}\hat{U}^\dagger = \sum_{n=0}^{+\infty}\frac{\hat{U}\cdot\overbrace{\hat{B}\cdot\hat{B}\cdot\ldots\cdot\hat{B}}^{n\text{ times}}\cdot\hat{U}^\dagger}{n!} = \\ &= \sum_{n=0}^{+\infty}\frac{\overbrace{\hat{U}\hat{B}\;\hat{U}^\dagger\cdot\hat{U}\hat{B}\hat{U}^\dagger\cdot\ldots\cdot\hat{U}\hat{B}\hat{U}^\dagger}^{n\text{ times}}}{n!} = e^{\hat{U}\,\hat{B}\,\hat{U}^\dagger}\end{aligned} \tag{6-28}$$

Going back to the transformation of the coherent state by the beamsplitter we write:

$$\begin{aligned}|\psi_{out}\rangle &= \hat{U}\,\hat{D}_a(\alpha)\,\hat{U}^\dagger\,|0\rangle = e^{\hat{U}\,\hat{B}\,\hat{U}^\dagger}\,|0\rangle = e^{\alpha\hat{U}\,\hat{a}^\dagger\,\hat{U}^\dagger - \alpha^*\hat{U}\,\hat{a}\,\hat{U}^\dagger}\,|0\rangle = \\ &= \exp\Big(\alpha(t\hat{a}^\dagger + \imath r^*\hat{b}^\dagger) - \alpha^*(t\hat{a} - \imath r\hat{b})\Big)\,|0\rangle = \exp\Big((t\alpha\hat{a}^\dagger - t\alpha^*\hat{a}) + (\imath r^*\alpha\hat{b}^\dagger - (\imath r^*\alpha)^*\hat{b})\Big)\,|0\rangle\end{aligned} \tag{6-29}$$

The operators associated to the different modes $a$ and $b$ commute, so we can write the exponential as a product of two, the first one containing $\hat{a}$ and $\hat{a}^\dagger$, the second containing $\hat{b}$ and $\hat{b}^\dagger$. We realize that these have indeed the form of displacement operators respectively on modes $a$ and $b$:

$$\begin{aligned}|\psi_{out}\rangle &= \exp\Big(t\alpha\hat{a}^\dagger - t\alpha^*\hat{a}\Big)\cdot\exp\Big(\imath r^*\alpha\hat{b}^\dagger - (\imath r^*\alpha)^*\hat{b}^\dagger\Big)\,|0\rangle = \\ &= \hat{D}_a(t\alpha)\cdot\hat{D}_b(\imath r^*\alpha)\,|0\rangle = |t\alpha\rangle_a\,|\imath r^*\alpha\rangle_b\end{aligned} \tag{6-30}$$

The two output modes result to be each in a coherent state, characterized respectively by the eigenvalues $t\alpha$ and $\imath r^*\alpha$. [14] Note that the overall state is **not entangled**. This result fully agrees with the classical description and is a confirmation of the validity of the quantum model that we have assumed for the beam splitter.

[14] If we consider Eqs. (6-2) and (6-3) with $\beta = 0$ we have indeed $\alpha' = t\alpha$ and $\beta' = \imath r^*\alpha$.

Box 6-3 - Beam splitter action on a Fock-state input

We consider an input state consisting in a Fock state $|n\rangle$ on mode $a$ and in the vacuum state on mode $b$:

$$|\psi_{in}\rangle = |n\rangle_a |0\rangle_b = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle$$

At the output we have:

$$|\psi_{out}\rangle = \frac{1}{\sqrt{n!}} \hat{U}(\hat{a}^\dagger)^n \hat{U}^\dagger |0\rangle = \frac{1}{\sqrt{n!}} \hat{U} \cdot \overbrace{\hat{a}^\dagger \cdot \hat{a}^\dagger \cdot \ldots \cdot a^\dagger}^{n \text{ times}} \cdot \hat{U}^\dagger |0\rangle =$$

$$= \frac{1}{\sqrt{n!}} \overbrace{\hat{U}\hat{a}^\dagger\hat{U}^\dagger \cdot \hat{U}\hat{a}^\dagger\hat{U}^\dagger \cdot \ldots \cdot \hat{U}\hat{a}^\dagger\hat{U}^\dagger}^{n \text{ times}} |0\rangle = \frac{1}{\sqrt{n!}} \left(\hat{U}\hat{a}^\dagger\hat{U}^\dagger\right)^n |0\rangle$$

which is:

$$|\psi_{out}\rangle = \frac{1}{\sqrt{n!}} (t\hat{a}^\dagger + \imath r^* b^\dagger)^n |0\rangle =$$

$$= \frac{1}{\sqrt{n!}} \sum_{k=0}^{n} \binom{n}{k} t^k (\imath r^*)^{n-k} (\hat{a}^\dagger)^k (\hat{b}^\dagger)^{n-k} |0\rangle$$

Now we observe that:

$$(\hat{a}^\dagger)^k |0\rangle = \sqrt{k!} |k\rangle_a |0\rangle_b \qquad (\hat{b}^\dagger)^{n-k} |0\rangle = \sqrt{(n-k)!} |0\rangle_a |n-k\rangle_b$$

in addition:

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

Therefore, we can write in conclusion:

$$|\psi_{out}\rangle = \sum_{k=0}^{n} \sqrt{\binom{n}{k}}\, t^k (\imath r^*)^{n-k}\, |k\rangle_a |n-k\rangle_b$$

which is a superposition of many terms $|k\rangle_a |n-k\rangle_b$; in general, the overall state is entangled. Note that the probability associated to the state $|k\rangle_a |n-k\rangle_b$, namely the squared modulus of its coefficient in the sum, is:

$$p_{k,n-k} = |c_{k,n-k}|^2 = \binom{n}{k} t^{2k} |r|^{2(n-k)} = \binom{n}{k} p_1^k (1-p_1)^{(n-k)}$$

where we have taken into account that $|r^2| = 1 - t^2$ and we have written $t^2 = p_1$.
The $p_{k,n-k}$ indeed follow a binomial distribution where the probability of the single event is $p_1$. It corresponds to the probability that one photon is transmitted by the beam-splitter. Reflections or transmissions of the photons in the Fock state may thus be seen as independent events.

Finally, we shall consider the case of a **symmetric input state of two photons**, one in mode $a$ and one in mode $b$:

$$|\psi_{in}\rangle = |1\rangle_a |1\rangle_b = \hat{a}^\dagger \hat{b}^\dagger |0\rangle \tag{6-31}$$

We adopt the usual procedure and we calculate:

$$\begin{aligned}|\psi_{out}\rangle &= \hat{U}\hat{a}^\dagger\hat{U}^\dagger\,\hat{U}\hat{b}^\dagger\hat{U}^\dagger\,|0\rangle = \left(t\hat{a}^\dagger + \imath r^*\hat{b}^\dagger\right)\left(\imath r\hat{a}^\dagger + t\hat{b}^\dagger\right)|0\rangle = \\ &= \imath tr\left((\hat{a}^\dagger)^2 + \imath tr^*(\hat{b}^\dagger)^2\right)|0\rangle + \left(t^2 - |r|^2\right)\hat{a}^\dagger\hat{b}^\dagger\,|0\rangle = \\ &= \imath 2tr\frac{|2\rangle_a|0\rangle_b + e^{-2\imath\varphi}|0\rangle_a|2\rangle_b}{\sqrt{2}} + \left(t^2 - |r|^2\right)|1\rangle_a|1\rangle_b\end{aligned} \tag{6-32}$$

where we have made explicit the phase term $\varphi$ of the beam splitter reflection coefficient $r = |r|e^{\imath\varphi}$. The denominator $\sqrt{2}$ was also added to make the state $\frac{|2\rangle_a|0\rangle_b + e^{-2\imath\varphi}|0\rangle_a|2\rangle_b}{\sqrt{2}}$ properly normalized.
The result in Eq. (6-32) is a superposition between the symmetric state (identical to the input one) $|1\rangle_a|1\rangle_b$ and a state where the two photons are either both on the $a$ mode or both on the $b$ mode. The weights in the combination depend on the beam splitter transmission and reflection coefficients $t$ and $r$.
A very significant situation occurs when the beam splitter is balanced, namely $t = |r|$, which also implies $t = |r| = \frac{\sqrt{2}}{2}$. In that case we read:

$$|\psi_{out}\rangle = \imath\frac{|2\rangle_a|0\rangle_b + e^{-2\imath\varphi}|0\rangle_a|2\rangle_b}{\sqrt{2}} \tag{6-33}$$

Surprisingly, the state $|1\rangle_a|1\rangle_b$ is no longer present at the output. This can be interpreted as a destructive **quantum interference** phenomenon, between the terms that build the resulting expression of Eq. (6-32). Two identical photons entering the balanced beam splitter at different inputs have vanishing probability of being detected, in coincidence, again on separate outputs. This phenomenon of two-photon quantum interference is often named **Hong-Ou-Mandel effect**.

## 6.3 Quantum description of a Mach-Zehnder interferometer

A **Mach-Zehnder interferometer** is modelled by three functional optical components operating one after the other on two optical modes. First, a beam splitter mixes the two modes; then, a suitable optical component, possibly tuneable, imparts a phase shift $\phi$ on one of the modes in the internal paths; finally, a second beam splitter mixes again the two modes. We neglect in this simple model all optical losses.
Note that propagation in the optical paths that form the interferometer also produces phase shifts. In addition, practical realizations of the interferometer require mirrors to fold the paths, so that the output beams from the first beam splitter can be injected in the second beam splitter with the proper alignment. Mirrors induce, too, additional phase shifts. However, phase shifts common to both paths amount to a global phase term that is not relevant experimentally; only differential phase shifts in the internal arms are important, which are modelled by the phase shift component mentioned above.
Before beginning with the quantum analysis, it is worth recalling the classical results: if a monochromatic light beam is injected in one input of the interferometer, interference fringes are observed at the outputs. Namely, the intensity at a given output varies as a sinusoidal function of $\phi$, with a periodicity in $\phi$ of $2\pi$.
In the following, we limit our study to the case of **balanced beam splitters** ($\theta = \pi/2, \varphi = 0$), which is the most significant. These are described by matrices:

$$M_{BS} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & \imath \\ \imath & 1\end{bmatrix} \tag{6-34}$$

The **phase shift** element acts on the mode $b$ with a phase $\phi$. Classically this is described by the relations $\alpha' = \alpha$, $\beta' = e^{\iota\phi}\beta$, involving the phasors. Input and output annihilation operators obey analogous linear relations:

$$\hat{a}' = \hat{a} \qquad\qquad \hat{b}' = e^{\iota\phi}\hat{b} \tag{6-35}$$

which can be expressed by the matrix:

$$M_\phi = \begin{bmatrix} 1 & 0 \\ 0 & e^{\iota\phi} \end{bmatrix} \tag{6-36}$$

While we have already discussed with some detail the quantum-mechanical operation of the beam splitter, it is worth spending a few words here about the phase shift element. Recalling that:

$$\begin{bmatrix} \hat{U}\hat{a}\hat{U}^\dagger \\ \hat{U}\hat{b}\hat{U}^\dagger \end{bmatrix} = M^\dagger \cdot \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} \qquad\qquad \begin{bmatrix} \hat{U}\hat{a}^\dagger\hat{U}^\dagger \\ \hat{U}\hat{b}^\dagger\hat{U}^\dagger \end{bmatrix} = M^T \cdot \begin{bmatrix} \hat{a}^\dagger \\ \hat{b}^\dagger \end{bmatrix}$$

The operation on Fock states of respectively $m$ and $n$ photons on the $a$ and $b$ modes is:

$$|\psi_{in}\rangle = |m\rangle_a |n\rangle_b = \frac{1}{\sqrt{m!\,n!}} (\hat{a}^\dagger)^m (\hat{b}^\dagger)^n |0\rangle \tag{6-37}$$

$$\begin{aligned} |\psi_{out}\rangle &= \frac{1}{\sqrt{m!\,n!}} \hat{U}(\hat{a}^\dagger)^m (\hat{b}^\dagger)^n \hat{U}^\dagger |0\rangle = \frac{1}{\sqrt{m!\,n!}} (\hat{U}\hat{a}^\dagger\hat{U}^\dagger)^m (\hat{U}\hat{b}^\dagger\hat{U}^\dagger)^n |0\rangle = \\ &= \frac{1}{\sqrt{m!\,n!}} (\hat{a}^\dagger)^m (e^{\iota\phi}\hat{b}^\dagger)^n |0\rangle = e^{\iota n\phi} |m\rangle_a |n\rangle_b \end{aligned} \tag{6-38}$$

Namely, this optical component produces a global phase shift on the state $|m\rangle_a |n\rangle_b$ that is proportional to the number of photons in the mode $b$.
Note that these relations on the Fock states give the expected classical limit in the case of coherent states:

$$|\psi_{in}\rangle = |\alpha\rangle_a |\beta\rangle_b = e^{-\frac{1}{2}|\alpha|^2} e^{-\frac{1}{2}|\beta|^2} \sum_{m=0}^{+\infty}\sum_{n=0}^{+\infty} \frac{\alpha^m\beta^n}{\sqrt{m!\,n!}} |m\rangle_a |n\rangle_b \tag{6-39}$$

$$\begin{aligned} |\psi_{out}\rangle &= \ldots = e^{-\frac{1}{2}|\alpha|^2} e^{-\frac{1}{2}|\beta|^2} \sum_{m=0}^{+\infty}\sum_{n=0}^{+\infty} e^{\iota n\phi} \frac{\alpha^m\beta^n}{\sqrt{m!\,n!}} |m\rangle_a |n\rangle_b = \\ &= e^{-\frac{1}{2}|\alpha|^2} e^{-\frac{1}{2}|\beta|^2} \sum_{m=0}^{+\infty}\sum_{n=0}^{+\infty} \frac{\alpha^m(e^{\iota\phi}\beta)^n}{\sqrt{m!\,n!}} |m\rangle_a |n\rangle_b \equiv |\alpha\rangle_a \left|e^{\iota\phi}\beta\right\rangle_b \end{aligned} \tag{6-40}$$

To study the action of the Mach-Zehnder interferometer on a given input state, we can apply in sequence the transformations associated to the optical components. As an instructive example, we can consider the evolution of an input state with $|\psi\rangle$ of single photon on mode $a$.

$$|\psi\rangle = |1\rangle_a |0\rangle_b = \hat{a}^\dagger |0\rangle \tag{6-41}$$

After the first beam splitter we have:

$$|\psi'\rangle = \hat{U}_{BS}\hat{a}^\dagger\hat{U}_{BS}^\dagger |0\rangle = \frac{1}{\sqrt{2}}(\hat{a}^\dagger + \iota\hat{b}^\dagger) |0\rangle \tag{6-42}$$

We then apply the phase shift:

$$|\psi''\rangle = \frac{1}{\sqrt{2}}(\hat{U}_\phi\hat{a}^\dagger\hat{U}_\phi^\dagger + \iota\hat{U}_\phi\hat{b}^\dagger\hat{U}_\phi^\dagger) |0\rangle = \frac{1}{\sqrt{2}}(\hat{a}^\dagger + \iota e^{\iota\phi}\hat{b}^\dagger) |0\rangle \tag{6-43}$$

and finally the last beam splitter:

$$\begin{aligned}
\left|\psi'''\right\rangle &= \frac{1}{\sqrt{2}}(\hat{U}_{BS}\hat{a}^\dagger\hat{U}_{BS}^\dagger + \iota e^{\iota\phi}\hat{U}_{BS}\hat{b}^\dagger\hat{U}_{BS}^\dagger)\left|0\right\rangle = \\
&= \frac{1}{2}(\hat{a}^\dagger + \iota\hat{b}^\dagger + \iota e^{\iota\phi}\cdot\iota\hat{a}^\dagger + \iota e^{\iota\phi}\hat{b}^\dagger)\left|0\right\rangle = \left(\frac{1-e^{\iota\phi}}{2}\hat{a}^\dagger + \iota\frac{1+e^{\iota\phi}}{2}\hat{b}^\dagger\right)\left|0\right\rangle = \\
&= \iota e^{\iota\phi/2}\left(\frac{e^{-\iota\phi/2}-e^{\iota\phi/2}}{2\iota}\left|1\right\rangle_a\left|0\right\rangle_b + \frac{e^{-\iota\phi/2}+e^{\iota\phi/2}}{2}\left|0\right\rangle_a\left|1\right\rangle_b\right) = \\
&= \iota e^{\iota\phi/2}\left(-\sin\frac{\phi}{2}\left|1\right\rangle_a\left|0\right\rangle_b + \cos\frac{\phi}{2}\left|0\right\rangle_a\left|1\right\rangle_b\right)
\end{aligned} \tag{6-44}$$

The two-mode output state is entangled, and one photon can be detected either on mode $a$ or on mode $b$. The probability to detect the single photon in either of the two outputs is given by:

$$p_a = \sin^2\frac{\phi}{2} = \frac{1-\cos\phi}{2} \qquad\qquad p_b = \cos^2\frac{\phi}{2} = \frac{1+\cos\phi}{2} \tag{6-45}$$

Interestingly, the detection probability show **interference fringes** as a function of the internal phase $\phi$, with precisely the same periodicity we would observe in the classical case. Interference indeed is not a collective effect due to the reciprocal interaction of several photons, but is due to the superposition of different possible paths (from input to output) that each single photon can travel.
This **single-photon interference** phenomenon has been reported nowadays in many experimental demonstrations. One of the first and most illustrative examples is the pioneering experiment by Grangier and coworkers.[15] In that experiment, interference fringes were observed in the photodetection counts at the output of a Mach-Zehnder interferometer fed by a single-photon source. Notably, the mean time between single-photon emission by the source was larger than the time required by a photon to travel in the interferometer, guaranteeing the presence of 'one photon at a time' inside the interferometer.
As a matter of fact, to study the action of the interferometer on generic input states, we may adopt a more convenient approach. In fact, each component operates as a **matrix multiplication** of the column vector of the annihilation (or creation) operators at its input. Performing a sequence of transformations on the input operators amounts to an (ordered) product of matrices.[16] If we denote with $\hat{U}_{\text{MZI}}$ the overall evolution operator of the interferometer, we can write:

$$\hat{U}_{\text{MZI}}^\dagger\begin{bmatrix}\hat{a}\\ \hat{b}\end{bmatrix}\hat{U}_{\text{MZI}}^\dagger = M_{BS}\cdot M_\phi\cdot M_{BS}\begin{bmatrix}\hat{a}\\ \hat{b}\end{bmatrix} = M_{\text{MZI}}\begin{bmatrix}\hat{a}\\ \hat{b}\end{bmatrix} \tag{6-47}$$

Thus, we can use directly $M_{\text{MZI}}$ to compute the effects on the operators. In detail, we have:

$$\begin{aligned}
M_{\text{MZI}} = M_{BS}\cdot M_\phi\cdot M_{BS} &= \frac{1}{2}\begin{bmatrix}1 & \iota\\ \iota & 1\end{bmatrix}\cdot\begin{bmatrix}1 & 0\\ 0 & e^{\iota\phi}\end{bmatrix}\cdot\begin{bmatrix}1 & \iota\\ \iota & 1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1-e^{\iota\phi} & \iota(1+e^{\iota\phi})\\ \iota(1+e^{\iota\phi}) & e^{\iota\phi}-1\end{bmatrix} = \\
&= \ldots = \iota e^{\iota\phi/2}\begin{bmatrix}-\sin\frac{\phi}{2} & \cos\frac{\phi}{2}\\ \cos\frac{\phi}{2} & \sin\frac{\phi}{2}\end{bmatrix}
\end{aligned} \tag{6-48}$$

[15] P. Grangier, G. Roger and A. Aspect, "Experimental Evidence for a Photon Anticorrelation Effect on a Beam Splitter: A New Light on Single-Photon Interferences" *Europhysics Letters* **1**, 173 (1986).

[16] As a matter of fact, this approach holds in general also for devices that involve a larger number of optical modes. If a set of optical modes travels through a sequence of linear components or blocks associated to the matrices $M_1, M_2, \ldots, M_N$, it is possible to associate to the overall propagation the matrix:

$$M_{tot} = M_N\cdot\ldots\cdot M_2\cdot M_1 \tag{6-46}$$

Note the (reverse) order in the matrix product, which is needed to describe the sequential application of the matrices to the vector of operators.

From inspection of Eq. (6-48) we acknowledge that we would have retrieved precisely the results of Eq. (6-44) by using directly $M_{\text{MZI}}$, instead of the three transformations in sequence. As global phase shifts do not produce relevant physical effects, we may use also the simplified matrix:

$$\boxed{M_{\text{MZI}} = \begin{bmatrix} -\sin\frac{\phi}{2} & \cos\frac{\phi}{2} \\ \cos\frac{\phi}{2} & \sin\frac{\phi}{2} \end{bmatrix}} \tag{6-49}$$

Actually, the whole Mach-Zehnder interferometer can be considered as a **tuneable beam splitter** with a reflection and transmission coefficients determined by the internal phase $\phi$. In fact, we can decompose $M_{\text{MZI}}$ also according to the following product of matrices:

$$M_{\text{MZI}} = \begin{bmatrix} -\sin\frac{\phi}{2} & \cos\frac{\phi}{2} \\ \cos\frac{\phi}{2} & \sin\frac{\phi}{2} \end{bmatrix} = \begin{bmatrix} -\iota & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \sin\frac{\phi}{2} & \iota\cos\frac{\phi}{2} \\ \iota\cos\frac{\phi}{2} & \sin\frac{\phi}{2} \end{bmatrix} \cdot \begin{bmatrix} -\iota & 0 \\ 0 & 1 \end{bmatrix} \tag{6-50}$$

This is a beam splitter in its standard form (with $\theta = \phi$) placed between two $\frac{3}{2}\pi$ phase elements on mode $a$ (indeed $e^{\frac{3}{2}\pi} = -\iota$). These external additional phase elements may not be relevant (or could be easily compensated for) in many applications, as they are equivalent to longer or shorter optical paths at the input or output of the device.
To discuss a significant example, we study the action of the interferometer on a coherent input $|\alpha\rangle$ on mode $a$. We transform the displacement operator using the $\hat{U}_{\text{MZI}}$ evolution:[17]

$$|\psi_{in}\rangle = |\alpha\rangle_a |0\rangle_b = \hat{D}_a(\alpha)|0\rangle = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}|0\rangle \tag{6-51}$$

$$\begin{aligned} |\psi_{out}\rangle &= e^{\alpha\hat{U}_{\text{MZI}}\hat{a}^\dagger\hat{U}^\dagger_{\text{MZI}} - \alpha^*\hat{U}_{\text{MZI}}\hat{a}\hat{U}^\dagger_{\text{MZI}}}|0\rangle = \\ &= \exp\left(-\alpha\sin\frac{\phi}{2}\hat{a}^\dagger + \alpha\cos\frac{\phi}{2}\hat{b}^\dagger + \alpha^*\sin\frac{\phi}{2}\hat{a} - \alpha^*\cos\frac{\phi}{2}\hat{b}\right)|0\rangle = \\ &= \exp\left(-\alpha\sin\frac{\phi}{2}\hat{a}^\dagger + \alpha^*\sin\frac{\phi}{2}\hat{a}\right)\cdot\exp\left(\alpha\cos\frac{\phi}{2}\hat{b}^\dagger - \alpha^*\cos\frac{\phi}{2}\hat{b}\right)|0\rangle = \\ &= \hat{D}_a\left(-\alpha\sin\frac{\phi}{2}\right)\hat{D}_b\left(\alpha\cos\frac{\phi}{2}\right)|0\rangle = \left|-\alpha\sin\frac{\phi}{2}\right\rangle_a \left|\alpha\cos\frac{\phi}{2}\right\rangle_b \end{aligned} \tag{6-52}$$

This is the separable (not-entangled) state given by the product of two coherent states on the two outputs. Being the average number of photons in the input state:

$$\bar{n}_{in} = |\alpha|^2 \tag{6-53}$$

the average number of photons in the outputs $a$ and $b$ is given by:

$$\bar{n}^a_{out} = \bar{n}_{in}\sin^2\frac{\phi}{2} \qquad\qquad \bar{n}^b_{out} = \bar{n}_{in}\cos^2\frac{\phi}{2} \tag{6-54}$$

These results indicate the appearance of interference fringes with periodicity in $\phi$ of $2\pi$, fully consistent with the expected classical result.

[17] In the case of a single coherent state input, the decomposition formalized in Eq. (6-50) can also provide a convenient route to calculate the output state. We start from the input state $|\psi_{in}\rangle = |\alpha\rangle_a |0\rangle_b$. Then, we apply in sequence the phase shift $\frac{3}{2}\pi$ on mode $a$, the beam splitter with $t = \sin\frac{\phi}{2}$ and $r = \cos\frac{\phi}{2}$, and again the phase shift $\frac{3}{2}\pi$ on mode $a$.

$$|\alpha\rangle_a|0\rangle_b \overset{phase}{\rightarrow} |-\iota\alpha\rangle_a|0\rangle_b \overset{BS}{\rightarrow} \left|-\iota\sin\frac{\phi}{2}\alpha\right\rangle_a \left|\cos\frac{\phi}{2}\alpha\right\rangle_b \overset{phase}{\rightarrow} \left|-\sin\frac{\phi}{2}\alpha\right\rangle_a \left|\cos\frac{\phi}{2}\alpha\right\rangle_b$$

More interestingly, we can study the case of two identical photons injected in the two inputs (one per each input). Again we drop for convenience the common phase term in $M_{\text{MZI}}$ and we proceed:

$$|\psi_{in}\rangle = |1\rangle_a |1\rangle_b = \hat{a}^\dagger \hat{b}^\dagger |0\rangle \tag{6-55}$$

$$\begin{aligned}
|\psi_{out}\rangle &= \hat{U}_{\text{MZI}} \hat{a}^\dagger \hat{U}_{\text{MZI}}^\dagger \hat{U}_{\text{MZI}} \hat{b}^\dagger \hat{U}_{\text{MZI}}^\dagger |0\rangle = \\
&= \left(-\sin\frac{\phi}{2}\hat{a}^\dagger + \cos\frac{\phi}{2}\hat{b}^\dagger\right)\left(\cos\frac{\phi}{2}\hat{a}^\dagger + \sin\frac{\phi}{2}\hat{b}^\dagger\right)|0\rangle = \\
&= -\underbrace{\sin\frac{\phi}{2}\cos\frac{\phi}{2}}_{=\frac{1}{2}\sin\phi}\left((\hat{a}^\dagger)^2 - (\hat{b}^\dagger)^2\right)|0\rangle + \underbrace{\left(\cos^2\frac{\phi}{2} - \sin^2\frac{\phi}{2}\right)}_{=\cos\phi}\hat{a}^\dagger\hat{b}^\dagger|0\rangle = \\
&= \sin\phi \cdot \frac{|0\rangle_a |2\rangle_b - |2\rangle_a |0\rangle_b}{\sqrt{2}} + \cos\phi\, |1\rangle_a |1\rangle_b
\end{aligned} \tag{6-56}$$

As we have understood that the interferometer works as a tuneable beam splitter with $\theta = \phi$, we are not too surprised to find an expression analogous to Eq. (6-32). The probability to detect two photons in coincidence on the two different outputs (i.e. the probability to detect the state $|1\rangle_a |1\rangle_b$) is given by:

$$p_{11} = \cos^2\phi = \frac{1 + \cos 2\phi}{2} \tag{6-57}$$

Note that the interference fringes on these detected events have a period of $\pi$ in $\phi$, namely half of the period we observe in the classical case. This is a purely quantum effect that cannot be explained in a classical framework: **two-photon fringes have double frequency**!
This effect arises from the quantum interference of two photons in different inputs and, not less importantly, from the two-photon coincidence detection. Indeed, the average number of photons is constantly equal to one in each output, irrespective of the phase $\phi$:

$$\langle n_a\rangle = \langle\psi_{out}|\, \hat{a}^\dagger\hat{a}\, |\psi_{out}\rangle = \langle n_b\rangle = \langle\psi_{out}|\, \hat{b}^\dagger\hat{b}\, |\psi_{out}\rangle = \sin^2\phi + \cos^2\phi = 1 \tag{6-58}$$

Thus, intensity measurements at the interferometer outputs in this case would not reveal any fringe.[18]
Finally, we should observe that the quantum interference effects are influenced *only* by the internal phase of the Mach-Zehnder interferometer (i.e. the equivalent beam splitter reflectivity or transmissivity), while phase terms applied to the input or output modes are not influent. Let's imagine for instance to place an additional phase shifter on mode $a$ before the interferometer. This means that the interferometer is fed with the state $|\psi\rangle = e^{\iota\varphi}\hat{a}^\dagger\hat{b}^\dagger|0\rangle$, which differs from the one in Eq. (6-55) only for a global phase term. This global phase term can be factored out along all the passages of Eq. (6-56) and thus do not influence significantly the result. Also phase shifts applied at the outputs of the interferometer are not relevant, as they just introduce phase terms to the coefficient that multiply $|0\rangle_a |2\rangle_b$, $|2\rangle_a |0\rangle_b$ or $|1\rangle_a |1\rangle_b$ without affecting their detection probability. These considerations hold naturally also for the Hong-Ou-Mandel effect observed on a simple beam-splitter.

[18] It is worth noting that also in the case of the Hong-Ou-Mandel effect previously discussed on the simple beam splitter, if we had just detected the average photon number we wouldn't have been able to discern any non-classical feature.

### 6.4 Phase sensitivity in interferometric measurements

Interferometers are widely employed as optical measurement instruments, in applications where the phase $\phi$ represents an unknown quantity that the experimenter would like to access and estimate. A quantum description of the typical **classical measurement scenario** may develop along the following lines:

- A laser beam with constant power is injected in input port $a$. This is described by a coherent state $|\alpha\rangle$ with $\bar{n}_0 = |\alpha|^2$ photons on average in each time bin. An optical power meter is placed in front of one output. In quantum mechanical terms this measurement is associated to the number operator $\hat{n}_a$ or $\hat{n}_b$ and is equivalent to an estimate of the average value $\langle n_a \rangle$ or $\langle n_b \rangle$.
- The relations between those averages and the phase $\phi$ are known (see Eqs. (6-54)), so that one can retrieve consequently an estimate of $\phi$. For instance, being $\bar{n}_a(\phi) = \bar{n}_0 \sin^2 \frac{\phi}{2} = \bar{n}_0 \frac{1-\cos\phi}{2}$ one can calculate:
$$\phi = \arccos\left(1 - 2\frac{\bar{n}_a}{\bar{n}_0}\right) \tag{6-59}$$
Note that some intrinsic ambiguity is present in this estimate, as the same $\bar{n}_a$ would results from both $+\phi$ and $-\phi$. Some prior information on the sign of $\phi$ is needed to resolve the issue.

Our measurement is affected, even in the ideal case, by the Poisson noise due to the photon statistics $\Delta n_a^2 \equiv \bar{n}_a$. The variance in the photon number reflects into an uncertainty in the phase estimate according to:
$$\Delta\phi^2 = \frac{\Delta n_a^2}{\left(\dfrac{d\bar{n}_a(\phi)}{d\phi}\right)^2} \tag{6-60}$$
The value of $\Delta\phi$ indicates the limit in precision that we can achieve with our measurement. In detail we get:
$$\frac{d\bar{n}_a(\phi)}{d\phi} = \bar{n}_0 \sin\frac{\phi}{2}\cos\frac{\phi}{2} \tag{6-61}$$
and as a consequence:
$$\Delta\phi^2 = \frac{\bar{n}_0 \sin^2\frac{\phi}{2}}{(\bar{n}_0)^2 \sin^2\frac{\phi}{2}\cos^2\frac{\phi}{2}} = \frac{1}{\bar{n}_0\,\cos^2\frac{\phi}{2}} \tag{6-62}$$
There are unfavorable points where the uncertainty becomes very large (up to infinite), which correspond to the peaks of the fringes ($\phi/2 = \pi/2 + m\pi$). Minimum variance $\Delta\phi^2 = \frac{1}{\bar{n}_0}$ is instead experienced for $\phi/2 = m\pi$ at the dips of the fringes. It may sound counterintuitive that one reaches optimal precision around the dip of a dark fringe, when no light is detected.[19] Actually, according the rigorous quantification of Eq. (6-60), one needs to compare the slope of the interference fringe $\frac{d\bar{n}_a(\phi)}{d\phi}$ to the variance in the photon number $\Delta n_a^2$: in $\phi/2 = m\pi$ they both tend to zero, but their ratio remains finite.
We now consider using for the measurement **non-classical states of light**. In such a case, it is better to restate the measurement procedure as follows:[20]

- A known quantum state $|\psi_{in}\rangle$ is repeatedly injected in the interferometer (say, $k$ times). At the output a detector D performs a measurement on the output state $|\psi_{out}\rangle$, which in the simplest case is equivalent to a projection measurement on a target state $|\psi_D\rangle$. Namely, this measurement gives as result 1 (the state $|\psi_D\rangle$ is detected), with probability:
$$p_D = |\langle\psi_D|\psi_{out}\rangle|^2 \tag{6-63}$$

[19] This, however, is also a consequence of the precise type of measurement we are conducting, that only considers output $a$. If we considered both outputs, taking as a signal $n_a - n_b$, a similar analysis would give as optimal points those with maximum fringe slope, which may be more intuitive.

[20] R. Okamoto, *et al.* "Beating the standard quantum limit: phase super-sensitivity of $N$-photon interferometers." *New Journal of Physics* **10**, 073033 (2008).

and returns 0 otherwise.

As an example we may consider the injection of a single-photon state $|\psi_{in}\rangle = |1\rangle_a |0\rangle_b$ in the Mach-Zehnder interferometer. At the output, the experimenter chooses to detect single photons on mode $a$, thus measuring the projection of the output state on $|\psi_D\rangle \equiv |1\rangle_a |0\rangle_b$.

- Having characterized the apparatus, the experimenter knows the relation between $p_D$ and $\phi$, i.e. the function $p_D(\phi)$. In our example of single-photon measurements this probability would be:

$$p_D = \sin^2 \frac{\phi}{2} = \frac{1 - \cos\phi}{2} \tag{6-64}$$

- The experimenter counts the $c_k$ positive results of the measurements occurred in $k$ trials, and from these they estimate $\phi$. In fact, in the limit of large $k$ it holds:

$$c_K = k\, p_D(\phi) \tag{6-65}$$

Thus, the probability $p_D(\phi)$ is estimated as $c_K/k$ and, by inverting the function $p_D(\phi)$, one gets $\phi$. Akin to the classical case, because of the non-monotonicity of Eq. (6-64) the inversion may present ambiguities that require complementary measurements or additional information to be resolved.

The measurement uncertainty requires to be treated with some attention. In the projection measurements, the collection of $c_k$ positive results in $k$ trials follows a binomial distribution. Its variance is given by:

$$\Delta c_k^2 = k\, p_D(1 - p_D) \tag{6-66}$$

The resulting uncertainty on $\phi$ is:

$$\Delta\phi^2 = \frac{\Delta c_k^2}{\left(\frac{dc_k}{d\phi}\right)^2} = \frac{k\, p_D\,(1 - p_D)}{k^2 \left(\frac{dp_D}{d\phi}\right)^2} = \frac{p_D\,(1 - p_D)}{k} \left(\frac{dp_D}{d\phi}\right)^{-2} \tag{6-67}$$

In our example of single-photon interferometry, we have $dp_D/d\phi = \sin\frac{\phi}{2}\cos\frac{\phi}{2}$

$$\Delta\phi^2 = \frac{\sin^2\frac{\phi}{2}\cos^2\frac{\phi}{2}}{k} \cdot \frac{1}{\sin^2\frac{\phi}{2}\cos^2\frac{\phi}{2}} = \frac{1}{k} \tag{6-68}$$

This variance is equal to best one we could obtain if we used a (classical) coherent state with $\bar{n} = k$, i.e. an equivalent average number of photons.[21] This inverse scaling with the amount of resources employed in the measurement is called **standard quantum limit**. The standard quantum limit is indeed the limit of precision of classical measurements.
A more interesting case is given by **two-photon interferometry**, i.e. a measurement procedure that consists in injecting the state $|1\rangle_a |1\rangle_b$ at the input and in detecting the same state $|1\rangle_a |1\rangle_b$ at the output. From our previous discussion we can calculate:

$$p_D = \cos^2\phi \qquad\qquad 1 - p_D = \sin^2\phi \tag{6-69}$$

$$\frac{dp_D}{d\phi} = 2\cos\phi\sin\phi \tag{6-70}$$

and then:

$$\Delta\phi^2 = \frac{\sin^2\phi\cos^2\phi}{k} \cdot \frac{1}{4\sin^2\phi\cos^2\phi} = \frac{1}{4k} \tag{6-71}$$

[21] Though, it is noteworthy that in the single-photon interferometer the variance does not depend on the point on the fringe (i.e., it does not depend on $\phi$)!

This variance should not be compared directly with $1/k$, as in the single photon interferometer with $k$ trials we had used $k$ photons, while here we have used $2k$ ones. A correct comparison should be made by keeping the same amount of employed resources.
To this purpose, we define the **sensitivity** parameter:

$$\mathcal{S}^2 = \frac{\Delta\phi^2_{\text{SQL}}}{\Delta\phi^2} \tag{6-72}$$

where $\Delta\phi^2$ is the phase variance of the measurement under study and $\Delta\phi^2_{\text{SQL}}$ is the phase variance in the standard quantum limit, having used the same amount of resources. A value $\mathcal{S}^2 > 1$ indicates an enhanced sensitivity with respect to the classical case. For the two-photon interferometer, we derive:

$$\mathcal{S}^2_{11} = \frac{\frac{1}{2k}}{\frac{1}{4k}} = 2 \tag{6-73}$$

which indeed testifies a quantum advantage in sensitivity with respect to the single photon case. It can be shown that, in general, for measurement with $N$ quantum particles the upper limit is $\mathcal{S}^2 = N$, which is known as the **Heisenberg limit**. Unfortunately, achieving this limit requires to prepare multi-particle states that are hard to obtain experimentally; in addition, losses (that we have here neglected) are shown to strongly compromise the sensitivity. We have indeed seen, for instance, that losses bring to an increase in the variance of the photocount statistics. These facts have prevented, up to now, to achieve an enhanced sensitivity in practical experiments using the two-photon states discussed above. However, an enhanced sensitivity has been demonstrated using squeezed light states, which we will discuss in the following.

# Notes on Quantum Optics - Section 7

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# 7 Squeezed states of light

## 7.1 Generalizing coherent states

Quasi-classical or coherent states are defined as the eigenstates of the annihilation operator $\hat{a}$. The concept of coherent state can be generalized by looking for the eigenstates of a suitable new operator, consisting in a linear combination of $\hat{a}$ and $\hat{a}^\dagger$:

$$\boxed{\hat{A}_R = \hat{a}\cosh R + \hat{a}^\dagger \sinh R} \tag{7-1}$$

In our discussion we shall consider the parameter $R$ as a real number. These generalized states, that we shall indicate as $|\alpha, R\rangle$, are called **squeezed coherent states** or simply **squeezed states**[22] for reasons that will be more clear in Section 7.2.

**Squeezed coherent states**

Squeezed coherent states $|\alpha, R\rangle$ of a single mode of the electromagnetic field are defined as the eigenstates of the operator $\hat{A}_R$:

$$\hat{A}_R\,|\alpha, R\rangle = \alpha\,|\alpha, R\rangle \tag{7-2}$$

where $\hat{A}_R = \hat{a}\cosh R + \hat{a}^\dagger \sinh R$ and $R$ is a real number.

We shall study at first these states in an abstract way, working out a few of their most interesting properties; then, in Sections 7.4 and 7.5 we will mention how it is actually possible generate them in the experiments. Note that in the case $R = 0$ we have $\hat{A}_R \equiv \hat{a}$, and from Eq. (7-2) we find the quasi-classical states $|\alpha, 0\rangle \equiv |\alpha\rangle$. This confirms that that this new family of states includes as a subset the known coherent ones.
The definition (7-1) brings along the definition of the conjugated operator:

$$\hat{A}_R^\dagger = \hat{a}\sinh R + \hat{a}^\dagger \cosh R \tag{7-3}$$

By taking the Hermitian conjugate of (7-2) we can write the following relations concerning the bras $\langle\alpha, R|$:

$$\langle\alpha, R|\,\hat{A}_R^\dagger = \langle\alpha, R|\,\alpha^* \tag{7-4}$$

The commutation relation of $\hat{A}_R^\dagger$ with $\hat{A}$ is analogous to that of $\hat{a}$ and $\hat{a}^\dagger$:

$$\boxed{[\hat{A}_R, \hat{A}_R^\dagger] = 1} \tag{7-5}$$

This is proven easily using the definitions Eqs. (7-1) and (7-3), and by recalling the identity valid for hyperbolic functions $\cosh^2 x - \sinh^2 x = 1$:

$$\begin{aligned}[\hat{A}_R, \hat{A}_R^\dagger] &= [\hat{a}\cosh R + \hat{a}^\dagger \sinh R, \hat{a}\sinh R + \hat{a}^\dagger \cosh R] = \\ &= \underbrace{[\hat{a}, \hat{a}]}_{=0}\cosh R \sinh R + \underbrace{[\hat{a}, \hat{a}^\dagger]}_{=1}\cosh^2 R + \underbrace{[\hat{a}^\dagger, \hat{a}]}_{=-1}\sinh^2 R + \underbrace{[\hat{a}^\dagger, \hat{a}^\dagger]}_{=0}\cosh R \sinh R = \\ &= \cosh^2 R - \sinh^2 R = 1 \end{aligned} \tag{7-6}$$

Directly from the definitions and relations discussed above, one retrieves immediately the following relations:

$$\langle\hat{A}_R\rangle = \langle\alpha, R|\,\hat{A}_R\,|\alpha, R\rangle = \alpha \quad \langle\hat{A}_R^\dagger\rangle = \langle\alpha, R|\,\hat{A}_R^\dagger\,|\alpha, R\rangle = \alpha^* \tag{7-7}$$

$$\langle\hat{A}_R^\dagger\hat{A}_R\rangle = \langle\alpha, R|\,\hat{A}_R^\dagger\hat{A}_R\,|\alpha, R\rangle = \alpha^*\alpha \tag{7-8}$$

[22] The squeezed states were first envisioned in a few seminal theoretical papers at the beginning of the 1970's, for example as a class of minimum-uncertainty states (D. Stoler, "Equivalence Classes of Minimum Uncertainty Packets," *Physical Review D* **1**, 3217–3219 (1970)) or as generalized coherent states (E.Y.C. Lu, "New coherent states of the Electromagnetic Field," *Lettere al Nuovo Cimento* **2**, 1241–1244 (1971)).

and more in general:

$$\langle(\hat{A}_R^\dagger)^m(\hat{A}_R)^n\rangle = \langle\alpha, R|\,(\hat{A}_R^\dagger)^m(\hat{A}_R)^n\,|\alpha, R\rangle = (\alpha^*)^m\alpha^n \tag{7-9}$$

Note, however, that the ordering of operators is always important. For instance,

$$\langle\hat{A}_R\hat{A}_R^\dagger\rangle = \langle\hat{A}_R^\dagger\hat{A}_R + 1\rangle = \alpha^*\alpha + 1 \tag{7-10}$$

The former equalities are not surprising because, due to the defining relation (7-2), the operators $\hat{A}_R$ and $\hat{A}_R^\dagger$ play for the squeezed states $|\alpha, R\rangle$ the same role that the operators $\hat{a}$ and $\hat{a}^\dagger$ play for the quasi-classical states.
Actually, we may also define generalized quadratures:

$$\hat{A}_{q,R} = \frac{\hat{A}_R + \hat{A}_R^\dagger}{2} = \frac{\hat{a} + \hat{a}^\dagger}{2}(\cosh R + \sinh R) = \hat{a}_q e^{+R} \tag{7-11}$$

$$\hat{A}_{p,R} = \frac{\hat{A}_R - \hat{A}_R^\dagger}{2\iota} = \frac{\hat{a} + \hat{a}^\dagger}{2\iota}(\cosh R - \sinh R) = \hat{a}_q e^{-R} \tag{7-12}$$

These look to be 'rescaled' versions of the usual quadratures, by a factor $e^{\pm R}$. Their expectation values are:

$$\langle\hat{A}_{q,R}\rangle = \frac{\langle\hat{A}_R\rangle + \langle\hat{A}_R^\dagger\rangle}{2} = \frac{\alpha + \alpha^*}{2} = \Re\{\alpha\} \tag{7-13}$$

$$\langle\hat{A}_{p,R}\rangle = \frac{\langle\hat{A}_R\rangle - \langle\hat{A}_R^\dagger\rangle}{2\iota} = \frac{\alpha - \alpha^*}{2\iota} = \Im\{\alpha\} \tag{7-14}$$

And the uncertainties are calculated as follows, in perfect analogy to what holds for regular quadratures and quasi-classical states:

$$\begin{aligned}\langle\hat{A}_{q,R}^2\rangle &= \frac{\langle(\hat{A}_R^2)^2\rangle + \langle\hat{(A_R^\dagger)}^2\rangle + \langle\hat{A}_R\hat{A}_R^\dagger\rangle + \langle\hat{A}_R^\dagger\hat{A}_R\rangle}{4} = \frac{\alpha^2 + (\alpha^*)^2 + 2\alpha^*\alpha + 1}{4} = \\ &= \left(\frac{\alpha + \alpha^*}{2}\right)^2 + \frac{1}{4} = (\Re\{\alpha\})^2 + \frac{1}{4}\end{aligned} \tag{7-15}$$

$$\langle\Delta A_{q,R}^2\rangle = \langle\hat{A}_{q,R}^2\rangle - \langle\hat{A}_{q,R}\rangle^2 = \frac{1}{4} \tag{7-16}$$

$$\begin{aligned}\langle\hat{A}_{p,R}^2\rangle &= -\frac{\langle(\hat{A}_R^2)^2\rangle + \langle\hat{(A_R^\dagger)}^2\rangle - \langle\hat{A}_R\hat{A}_R^\dagger\rangle - \langle\hat{A}_R^\dagger\hat{A}_R\rangle}{4} = -\frac{\alpha^2 + (\alpha^*)^2 - 2\alpha^*\alpha - 1}{4} = \\ &= \left(\frac{\alpha - \alpha^*}{2\iota}\right)^2 + \frac{1}{4} = (\Im\{\alpha\})^2 + \frac{1}{4}\end{aligned} \tag{7-17}$$

$$\langle\Delta A_{p,R}^2\rangle = \langle\hat{A}_{p,R}^2\rangle - \langle\hat{A}_{p,R}\rangle^2 = \frac{1}{4} \tag{7-18}$$

## 7.2 Properties of squeezed states

Relations (7-1) and (7-3) can be inverted and the operators $\hat{a}$ and $\hat{a}^\dagger$ can be written as:[23]

$$\hat{a} = \hat{A}_R \cosh R - \hat{A}_R^\dagger \sinh R \tag{7-19}$$

$$\hat{a}^\dagger = -\hat{A}_R \sinh R + \hat{A}_R^\dagger \cosh R \tag{7-20}$$

From these relations one has immediately:

$$\langle \hat{a} \rangle = \langle \alpha, R| \left(\hat{A}_R \cosh R - \hat{A}_R^\dagger \sinh R\right) |\alpha, R\rangle = \alpha \cosh R - \alpha^* \sinh R = \alpha' \tag{7-21}$$

$$\langle \hat{a}^\dagger \rangle = \langle \alpha, R| \left(-\hat{A}_R \sinh R + \hat{A}_R^\dagger \cosh R\right) |\alpha, R\rangle = -\alpha \sinh R + \alpha^* \cosh R = \alpha'^* \tag{7-22}$$

where we have defined:

$$\boxed{\alpha' = \alpha \cosh R - \alpha^* \sinh R} \tag{7-23}$$

The expectation value of the electric field thus becomes:

$$\boxed{\langle \hat{E} \rangle = \iota \mathcal{E} \left[\langle \hat{a} \rangle e^{\iota \vec{k}\cdot\vec{r}} - \langle \hat{a}^\dagger \rangle e^{-\iota \vec{k}\cdot\vec{r}}\right] = \iota \mathcal{E} \left[\alpha' e^{\iota \vec{k}\cdot\vec{r}} - \alpha'^* e^{-\iota \vec{k}\cdot\vec{r}}\right]} \tag{7-24}$$

By comparing the latter expression with the one valid for a coherent state, we see that **a squeezed state yields the same expectation value of the electric field as a coherent state** with the eigenvalue $\alpha'$. This is confirmed also by looking at the quadratures, which are indeed adimensional snapshots of the electric field:

$$\boxed{\begin{aligned} \langle \hat{a}_q \rangle &= \frac{\langle \hat{a} \rangle + \langle \hat{a}^\dagger \rangle}{2} = \frac{\alpha' + \alpha'^*}{2} = \Re\{\alpha'\} \\ \langle \hat{a}_p \rangle &= \frac{\langle \hat{a} \rangle - \langle \hat{a}^\dagger \rangle}{2\iota} = \frac{\alpha' - \alpha'^*}{2\iota} = \Im\{\alpha'\} \end{aligned}} \tag{7-25}$$

Again, Eqs. (7-25) show the same form as in the case of a quasi-classical state, where $\alpha'$ has been used instead of $\alpha$.
However, this similarity cannot be extended too much. For instance, if we calculate the expectation value of the number operator, we obtain:

$$\begin{aligned} \langle \hat{n} \rangle &= \langle \hat{a}^\dagger \hat{a} \rangle = \langle (-\hat{A}_R \sinh R + \hat{A}_R^\dagger \cosh R)(\hat{A}_R \cosh R - \hat{A}_R^\dagger \sinh R) \rangle = \\ &= -\langle (\hat{A}_R)^2 \rangle \sinh R \cosh R + \langle \hat{A}_R \hat{A}_R^\dagger \rangle \sinh^2 R + \langle \hat{A}_R^\dagger \hat{A}_R \rangle \cosh^2 R - \langle (\hat{A}_R^\dagger)^2 \rangle \sinh R \cosh R = \\ &= -\alpha^2 \sinh R \cosh R + \alpha^* \alpha \sinh^2 R + \sinh^2 R + \alpha^* \alpha \cosh^2 R - (\alpha^*)^2 \sinh R \cosh R = \\ &= \underbrace{(-\alpha \cosh R + \alpha^* \sinh R)}_{\alpha'^*} \underbrace{(\alpha \sinh R - \alpha^* \cosh R)}_{\alpha'} + \sinh^2 R = \\ &= |\alpha'|^2 + \sinh^2 R \end{aligned} \tag{7-26}$$

The squeezed state yield a different average number of photons with respect to the somehow equivalent coherent state with eigenvalue $\alpha'$, as there is an additive term $\sinh^2 R$.

[23] In fact, multiplying Eq. (7-1) by $\cosh R$ and multiplying Eq. (7-3) by $\sinh R$ one writes:

$$\hat{A}_R \cosh R = \hat{a} \cosh^2 R + \hat{a}^\dagger \sinh R \cosh R$$

$$\hat{A}_R^\dagger \sinh R = \hat{a} \sinh^2 R + \hat{a}^\dagger \sinh R \cosh R$$

Subtracting the second equation from the first one:

$$\hat{A}_R \cosh R - \hat{A}_R^\dagger \sinh = \hat{a} \cosh^2 R - \hat{a} \sinh^2 R = \hat{a}(\cosh^2 R - \sinh^2 R) = \hat{a}$$

The relation for $\hat{a}^\dagger$ is retrieved by conjugation.

Squeezed states indeed present important features that make them extremely *non-classical*. One of the most interesting aspects involves the uncertainty in the quadrature amplitudes. In detail, we can exploit Eqs. (7-12) and (7-11), which are simple 'rescaling' relations between standard and generalized quadratures, and write for $\langle \Delta a_q^2 \rangle$:

$$\langle \Delta a_q^2 \rangle = \langle \hat{a}_q^2 \rangle - (\langle \hat{a}_q \rangle)^2 = e^{-2R} \langle \hat{A}_{q,R}^2 \rangle - \left( e^{-R} \langle \hat{A}_{q,R} \rangle \right)^2 = e^{-2R} \left[ \langle \hat{A}_{q,R}^2 \rangle - (\langle \hat{A}_{q,R} \rangle)^2 \right] = e^{-2R} \langle \Delta A_{q,R}^2 \rangle \tag{7-27}$$

$$\boxed{\langle \Delta a_q^2 \rangle = \frac{1}{4} e^{-2R}} \tag{7-28}$$

and for $\langle \Delta a_p^2 \rangle$, proceeding in the same way:

$$\langle \Delta a_p^2 \rangle = \ldots = e^{+2R} \langle \Delta A_{p,R}^2 \rangle \tag{7-29}$$

$$\boxed{\langle \Delta a_p^2 \rangle = \frac{1}{4} e^{+2R}} \tag{7-30}$$

These results are striking. On the one side squeezed states are still **minimum-uncertainty states**, because it holds:

$$\Delta a_q \, \Delta a_p = \frac{1}{2} e^{-R} \, \frac{1}{2} e^{+R} = \frac{1}{4} \tag{7-31}$$

On the other side, and very differently from the case of quasi-classical states, different quadratures have different variances. The amount of this difference depends on the parameter $R$. This feature gives indeed the name to these states: in a squeezed state the quadrature variance is *squeezed*, with a sort of 'compression' in one quadrature and an 'expansion' in the other.
This has important consequences on the quantum fluctuations of the electric field. We recall that the electric field operator can be written as:

$$\hat{E}(\vec{r}, t) = -2\mathcal{E} \left[ \hat{a}_q \sin\left( \vec{k} \cdot \vec{r} - \omega t \right) + \hat{a}_p \cos\left( \vec{k} \cdot \vec{r} - \omega t \right) \right]$$

This means that in $\vec{k} \cdot \vec{r} - \omega t = 0 + m\pi$ one writes the electric field as $\hat{E}_0 \equiv \pm 2\mathcal{E} \hat{a}_q$ and the variance is:

$$\langle \Delta E_0^2 \rangle = 4\mathcal{E}^2 \langle \Delta a_q^2 \rangle = \mathcal{E}^2 e^{-2R} \tag{7-32}$$

Instead, in $\vec{k} \cdot \vec{r} - \omega t = \frac{\pi}{2} + m\pi$ one writes the electric field as $\hat{E}_{\frac{\pi}{2}} \equiv \pm 2\mathcal{E} \hat{a}_p$ and the variance is:

$$\langle \Delta E_{\frac{\pi}{2}}^2 \rangle = 4\mathcal{E}^2 \langle \Delta a_p^2 \rangle = \mathcal{E}^2 e^{+2R} \tag{7-33}$$

The variance of the electric field changes periodically along the wave propagation. There are points with more variance and points with less variance. These can be linked to a potentially greater or smaller *noise* in interferometric measurements.

## 7.3 The squeezed vacuum state

The **squeezed vacuum** is the state $|0, R\rangle$, namely it is the state satisfying the eigenvalue equation (7-2) with the eigenvalue $\alpha = 0$. Note that in this case, also $\alpha' = \alpha \cosh R - \alpha^* \sinh R = 0$. This means that, according to Eq. (7-24), the expectation value of the electric field is uniformly and constantly zero.
However, the squeezed vacuum state *is not the same thing as the vacuum state we have encountered previously.*[24] Put in other words, it is not the ground state of the field oscillator.

$$|0, R\rangle \neq |0\rangle \tag{7-34}$$

[24] Apart from the trivial case in which also $R = 0$: in this condition indeed the squeezed states coincide with the quasi-classical ones.

Firstly, the expectation value of the number operator is *not* zero.

$$\langle 0, R| \hat{n} |0, R\rangle = |\alpha'|^2 + \sinh^2 R = \sinh^2 R \neq 0 \tag{7-35}$$

We can have a deeper insight on the features of squeezed vacuum state by studying its decomposition in the Fock state basis.

$$|0, R\rangle = \sum_{n=0}^{+\infty} c_n |n\rangle \tag{7-36}$$

We begin with imposing that $|0, R\rangle$ should satisfy the eigenvalue equation $\hat{A}_R |0, R\rangle = 0 |0, R\rangle$:

$$\hat{A}_R |0, R\rangle = 0 \tag{7-37}$$

$$\left(\cosh R\, \hat{a} + \sinh R\, \hat{a}^\dagger\right) \sum_{n=0}^{+\infty} c_n |n\rangle = 0 \tag{7-38}$$

$$\sum_{n=0}^{+\infty}\left(c_n \cosh R\, \hat{a} |n\rangle\right) + \sum_{n=0}^{+\infty}\left(c_n \sinh R\, \hat{a}^\dagger |n\rangle\right) = 0 \tag{7-39}$$

Then we remember that:

$$\hat{a} |0\rangle = 0 \qquad \hat{a} |n\rangle = \sqrt{n} |n-1\rangle \qquad \hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

and we elaborate:

$$\sum_{n=1}^{+\infty}\left(c_n \cosh R \sqrt{n} |n-1\rangle\right) + \sum_{n=0}^{+\infty}\left(c_n \sinh R \sqrt{n+1} |n+1\rangle\right) = 0 \tag{7-40}$$

Note that the index in the first summation now starts from 1. We can change indices to both summations, so that they are written in terms of $|n\rangle$:

$$\sum_{n=0}^{+\infty}\left(c_{n+1} \cosh R \sqrt{n+1} |n\rangle\right) + \sum_{n=1}^{+\infty}\left(c_{n-1} \sinh R \sqrt{n} |n\rangle\right) = 0 \tag{7-41}$$

In this way the first summation starts from 0 and the second starts from 1. We can take out the term with $n = 0$ from the first one and join them together:

$$c_1 \cosh R\, |0\rangle + \sum_{n=1}^{+\infty}\left(c_{n+1} \cosh R \sqrt{n+1} + c_{n-1} \sinh R \sqrt{n}\right) |n\rangle = 0 \tag{7-42}$$

This is a summation of Fock states that should give identically zero.[25] This means that all coefficients multiplying these Fock states are indeed zero:

$$c_1 = 0 \tag{7-43}$$

$$c_{n+1} \cosh R \sqrt{n+1} + c_n \sinh R \sqrt{n} = 0 \qquad \forall n \geq 1 \tag{7-44}$$

The second relation is a recursive relation that allows to retrieve all $c_n$ values as we may rewrite it as:

$$c_{n+1} = -\frac{\sinh R}{\cosh R}\sqrt{\frac{n}{n+1}} c_{n-1} \qquad n \geq 1 \tag{7-45}$$

[25] Please note that, even though in the passages of this proof the $c_n$ multiply different $|n\rangle$ states, together with other multiplicative coefficients (for instance in Eq. (7-42) the $c_1$ coefficient multiplies the $|0\rangle$ state together with the further factor $\cosh R$), the physical meaning of the $c_n$ remains the one given by Eq. (7-36). Each $c_n$ is the multiplicative coefficient of $|n\rangle$ in the Fock-state expansion of the squeezed vacuum state.

In a more clear summary, the coefficients $c_n$ are governed by the following laws:

$$\boxed{\begin{aligned} &c_1 = 0 \\ &c_n = -\tanh R \sqrt{\frac{n-1}{n}}\, c_{n-2} \qquad\qquad n \geq 2 \end{aligned}} \tag{7-46}$$

As $c_1 = 0$, the iterative rule forces all $c_n$ with odd $n$ to be zero. Instead, the coefficients of the $c_n$ with even $n$ are non-vanishing, and present alternated signs. Akin to the procedure adopted for writing the Fock-state expansion of coherent states, the iterative rule could be used to find all coefficients $c_n$ (with $n$ even) as functions of $c_0$; finally, $c_0$ could be found by imposing the correct normalization of the state.
In addition, the iterative rule implies $|c_n|^2 > |c_{n-2}|^2$: the coefficients are decreasing in modulus with larger $n$. We recall that the squared moduli of the coefficients $|c_n|^2$ coincide with the probabilities to measure $n$ photons in the optical mode, namely describe the photon statistics. It can be shown that the statistics of the squeezed state may be super-Poissonian or sub-Poissonian, depending on the value of $R$. A bar graph showing the values of $|c_n|^2$, for an example of squeezed vacuum state with $R = 1$, is reported in Figure 8.

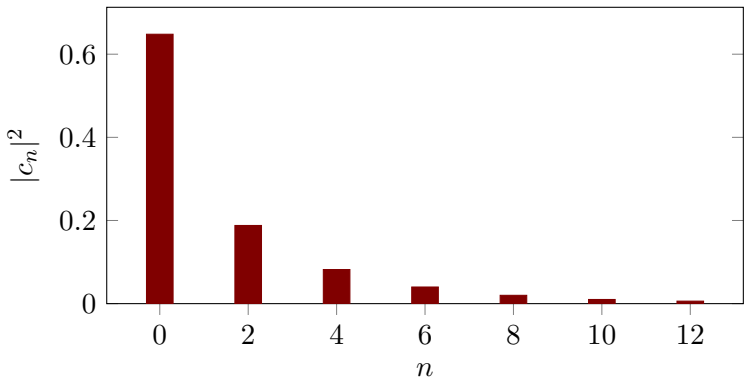


Figure 8: Squared modulus of the coefficients $c_n$ of the Fock state expansion (for $n \leq 12$), in the case of a squeezed vacuum state $|0, R\rangle$ with $R = 1$.

## 7.4 The squeezing operator and the squeezing Hamiltonian

It would be useful to find an operator $\hat{S}$ that *squeezes* the coherent state, namely that transforms the coherent state $|\alpha, 0\rangle \equiv |\alpha\rangle$ into the squeezed coherent state $|\alpha, R\rangle$:

$$\boxed{\hat{S}(R)\,|\alpha, 0\rangle = |\alpha, R\rangle} \tag{7-47}$$

It can be shown that this is accomplished by taking an operator with the following form.

**Squeezing operator**

The squeezing operator $\hat{S}(R)$ is defined as:

$$\hat{S}(R) = e^{\frac{R}{2}\left(\hat{a}^2 - (\hat{a}^\dagger)^2\right)} \tag{7-48}$$

Inspection shows that $\hat{S}(R)$ is unitary and that $\hat{S}^\dagger(R) = \hat{S}^{-1}(R) = \hat{S}(-R)$.

In order to prove that this operator does the required job, we shall first calculate $\hat{S}_R \hat{a} \hat{S}_R^\dagger$. This is done by exploiting the Baker-Hausdorff lemma, which states that for any two operators $\hat{A}$ and $\hat{B}$:

$$e^{\hat{B}}\,\hat{A}\,e^{-\hat{B}} = \hat{A} + \sum_{n=1}^{+\infty} \frac{\overbrace{[\hat{B},[\hat{B},\dots[\hat{B},\hat{A}]]]}^{n \text{ times}}}{n!}$$

Here $\hat{A} \equiv \hat{a}$ and $\hat{B} = \frac{R}{2}\left(\hat{a}^2 - (\hat{a}^\dagger)^2\right)$. It is easily shown[26] that:

$$[\hat{B},\hat{a}] = R\hat{a}^\dagger \qquad\qquad [\hat{B},\hat{a}^\dagger] = R\hat{a} \tag{7-49}$$

This implies $[\hat{B},[\hat{B},\hat{a}]] = R^2\hat{a}$, then $[\hat{B},[\hat{B},[\hat{B},\hat{a}]]] = R^3\hat{a}^\dagger$ and so on. In short:

$$\overbrace{[\hat{B},[\hat{B},\dots[\hat{B},\hat{A}]]]}^{n \text{ times}} = \begin{cases} R^n\hat{a}^\dagger & n \text{ odd} \\ R^n\hat{a} & n \text{ even} \end{cases} \tag{7-50}$$

Therefore we have:

$$\hat{S}_R\hat{a}\hat{S}_R^\dagger = \sum_{n \text{ even}} \frac{R^n}{n!}\hat{a} + \sum_{n \text{ odd}} \frac{R^n}{n!}\hat{a}^\dagger = \cosh R\,\hat{a} + \sinh R\,\hat{a}^\dagger \tag{7-51}$$

In addition, by taking the Hermitian conjugate:

$$\hat{S}_R\hat{a}^\dagger\hat{S}_R^\dagger = \sinh R\,\hat{a} + \cosh R\,\hat{a}^\dagger \tag{7-52}$$

Considering also the inverse relations, we can write in summary:

$$\boxed{\begin{array}{ll} \hat{S}_R\hat{a}\hat{S}_R^\dagger = \hat{A}_R & \hat{S}_R^\dagger\hat{A}_R\hat{S}_R = \hat{a} \\ \hat{S}_R\hat{a}^\dagger\hat{S}_R^\dagger = \hat{A}_R^\dagger & \hat{S}_R^\dagger\hat{A}_R^\dagger\hat{S}_R = \hat{a}^\dagger \end{array}} \tag{7-53}$$

On the basis of these results, let us define $|\psi\rangle = \hat{S}_R\,|\alpha,0\rangle$. We can show that this state satisfies:

$$\hat{A}_R\,|\psi\rangle = \hat{A}_R\hat{S}_R\,|\alpha,0\rangle = \underbrace{\hat{S}_R\hat{S}_R^\dagger}_{=\mathbb{I}}\hat{A}_R\hat{S}_R\,|\alpha,0\rangle = \hat{S}_R\hat{a}\,|\alpha,0\rangle = \alpha\hat{S}_R\,|\alpha,0\rangle = \alpha\,|\psi\rangle \tag{7-54}$$

where we have used the fact that $|\alpha,0\rangle$ is by definition an eigenstate of the $\hat{a}$ operator with eigenvalue $\alpha$. Equation (7-54) then shows that the state $|\psi\rangle = \hat{S}_R\,|\alpha,0\rangle$ is indeed an eigenstate of $\hat{A}_R$, with eigenvalue ‘$\alpha$. The latter is, according to Eq. (7-2), the definition of the squeezed state $|\alpha,R\rangle$. Hence, our operator $\hat{S}_R$ is the one we are looking for, producing a squeezed state out of a coherent one as in Eq. (7-47).

[26] The calculation is rather straightforward but requires a handful of passages. In the case of $[\hat{B},\hat{a}]$:

$$[\hat{B},\hat{a}] = \left[\frac{R}{2}\left(\hat{a}^2 - (\hat{a}^\dagger)^2\right),\hat{a}\right] = \frac{R}{2}\Big(\underbrace{[\hat{a}^2,\hat{a}]}_{=0} - [(\hat{a}^\dagger)^2,\hat{a}]\Big) =$$

$$= -\frac{R}{2}(\hat{a}^\dagger\underbrace{\hat{a}^\dagger\hat{a}}_{=\hat{a}\hat{a}^\dagger-1} - \underbrace{\hat{a}\hat{a}^\dagger}_{=\hat{a}\hat{a}^\dagger+1}\hat{a}^\dagger) = -\frac{R}{2}(\hat{a}^\dagger\hat{a}\hat{a}^\dagger - \hat{a}^\dagger - \hat{a}^\dagger\hat{a}\hat{a}^\dagger - \hat{a}^\dagger) = R\hat{a}^\dagger$$

In the case of $[\hat{B},\hat{a}^\dagger]$, by observing that $\hat{B}^\dagger = -\hat{B}$:

$$[\hat{B},\hat{a}^\dagger] = \left([\hat{a},\hat{B}^\dagger]\right)^\dagger = \left([\hat{a},-\hat{B}]\right)^\dagger = \left([\hat{B},\hat{a}]\right)^\dagger = (R\hat{a}^\dagger)^\dagger = R\hat{a}$$

Considering together the squeezing and the displacement operators, one rewrites Eq. (7-47) as:

$$|\alpha, R\rangle = \hat{S}(R)\hat{D}(\alpha)\,|0\rangle \tag{7-55}$$

Note that $\hat{S}(R)$ and $\hat{D}(\alpha)$ do not commute, so $\hat{D}(\alpha)\hat{S}(R)\,|0\rangle \neq \hat{S}(R)\hat{D}(\alpha)\,|0\rangle$. However, the following relation holds:[27]

$$\hat{S}(R)\hat{D}(\alpha)\,|0\rangle = \hat{D}(\alpha')\hat{S}(R)\,|0\rangle \tag{7-56}$$

with $\alpha' = \alpha\cosh R - \alpha^*\sinh R$.
To understand the physical meaning of the squeezing operator, and to get an intuition on a viable method to generate squeezed states in experiments, it is useful to put $\hat{S}(R)$ in the form:

$$\hat{S}(R) = e^{-\iota\frac{\hat{H}_S}{\hbar}\tau} \tag{7-57}$$

where $\hat{H}_S$ is a Hermitian operator. In this way, the operator $\hat{H}_S$ may correspond to a Hamiltonian associated to some physical process, and the operator $\hat{S}(R)$ would be the time-evolution operator associated to this Hamiltonian, acting precisely for a time $\tau$. This is readily accomplished by taking:

$$\boxed{\hat{H}_S = -\iota\frac{\hbar R}{2\tau}\left((\hat{a}^\dagger)^2 - \hat{a}^2\right)} \tag{7-58}$$

This operator is surely Hermitian and corresponds to the **squeezing Hamiltonian**.
The Hamiltonian $\hat{H}_S$ is describing a process where two photons are simultaneously created or annihilated. This corresponds for instance to a process known in nonlinear optics as (degenerate) parametric mixing. Actually, this is nowadays among the most used methods to generate squeezed light in experiments.
In this process an intense coherent wave at frequency $2\omega$, called *pump*, propagates in a nonlinear crystal. Due to nonlinear polarization of the medium, a photon at frequency $2\omega$ can be converted with some probability into two photons at frequency $\omega$. This probability practically depends on the intensity of the pump field and on the characteristics of the medium, which concur to determine the actual values of $R$ and $\tau$. The model holds in case the pump wave is sufficiently intense to be treated classically, and in case the converted photons are a negligible fraction of the initial ones.
If no photons at frequency $\omega$ are present at the beginning, namely there is no seed beam at this frequency impinging on the nonlinear crystal together with the pump, the Hamiltonian is acting on an initial state equal to the vacuum $|0\rangle$. In this case, *squeezed vacuum* will be generated by the parametric process.

[27] For two operators $\hat{A}$ and $\hat{U}$, with $\hat{U}$ unitary, it holds $\hat{U}e^{\hat{A}}\hat{U}^\dagger = e^{\hat{U}\hat{A}\hat{U}^\dagger}$. This implies that:

$$\hat{S}(R)\hat{D}(\alpha)\hat{S}^\dagger(R) = \hat{S}(R)\hat{e}^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}\hat{S}^\dagger(R) = \exp\left\{\alpha\hat{S}(R)\hat{a}^\dagger S^\dagger(R) - \alpha^*\hat{S}(R)\hat{a}S^\dagger(R)\right\} =$$
$$= \exp\left\{\alpha\hat{a}\sinh R + \alpha\hat{a}^\dagger\cosh R - \alpha^*\hat{a}\cosh R - \alpha^*\hat{a}^\dagger\sinh R\right\} = \exp\left\{\alpha'\hat{a}^\dagger - \alpha'^*\hat{a}\right\}$$

with $\alpha' = \alpha\cosh R - \alpha^*\sinh R$. We can thus prove Eq. (7-56):

$$\hat{S}(R)\hat{D}(\alpha)\,|0\rangle = \hat{S}(R)\hat{D}(\alpha)\hat{S}^\dagger(R)\hat{S}(R)\,|0\rangle = \hat{D}(\alpha')\hat{S}(R)\,|0\rangle$$

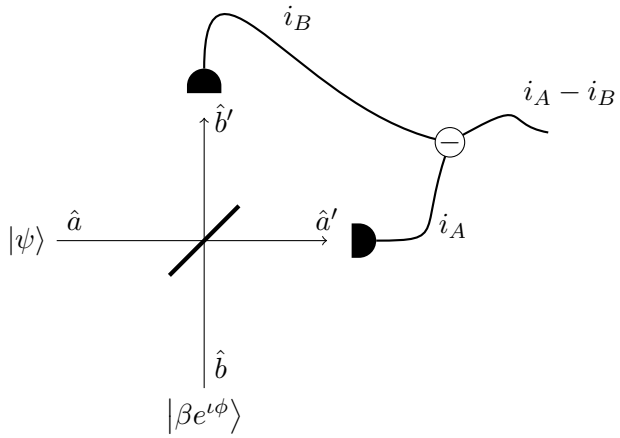


Figure 9: Schematic of an experimental apparatus for homodyne detection. The field state to be studied $|\psi\rangle$ impinges on one of the input ports of a balanced beam splitter. An intense coherent state $|\beta e^{\iota\phi}\rangle$ is injected in the other input. Detectors are placed at the two outputs and the difference of the photo-currents $i_A - i_B$ is measured.

### 7.5 The homodyne detection technique

A powerful experimental technique to investigate the properties of non-classical states of light, including squeezed states, is the **homodyne detection**. The experimental apparatus is sketched in Figure 9. Basically, this technique consists in impinging the state $|\psi\rangle$ that we want to study, together with a strong coherent field at the same wavelength, respectively on the two input ports of a balanced beam splitter. Two photo-diode detectors are placed on the two outputs and the difference of the electric signals is operated.

In our discussion we associate the two input modes of the beam splitter to the annihilation operators $\hat{a}$ and $\hat{b}$; the output modes to the operators $\hat{a}'$ and $\hat{b}'$. The mode $a$ is on the potentially unknown field state $|\psi\rangle$, which we want to investigate and characterize. The mode $b$ is in a coherent state with complex eigenvalue $\tilde{\beta} = \beta e^{\iota\phi}$, where $\beta$ is large. We will see later why and how big $\beta$ needs to be, for the technique to be effective.

The detectors at the output *are not* of the photon-counting type. They do not need to produce 'clicks' upon the detection of the individual photons. Instead they provide continuous current signals (*photo-currents*), proportional to the average photon number present in the modes.

$$i_A \propto \langle \hat{n}_{a'} \rangle = \langle \hat{a}'^\dagger \hat{a}' \rangle \tag{7-59}$$

$$i_B \propto \langle \hat{n}_{b'} \rangle = \langle \hat{b}'^\dagger \hat{b}' \rangle \tag{7-60}$$

In this experimental technique, suitable electronics evaluates the difference in the two photo-currents:

$$\Delta i = i_A - i_B \propto \langle \Delta \hat{n} \rangle = \langle \hat{n}_{a'} \rangle - \langle \hat{n}_{b'} \rangle \tag{7-61}$$

Hence, the relevant physical quantity that we will discuss in the following is the difference in the photon number $\Delta \hat{n}$.

Let us study this physical situation using the knowledge on the balanced beam splitter transformation. The operators $\hat{a}'^\dagger$ and $\hat{a}'$ are the evolved operators from $\hat{a}^\dagger$ and $\hat{a}$, while the operators $\hat{b}'^\dagger$ and $\hat{b}'$ are

the evolved operators from $\hat{b}^\dagger$ and $\hat{b}$, following the unitary transformation of the beam splitter.

$$\hat{a}' = \hat{U}_{\rm BS}^\dagger \hat{a} \hat{U}_{\rm BS} = \frac{\hat{a} + \iota \hat{b}}{\sqrt{2}} \qquad \hat{a}'^\dagger = \hat{U}_{\rm BS}^\dagger \hat{a}^\dagger \hat{U}_{\rm BS} = \frac{\hat{a}^\dagger - \iota \hat{b}^\dagger}{\sqrt{2}} \tag{7-62}$$

$$\hat{b}' = \hat{U}_{\rm BS}^\dagger \hat{b} \hat{U}_{\rm BS} = \frac{\iota \hat{a} + \hat{b}}{\sqrt{2}} \qquad \hat{b}'^\dagger = \hat{U}_{\rm BS}^\dagger \hat{b}^\dagger \hat{U}_{\rm BS} = \frac{-\iota \hat{a}^\dagger + \hat{b}^\dagger}{\sqrt{2}} \tag{7-63}$$

We therefore obtain:

$$\langle \hat{n}_{a'} \rangle = \left\langle \frac{\hat{a}^\dagger - \iota \hat{b}^\dagger}{\sqrt{2}} \; \frac{\hat{a} + \iota \hat{b}}{\sqrt{2}} \right\rangle = \frac{1}{2} \langle \hat{a}^\dagger \hat{a} \rangle + \frac{1}{2} \langle \hat{b}^\dagger \hat{b} \rangle + \frac{\iota}{2} \langle \hat{a}^\dagger \hat{b} - \hat{b}^\dagger \hat{a} \rangle \tag{7-64}$$

$$\langle \hat{n}_{b'} \rangle = \left\langle \frac{-\iota \hat{a}^\dagger + \hat{b}^\dagger}{\sqrt{2}} \; \frac{+\iota \hat{a} + \hat{b}}{\sqrt{2}} \right\rangle = \frac{1}{2} \langle \hat{a}^\dagger \hat{a} \rangle + \frac{1}{2} \langle \hat{b}^\dagger \hat{b} \rangle - \frac{\iota}{2} \langle \hat{a}^\dagger \hat{b} - \hat{b}^\dagger \hat{a} \rangle \tag{7-65}$$

And the expectation value of the difference in the photon number is:

$$\langle \Delta \hat{n} \rangle = \iota \langle \hat{a}^\dagger \hat{b} - \hat{b}^\dagger \hat{a} \rangle \tag{7-66}$$

We recall that in our experimental setting the expectation value has to be calculated on a composite state, consisting in the $|\psi\rangle$ state on mode $a$ and in the $\left|\beta e^{\iota\phi}\right\rangle$ state on mode $b$. In addition, for the coherent state we have:

$$\hat{b} \left|\beta e^{\iota\phi}\right\rangle = \beta e^{\iota\phi} \left|\beta e^{\iota\phi}\right\rangle$$

$$\left\langle \beta e^{\iota\phi} \right| \hat{b}^\dagger = \left\langle \beta e^{\iota\phi} \right| \beta e^{-\iota\phi}$$

Considering further that operators on modes $a$ and $b$ commute, we calculate

$$\begin{aligned} \langle \Delta \hat{n} \rangle &= \left\langle \psi, \beta e^{\iota\phi} \right| \Delta \hat{n} \left| \psi, \beta e^{\iota\phi} \right\rangle = \iota \left\langle \psi, \beta e^{\iota\phi} \right| \hat{a}^\dagger \hat{b} - \hat{b}^\dagger \hat{a} \left| \psi, \beta e^{\iota\phi} \right\rangle = \\ &= \iota \left\langle \psi \right| \beta e^{\iota\phi} \hat{a}^\dagger - \beta e^{-\iota\phi} \hat{a} \left| \psi \right\rangle = 2\beta \left\langle \psi \right| \frac{\hat{a} e^{-\iota\phi} - \hat{a}^\dagger e^{\iota\phi}}{2\iota} \left| \psi \right\rangle \end{aligned} \tag{7-67}$$

Now, we compare this result with the expression of the generic quadrature operator

$$\hat{a}_\theta = \frac{1}{2} \left( \hat{a} e^{-\iota\theta} + \hat{a}^\dagger e^{+\iota\theta} \right)$$

and we realize that inside the bra-ket we have the generic quadrature for an angle $\theta = \phi + \pi/2$, namely:

$$\boxed{\Delta i \propto \langle \Delta \hat{n} \rangle = 2\beta \langle \hat{a}_{\phi + \frac{\pi}{2}} \rangle} \tag{7-68}$$

The difference in the photo-current is thus proportional to the expectation value of the generic quadrature on the unknown state $|\psi\rangle$. The difference in the photo-currents is also proportional to the amplitude $\beta$ of the coherent state. By sweeping the phase $\phi$ of the latter *it is possible to scan all the quadratures and study experimentally their expectation values on the unknown state.*

This is very interesting. However we have seen that, in the case of a squeezed state, looking at the expectation values of the quadratures we would not be able to distinguish it from a coherent state. In fact, it is even more interesting to study the noise in the $\Delta i$ signal, i.e. the variance of the difference of the photo-currents:

$$\mathrm{Var}(\Delta i) \propto \mathrm{Var}(\Delta n) = \langle (\Delta \hat{n})^2 \rangle - (\langle \Delta n \rangle)^2 \tag{7-69}$$

The operator $(\Delta \hat{n})^2$ can be elaborated as follows:

$$\begin{aligned} (\Delta \hat{n})^2 &= -(\hat{a}^\dagger \hat{b} - \hat{b}^\dagger \hat{a})(\hat{a}^\dagger \hat{b} - \hat{b}^\dagger \hat{a}) = -\Big( (\hat{a}^\dagger)^2 \hat{b}^2 + \hat{a}^2 (\hat{b}^\dagger)^2 - \hat{b}^\dagger \hat{b} \hat{a} \hat{a}^\dagger - \underbrace{\hat{b}\hat{b}^\dagger}_{=\hat{b}^\dagger \hat{b} + 1} \hat{a}^\dagger \hat{a} \Big) = \\ &= -\Big( (\hat{a}^\dagger)^2 \hat{b}^2 + \hat{a}^2 (\hat{b}^\dagger)^2 - \hat{b}^\dagger \hat{b} (\hat{a} \hat{a}^\dagger + \hat{a}^\dagger \hat{a}) - \hat{a}^\dagger \hat{a} \Big) \end{aligned} \tag{7-70}$$

We write now the expectation value. We take the mean on the state $\left|\beta e^{\iota\phi}\right\rangle$ and we leave indicated $\langle\ldots\rangle = \langle\psi|\ldots|\psi\rangle$ on the unknown state.

$$\begin{aligned}\langle(\Delta\hat{n})^2\rangle &= -\langle\beta^2(\hat{a}^\dagger)^2e^{2\iota\phi} + \beta^2\hat{a}^2e^{-2\iota\phi} - \beta^2(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a})\rangle + \langle\hat{a}^\dagger\hat{a}\rangle = \\ &= 4\beta^2\left\langle\left(\frac{\hat{a}e^{-\iota\phi} - \hat{a}^\dagger e^{\iota\phi}}{2\iota}\right)^2\right\rangle + \langle\hat{a}^\dagger\hat{a}\rangle = \\ &= 4\beta^2\langle(\hat{a}_{\phi+\frac{\pi}{2}})^2\rangle + \langle\hat{n}_a\rangle \end{aligned} \tag{7-71}$$

Interestingly, this means that in case $\beta$ is sufficiently large (in detail, $\beta \gg \langle\hat{n}_a\rangle$):

$$\langle(\Delta\hat{n})^2\rangle \simeq 4\beta^2\langle(\hat{a}_{\phi+\frac{\pi}{2}})^2\rangle \tag{7-72}$$

To conclude, in the hypothesis that $\beta$ is sufficiently large, namely in case the coherent state is much more intense that the unknown beam we want to study:

$$\mathrm{Var}(\Delta i) \propto \mathrm{Var}(\Delta n) = \langle(\Delta\hat{n})^2\rangle - (\langle\Delta n\rangle)^2 \simeq 4\beta^2\left[\langle(\hat{a}_{\phi+\frac{\pi}{2}})^2\rangle - (\langle\hat{a}_{\phi+\frac{\pi}{2}}\rangle)^2\right] \tag{7-73}$$

$$\boxed{\mathrm{Var}(\Delta i) \propto \mathrm{Var}(\Delta n) \simeq 4\beta^2\mathrm{Var}(a_{\phi+\frac{\pi}{2}})} \tag{7-74}$$

By analyzing the electronic noise in the difference in the photo-currents $\Delta i$, in correspondence of the different phases $\phi$ of the coherent beam, it is possible to study the variance of all quadratures of the unknown field state. The homodyne detection technique is indeed extremely useful for the study of squeezed states, because it directly allows to measure all quadratures and their variance.

# Notes on Quantum Optics - Section 8

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# 8 The Jaynes-Cummings model

## 8.1 Charge-field interactions

The interaction of quantized radiation modes with atoms is of utmost interest in Physics. In the following we will address this problem in a simplified perspective. We will consider a charged particle with mass $m$ and charge $q$ moving in region of space where radiation modes can be present (actually, for simplicity we shall consider only one radiation mode when it comes to quantization), together with other charges that are fixed in constant positions. This is indeed akin to the condition of an electron bound to an atom: it is both exposed to radiation that comes from outside and subject to the Coulombian potential due to the nucleus and to the averaged effect of the other electrons.
In this framework, the electromagnetic field is the resultant of different contributions. In the Coulomb gauge ($\nabla \cdot \vec{A} = 0$), oscillatory contributions of the radiation modes are described by means of a wave-like vector potential $\vec{A}(\vec{r}, t)$, while the static charges are responsible for the scalar potential $V_{coul}(\vec{r})$ constant in time (see Box 8-1). In detail, the electric and magnetic field would be:c

$$\vec{E}(\vec{r}, t) = -\nabla V_{coul}(\vec{r}) - \frac{\partial \vec{A}(\vec{r}, t)}{\partial t} \qquad \vec{B}(\vec{r}, t) = \nabla \times \vec{A}(\vec{r}, t) \tag{8-1}$$

Quantum mechanics, however, is built upon the Hamiltonian approach. The classical Hamiltonian for a charged particle interacting with an electromagnetic field is written as:

$$H = \int_V \left( \frac{\varepsilon_0}{2} |\vec{E}|^2 + \frac{1}{2\mu_0} |\vec{B}|^2 \right) dV + \frac{1}{2m} \left| \vec{p} - q\vec{A} \right|^2 + qV_{coul} \tag{8-2}$$

This brings us to postulate the corresponding quantum Hamiltonian for one charged particle that interacts, on the one hand, with one quantized electromagnetic mode and, on the other hand, with the static field given by other charges:

$$\boxed{\hat{H} = \hbar\omega \left( \hat{a}^\dagger \hat{a} + \frac{1}{2} \right) + \frac{1}{2m} \left( \hat{\vec{p}} - q\hat{\vec{A}} \right)^2 + q\hat{V}_{coul}} \tag{8-3}$$

In Eq. (8-3) $\hat{\vec{A}}$ and $\hat{V}_{coul}$ are the vector and scalar potential respectively, evaluated at the particle's coordinate. $\hat{\vec{A}}$ is associated to the quantized electromagnetic mode, while $V_{coul}$ is due the mentioned static charges. The operator $\hat{\vec{p}}$ is the momentum of the particle.
In case the charged particle is bound to an atom, the particle position $\vec{r}$ cannot vary much around a central position $\vec{r}_0$. If the wavelength of quantized field is, in addition, much larger than the uncertainty in the particle's position, the so-called **electric-dipole approximation** can be employed and the Hamiltonian can take the more convenient form (see Box 8-2):

$$\hat{H} = \hbar\omega \left( \hat{a}^\dagger \hat{a} + \frac{1}{2} \right) + \frac{1}{2m} \hat{\vec{p}}^2 + q\hat{V}_{coul} - \hat{\vec{D}} \cdot \hat{\vec{E}} \tag{8-4}$$

where $\hat{\vec{D}}$ is the electric dipole operator:

$$\hat{\vec{D}} = q(\hat{\vec{r}} - \vec{r}_0) \tag{8-5}$$

and $\hat{\vec{E}}$ is the electric field of the quantized mode.
We recognize three parts of the Hamiltonian:

$$\boxed{\hat{H} = \hat{H}_R + \hat{H}_A + \hat{A}_I} \tag{8-6}$$

with:

$$\hat{H}_R = \hbar\omega \left( \hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \qquad \hat{H}_A = \frac{1}{2m} \hat{\vec{p}}^2 + q\hat{V}_{coul} \qquad \hat{H}_I = -\hat{\vec{D}} \cdot \hat{\vec{E}} \tag{8-7}$$

The term $\hat{H}_R$ is indeed the Hamiltonian of the quantized radiation mode, the term $\hat{H}_A$ regards the bound charge and, finally, the term $\hat{H}_I$ regards their relative interaction.

Box 8-1 - Electromagnetic potentials in the presence of static charges

In discussing the modal decomposition of the electromagnetic field, we considered the case of absence of charges and currents. In that case, and by assuming the Coulomb gauge, the scalar potential can be taken as uniformly null, and both the electric and magnetic fields are determined by the vector potential only, which has oscillating solutions in time.
We consider here the case in which there may be static charges with density $\rho$ in the region of space of our interest, but no electric currents. The Maxwell equations read:

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \qquad \nabla \cdot \vec{B} = 0$$
$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \nabla \times \vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

and the potentials are, as usual, defined by:

$$\nabla \times \vec{A} = \vec{B} \qquad \nabla V = -\vec{E} - \frac{\partial \vec{A}}{\partial t}$$

We maintain the Coulomb gauge $\nabla \cdot \vec{A} = 0$, and from the first Maxwell equation we get:

$$\frac{\rho}{\varepsilon_0} = \nabla \cdot \vec{E} = \nabla \cdot \left( -\nabla V - \frac{\partial \vec{A}}{\partial t} \right) = -\nabla^2 V - \frac{\partial}{\partial t}(\nabla \cdot \vec{A}) = -\nabla^2 V$$

Note that, as we are considering static charges, the potential $V$ obtained by this equation does not depend on time.
From the fourth Maxwell equations instead we derive:

$$\nabla \times \nabla \times \vec{A} = \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left( -\nabla V - \frac{\partial \vec{A}}{\partial t} \right) = -\mu_0 \varepsilon_0 \left( \nabla \frac{\partial V}{\partial t} + \frac{\partial^2 \vec{A}}{\partial t^2} \right)$$

and since $\frac{\partial V}{\partial t} = 0$ and $\nabla \cdot \vec{A} = 0$ we conclude $\nabla^2 \vec{A} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2}$. The vector potential $\vec{A}$ has wave-like solutions, precisely as those studied in the decomposition of the modes.
In summary the potentials are governed by the equations:

$$\boxed{\nabla \cdot \vec{A} = 0 \qquad \nabla^2 \vec{A} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{A}}{\partial t^2} \qquad \nabla^2 V = -\frac{\rho}{\varepsilon_0}}$$

As a consequence, the two terms that produce the electric field acquire a distinct physical significance:

$$\vec{E} = -\nabla V - \frac{\partial \vec{A}}{\partial t} = \vec{E}_{coul} + \vec{E}_{wave}$$

The first term $\vec{E}_{coul} = -\nabla V$ accounts for the contribution of the static charges, while $\vec{E}_{wave} = -\frac{\partial \vec{A}}{\partial t}$ accounts for the contribution of electromagnetic waves. The magnetic field $\vec{B} = \nabla \times \vec{A}$, on the contrary is only determined by the wave contribution and cannot depend on the static charges.

Box 8-2 - The electric dipole approximation

We start from the classical Hamiltonian:

$$H = \frac{1}{2m}(\vec{p} - q\vec{A})^2 + qV$$

where we neglect for simplicity the intrinsic energy of the electromagnetic field, which could be anyway added later. As explained already, we place ourselves in the Coulomb gauge $\nabla \cdot \vec{A} = 0$. The scalar potential $V = V_{coul}$ is describing the contributions coming from static charges, while the vector potential $\vec{A} = \vec{A}_{wave}$ is describing the oscillating fields of the electromagnetic waves. No currents are considered.
The gauge invariance relations for the electromagnetic potentials allow to choose freely not only the divergence $\nabla \cdot \vec{A}$, but also and a scalar function $\chi(\vec{r}, t)$. The potentials can be freely changed according to:

$$\vec{A}' = \vec{A} + \nabla\chi \qquad V' = V - \frac{\partial \chi}{\partial t}$$

We adopt at this point the Goppert-Mayer gauge, which consists in choosing:

$$\chi(\vec{r}, t) = -(\vec{r} - \vec{r}_0) \cdot \vec{A}(\vec{r}_0, t)$$

while maintaining the Coulomb-gauge divergence $\nabla \cdot \vec{A} = 0$. Note that $\vec{r}_0$ is a fixed position vector, relevant for the problem; thus the first part $(\vec{r} - \vec{r}_0)$ depends on the position $\vec{r}$ but not on $t$, while the second part $\vec{A}(\vec{r}_0, t)$ only depends on $t$.
Interestingly, the gradient of $\chi(\vec{r}, t)$ is:

$$\begin{aligned}\nabla\chi(\vec{r}, t) &= \nabla\left[-(\vec{r} - \vec{r}_0) \cdot \vec{A}(\vec{r}_0, t)\right] = \\ &= \nabla\left[-(x - x_0)A_x(\vec{r}_0, t) - (y - y_0)A_y(\vec{r}_0, t) - (z - z_0)A_z(\vec{r}_0, t)\right] = \\ &= -\vec{u}_x\, A_x(\vec{r}_0, t) - \vec{u}_y\, A_y(\vec{r}_0, t) - \vec{u}_z\, A_z(\vec{r}_0, t) = -\vec{A}(\vec{r}_0, t)\end{aligned}$$

while its time derivative is:

$$\frac{\partial \chi(\vec{r}, t)}{\partial t} = -(\vec{r} - \vec{r}_0) \cdot \frac{\partial \vec{A}(\vec{r}_0, t)}{\partial t}$$

Since the vector potential does not contain contributions from the static charges, we may write $\frac{\partial \vec{A}(\vec{r}_0, t)}{\partial t} = -\vec{E}_{wave}(\vec{r}, t)$ and the new potentials become:

$$\vec{A}'(\vec{r}, t) = \vec{A}(\vec{r}, t) + \nabla\chi = \vec{A}(\vec{r}, t) - \vec{A}(\vec{r}_0, t)$$

$$V' = V_{coul} - \frac{\partial \chi}{\partial t} = V_{coul} - (\vec{r} - \vec{r}_0) \cdot \vec{E}_{wave}(\vec{r}, t)$$

Finally, we assume that the charge $q$ remains confined in a region around $\vec{r}_0$ that is much smaller than the wavelength of the radiation described by $\vec{A}$. This means that in all positions $\vec{r}$ relevant for our problem, we have $\vec{A}'(\vec{r}, t) = \vec{A}(\vec{r}, t) - \vec{A}(\vec{r}_0, t) \simeq 0$.
Having defined the electric dipole operator $\hat{\vec{D}} = q(\hat{\vec{r}} - \vec{r}_0)$ we can write the Hamiltonian in the new gauge as:

$$\begin{aligned}H' &= \frac{1}{2m}(\vec{p} - q\vec{A}')^2 + qV' = \frac{1}{2m}\vec{p}^2 + qV_{coul} - q(\vec{r} - \vec{r}_0) \cdot \vec{E}_{wave} = \\ &= \frac{1}{2m}\vec{p}^2 + qV_{coul} - \vec{D} \cdot \vec{E}_{wave}\end{aligned}$$

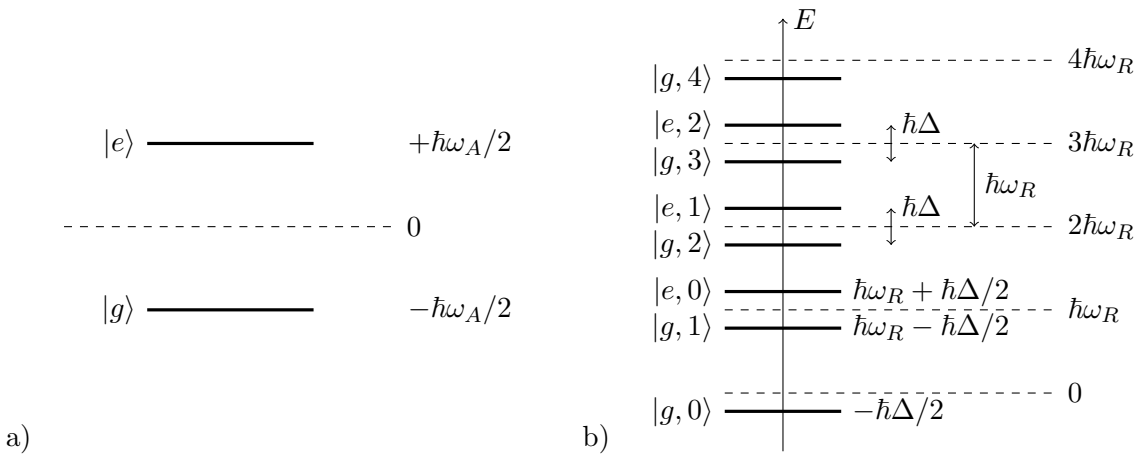


Figure 10: (a) Scheme of the energies of the two-level atom considered in $\hat{H}_A$. The ground $|g\rangle$ and excited states $|e\rangle$ have an energy difference $E_e - E_g = \hbar\omega_A$ and the zero level of the energy is placed in the middle. (b) Scheme of the energy levels of the eigenstates of the uncoupled Hamiltonian $\hat{H}_0 = \hat{H}_R + \hat{H}_A$. The states $|g,n\rangle$ and $|e,n\rangle$ are shown. $\omega_R$ is the frequency of the radiation mode and $\Delta = \omega_A - \omega_R$.

## 8.2 The uncoupled Hamiltonian

Before discussing the dynamics produced by the full Hamiltonian (8-4) it is instructive to study the *uncoupled* Hamiltonian $\hat{H}_0 = \hat{H}_A + \hat{H}_R$ where the interaction term is being neglected.
The Hamiltonian $\hat{H}_A$ deals with the electron within an atom. Any operator $\hat{Q}$ with eigenstates $|e_{\lambda_i}\rangle$, associated to eigenvalues $\lambda_i$, can be written in the so-called diagonal form as $\hat{Q} = \sum_i \lambda_i |e_{\lambda_i}\rangle \langle e_{\lambda_i}|$. The diagonal form of the atomic Hamiltonian $\hat{H}_A = \frac{1}{2m}\hat{\vec{p}}^2 + q\hat{V}_{coul}$ employs the atomic energy levels $|a_i\rangle$ and their energies $E_i$:

$$\hat{H}_A = \sum_i E_i |a_i\rangle \langle a_i| \tag{8-8}$$

In the following we will consider a strong simplification of this atomic Hamiltonian, which will consider only **two levels**. This approximation is justified in many applications, where the radiation is almost resonant to one transition, so that only two atomic levels are basically involved in the physical processes. In detail, we will consider a *ground state* $|g\rangle$ with energy $E_g$ and an *excited state* $|e\rangle$ with energy $E_e$. Therefore, the simplified atomic Hamiltonian is:

$$\hat{H}_A = E_g |g\rangle \langle g| + E_f |e\rangle \langle e| \tag{8-9}$$

As energy zeros may be chosen with some freedom, we shall also impose $\frac{E_e+E_g}{2} = 0$ and rewrite $\hat{H}_A$ as:

$$\hat{H}_A = \frac{\hbar\omega_A}{2} (|e\rangle \langle e| - |g\rangle \langle g|) \tag{8-10}$$

where we have also defined the atomic transition frequency $\omega_A$:

$$\omega_A = \frac{1}{\hbar} (E_e - E_g) \tag{8-11}$$

These energy levels are represented in Fig. 10a.
For the following discussion, it is useful to employ the **atomic raising and lowering operators**:

$$\hat{\sigma} = |g\rangle \langle e| \qquad \hat{\sigma}^\dagger = |e\rangle \langle g| \tag{8-12}$$

that yield the following properties:

$$\hat{\sigma} |g\rangle = 0 \qquad \hat{\sigma} |e\rangle = |g\rangle \qquad \hat{\sigma}^2 = 0 \tag{8-13}$$

$$\hat{\sigma}^\dagger |g\rangle = |e\rangle \qquad \hat{\sigma}^\dagger |e\rangle = 0 \qquad (\hat{\sigma}^\dagger)^2 = 0 \tag{8-14}$$

In addition, one has:

$$\hat{\sigma}\hat{\sigma}^\dagger = |g\rangle\langle g| \qquad\qquad \hat{\sigma}^\dagger\hat{\sigma} = |e\rangle\langle e| \tag{8-15}$$

so that the commutator and anticommutator read:

$$[\hat{\sigma}, \hat{\sigma}^\dagger] = \hat{\sigma}\hat{\sigma}^\dagger - \hat{\sigma}^\dagger\hat{\sigma} = -\hat{\sigma}_z \tag{8-16}$$

$$\{\hat{\sigma}, \hat{\sigma}^\dagger\} = \hat{\sigma}\hat{\sigma}^\dagger + \hat{\sigma}^\dagger\hat{\sigma} = 1 \tag{8-17}$$

In the commutator we have further exploited the definition of the Pauli $z$ operator:[28]

$$\hat{\sigma}_z = |e\rangle\langle e| - |g\rangle\langle g| \tag{8-18}$$

Note the difference between these operators and the bosonic $\hat{a}, \hat{a}^\dagger$ operators employed in the harmonic oscillator model. Indeed, while the oscillator can take many excitation quanta, the two-level atom can take only one excitation.
Harnessing the relations written above, in particular the definition of $\hat{\sigma}_z$ and the expressions of the commutator and anticommutator of the atoic operator, the **atomic Hamiltonian** $\hat{H}_A$ is finally rewritten as:[29]

$$\boxed{\hat{H}_A = \frac{\hbar\omega_A}{2}\hat{\sigma}_z = \hbar\omega_A\left(\hat{\sigma}^\dagger\hat{\sigma} - \frac{1}{2}\right)} \tag{8-19}$$

The **radiation Hamiltonian** $\hat{H}_R$, on the other hand, takes the form that was vastly discussed in the previous Sections. In case a single mode is considered, it reads:

$$\hat{H}_R = \hbar\omega_R\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$$

We have now all the elements to discuss the eigenstates of $\hat{H}_0$:

$$\hat{H}_0 = \hat{H}_R + \hat{H}_A = \hbar\omega_R\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right) + \hbar\omega_A\left(\hat{\sigma}^\dagger\hat{\sigma} - \frac{1}{2}\right) \tag{8-20}$$

The two parts $\hat{H}_R$ and $\hat{H}_A$ of $\hat{H}_0$ commute one with the other, as they regard different systems. Therefore, the eigenstates of the overall Hamiltonian are the tensor products of the eigenstates of the individual parts. Namely, the eigenstates of $\hat{H}_0$ are states with the form:

$$\boxed{|g, n\rangle \qquad\qquad |e, n\rangle} \tag{8-21}$$

where $g$ and $e$ are labels for the atomic ground and excited states respectively, while $n$ indicates the $|n\rangle$ Fock state of the radiation mode.
The energy structure of the eigenstates is studied more conveniently by rewriting the Hamiltonian in a slightly different form. First, we introduce the **frequency detuning** parameter:

$$\Delta = \omega_A - \omega_R \tag{8-22}$$

[28] The two-level atom is formally analogous to a spin-1/2 system and it is possible to define the related Pauli operators.
[29] One could indeed work out several equivalent forms:

$$\hat{H}_A = \frac{\hbar\omega_A}{2}(\hat{\sigma}^\dagger\hat{\sigma} - \hat{\sigma}\hat{\sigma}^\dagger) = \frac{\hbar\omega_A}{2}\left(\hat{\sigma}^\dagger\hat{\sigma} - (1 - \hat{\sigma}^\dagger\hat{\sigma})\right) = \hbar\omega_A\left(\hat{\sigma}^\dagger\hat{\sigma} - \frac{1}{2}\right)$$

$$\hat{H}_A = \frac{\hbar\omega_A}{2}(\hat{\sigma}^\dagger\hat{\sigma} - \hat{\sigma}\hat{\sigma}^\dagger) = \frac{\hbar\omega_A}{2}\left((1 - \hat{\sigma}\hat{\sigma}^\dagger) - \hat{\sigma}\hat{\sigma}^\dagger\right) = \hbar\omega_A\left(\frac{1}{2} - \hat{\sigma}\hat{\sigma}^\dagger\right)$$

and write:

$$\hat{H}_0 = \hbar\omega_R \left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right) + \hbar(\omega_R + \Delta)\left(\hat{\sigma}^\dagger \hat{\sigma} - \frac{1}{2}\right) =$$
$$= \hbar\omega_R \left(\hat{a}^\dagger \hat{a} + \hat{\sigma}^\dagger \hat{\sigma}\right) + \hbar\Delta \left(\hat{\sigma}^\dagger \hat{\sigma} - \frac{1}{2}\right) \tag{8-23}$$

Second, we define an operator $\hat{M}$ that represents the **total number of excitations**:

$$\hat{M} = \hat{a}^\dagger \hat{a} + \hat{\sigma}^\dagger \hat{\sigma} \tag{8-24}$$

Note that the eigenstates of $\hat{H}_0$ are also eigenstates of $\hat{M}$:

$$\hat{M}\,|g,n\rangle = n\,|g,n\rangle \qquad\qquad \hat{M}\,|e,n\rangle = (n+1)\,|e,n\rangle \tag{8-25}$$

and that the states $|g,n+1\rangle$ and $|e,n\rangle$ share the same eigenvalue $(n+1)$. By further recalling the properties of the $\hat{\sigma}$, $\hat{\sigma}^\dagger$ operators, we reach a particularly simple expression for $\hat{H}_0$:

$$\boxed{\hat{H}_0 = \hbar\omega_R \hat{M} + \frac{\hbar\Delta}{2}\hat{\sigma}_z} \tag{8-26}$$

The eigenenergies are immediately found:

$$\hat{H}_0\,|g,n\rangle = E_{g,n}\,|g,n\rangle \qquad\qquad E_{g,n} = \hbar\omega_R n - \hbar\frac{\Delta}{2} \tag{8-27}$$

$$\hat{H}_0\,|e,n\rangle = E_{e,n}\,|e,n\rangle \qquad\qquad E_{e,n} = \hbar\omega_R (n+1) + \hbar\frac{\Delta}{2} \tag{8-28}$$

A more careful inspection reveals that the eigenstates are organized as a ladder of doublets, namely pairs of states $|g,n+1\rangle$ and $|e,n\rangle$ , as shown in Figure 10b. From center to center, the doublets are separated by $\hbar\omega_R$, while the two levels of each doublet are separated by $\hbar\Delta$:

$$E_{g,n+1} = \hbar\omega_R (n+1) - \hbar\frac{\Delta}{2} \tag{8-29}$$

$$E_{e,n} = \hbar\omega_R (n+1) + \hbar\frac{\Delta}{2} \tag{8-30}$$

If $\Delta = 0$ the states in each doublet become degenerate. Note that Figure 10b represents the case $\Delta > 0$ which implies $E_{e,n} > E_{g,n+1}$, otherwise they are exchanged in the ladder. The ground state $|g,0\rangle$ is the only unpaired one, with energy $E_{g,0} = -\hbar\Delta/2$.
As a final remark, we note that the set of states $\{|g,n\rangle, |e,n\rangle\}$ is a complete basis for the Hilbert space of this physical system, consisting in a two-level atom and a single radiation mode. The introduction of an interaction term does not add other components to the physical system. Thus, these states will still be a complete basis to describe the system, even though they will not be eigenstates of the full Hamiltonian.

## 8.3 The Jaynes-Cummings Hamiltonian

The **Jaynes-Cummings** model is a cornerstone in Quantum Optics, describing the interaction between the quantized electromagnetic field and a two-level atom. It is indeed a particularly simple framework to understand basic phenomena of light-matter interactions in optical cavities, where a single quantum emitter (atom) is strongly coupled to the radiation field. This model employs the Hamiltonian $\hat{H}_0 = \hat{H}_A + \hat{H}_R$ studied in the previous section, and some approximations for the interaction Hamiltonian $\hat{H}_I$ that we shall now discuss.
We assume as a first thing the electric dipole approximation, in which the interaction energy is the scalar product between the electric dipole and the electric field (with a minus sign):

$$\hat{H}_I = -\hat{\vec{D}} \cdot \hat{\vec{E}}$$

Implicit in this condition is also the long-wavelength approximation, for which the wavelength of the quantized field is much larger than the atomic size. We shall also adopt a reference frame with the origin in the center of the atom, namely with $\vec{r}_0 = 0$. This allows us to write the electric dipole operator as:

$$\hat{\vec{D}} = q\hat{\vec{r}} \tag{8-31}$$

where $q$ is the electron charge and $\hat{\vec{r}}$ its position operator.
The electric dipole operator can be expanded in terms of the matrix elements in the space that describes the state of the atom (we drop for a moment the operator symbol '^' to simplify the notation):

$$\frac{1}{q}\vec{D} = \langle g|\,\vec{r}\,|g\rangle\ |g\rangle\langle g| + \langle g|\,\vec{r}\,|e\rangle\ |g\rangle\langle e| + \langle e|\,\vec{r}\,|g\rangle\ |e\rangle\langle g| + \langle e|\,\vec{r}\,|e\rangle\ |e\rangle\langle e| \tag{8-32}$$

The atom states $|g\rangle$ and $|e\rangle$ are symmetric in the space, which implies that diagonal elements are null:

$$\langle g|\,\vec{r}\,|g\rangle = \langle e|\,\vec{r}\,|e\rangle = 0 \tag{8-33}$$

The other elements, instead, may not vanish and we can define a real constant $d$ and a complex unit vector $\vec{\epsilon}_A$ such that:

$$\langle g|\,\vec{r}\,|e\rangle = \frac{1}{q}d\,\vec{\epsilon}_A \tag{8-34}$$

$$\langle e|\,\vec{r}\,|g\rangle = \Big(\langle g|\,\vec{r}\,|e\rangle\Big)^* = \frac{1}{q}d\,\vec{\epsilon}_A^{\,*} \tag{8-35}$$

Hence, by further recalling the definitions of the atomic operators $\hat{\sigma} = |g\rangle\langle e|$ and $\hat{\sigma}^\dagger = |e\rangle\langle g|$, we write:

$$\boxed{\hat{\vec{D}} = d\left(\hat{\sigma}\vec{\epsilon}_A + \hat{\sigma}^\dagger\vec{\epsilon}_A^{\,*}\right)} \tag{8-36}$$

In treating the interaction of a single electromagnetic mode with an atom, for reasons that will be clear later, it is convenient to use a modal form with a generic state of polarization. In practice, we need to form a combination, with complex coefficients, of two electromagnetic modes with the same wavevector and orthogonal polarizations. This amounts to using a complex unit vector $\vec{\epsilon}_R$ that defines the polarization state of the wave, and to writing the electric field operator as:

$$\hat{\vec{E}} = \iota\mathcal{E}(\hat{a}e^{\iota\vec{k}\cdot\vec{r}}\,\vec{\epsilon}_R\ -\ \hat{a}^\dagger e^{-\iota\vec{k}\cdot\vec{r}}\,\vec{\epsilon}_R^{\,*}) \tag{8-37}$$

If $\vec{k} \parallel \vec{u}_z$, the polarization vector is $\vec{\epsilon}_R = c_X\vec{u}_x + c_Y\vec{u}_y$, with $|c_X|^2 + |c_Y|^2 = 1$. For instance, circular polarization states in this framework correspond to $\vec{\epsilon}_R = (\vec{u}_x \pm \iota\vec{u}_y)/\sqrt{2}$.
In addition, we assume that all relevant phenomena are occurring close to $\vec{r} = 0$ (the atomic center) and that the radiation wavelength is long, so that $e^{\iota\vec{k}\cdot\vec{r}} = e^{-\iota\vec{k}\cdot\vec{r}} = 1$. The electric field operator gains a more compact expression:

$$\boxed{\hat{\vec{E}} = \iota\mathcal{E}(\hat{a}\vec{\epsilon}_R\ -\ \hat{a}^\dagger\vec{\epsilon}_R^{\,*})} \tag{8-38}$$

while the interaction Hamiltonian has now assumed the form:

$$\hat{H}_I = -\iota d\mathcal{E}\left(\hat{\sigma}\vec{\epsilon}_A + \hat{\sigma}^\dagger\vec{\epsilon}_A^{\,*}\right)\cdot\left(\hat{a}\vec{\epsilon}_R - \hat{a}^\dagger\vec{\epsilon}_R^{\,*}\right) \tag{8-39}$$

By expanding this product we see four terms respectively proportional to $\hat{\sigma}\hat{a}$, $\hat{\sigma}\hat{a}^\dagger$, $\hat{\sigma}^\dagger\hat{a}$ and $\hat{\sigma}^\dagger\hat{a}^\dagger$. Using the relations:

$$\hat{\sigma} = |g\rangle\langle e| \qquad\qquad \hat{\sigma}^\dagger = |e\rangle\langle g|$$

$$\hat{a} = \sum_{n=0}^{+\infty}\sqrt{n+1}\,|n\rangle\langle n+1| \qquad\qquad \hat{a}^\dagger = \sum_{n=0}^{+\infty}\sqrt{n+1}\,|n+1\rangle\langle n|$$

we can identify their operation on states of the kind $|g,n\rangle$ and $|e,n\rangle$. It is worth discussing them with some detail.

- $\hat{\sigma}\hat{a}^\dagger = \sum_{n=0}^{+\infty} \sqrt{n+1}\, |g, n+1\rangle \langle e, n|$ corresponds to the standard **emission process** of one photon, with the field passing from the state $|n\rangle$ to the state $|n+1\rangle$ and the atom passing from the excited state to the ground state;
- $\hat{\sigma}^\dagger\hat{a} = \sum_{n=0}^{+\infty} \sqrt{n+1}\, |e, n\rangle \langle g, n+1|$ corresponds to the standard **absorption process** of one photon, with the field passing from the state $|n+1\rangle$ to the state $|n\rangle$, while the atom passes from the ground to the excited state;
- $\hat{\sigma}\hat{a} = \sum_{n=0}^{+\infty} \sqrt{n+1}\, |g, n\rangle \langle e, n+1|$ and $\hat{\sigma}^\dagger\hat{a}^\dagger = \sum_{n=0}^{+\infty} \sqrt{n+1}\, |e, n+1\rangle \langle g, n|$ are respectively processes in which the atom is de-excited and simultaneously a photon is annihilated, or the atom is excited and a photon is created. These processes do not have an intuitive picture as they do not conserve the number of excitations.

We note now that the first two processes $\hat{\sigma}\hat{a}^\dagger$ and $\hat{\sigma}^\dagger\hat{a}$ correspond to transitions within the same doublet. Namely, they connect in either direction the two states $|g, n+1\rangle$ and $|e, n\rangle$, which are separated by an energy difference $\hbar\Delta$. This energy difference tends to zero at resonance ($\omega_A \simeq \omega_R$). On the other hand, the processes $\hat{\sigma}\hat{a}$ and $\hat{\sigma}^\dagger\hat{a}^\dagger$ correspond to transitions between different doublets, which are highly non-resonant transitions. They connect the states $|g, n\rangle$ and $|e, n+1\rangle$ separated by a large energy difference $2\hbar\omega_R + \hbar\Delta$. In the Jaynes-Cummings model an approximation is made and *the non-resonant terms are neglected.*[30] This is called the **rotating wave approximation**, which allows to write the interaction Hamiltonian as:

$$\hat{H}_I = -\iota d\mathcal{E} \left( \hat{\sigma}^\dagger \hat{a}\, \vec{\epsilon}_A^* \cdot \vec{\epsilon}_R - \hat{\sigma}\hat{a}^\dagger\, \vec{\epsilon}_A \cdot \vec{\epsilon}_R^* \right) \tag{8-40}$$

For simplicity we assume $\vec{\epsilon}_A^* \cdot \vec{\epsilon}_R$ to be a real number. This is quite a general case indeed, because the complex polarization vector $\vec{\epsilon}_R$ may easily include a phase factor on all its components, which makes this scalar product real without changing significantly the physics of the system described. This is also an indication for the opportunity of using a complex vector to describe polarization of the mode. In addition, we define the **vacuum Rabi frequency**:

$$\boxed{\Omega_0 = \frac{2d\mathcal{E}}{\hbar} \vec{\epsilon}_A^* \cdot \vec{\epsilon}_R} \tag{8-41}$$

and the interaction Hamiltonian becomes finally:

$$\boxed{\hat{H}_I = -\iota\hbar \frac{\Omega_0}{2} \left( \hat{\sigma}^\dagger \hat{a} - \hat{\sigma}\hat{a}^\dagger \right)} \tag{8-42}$$

Therefore, the resulting overall **Hamiltonian of the Jaynes-Cummings model** takes the form:

$$\boxed{\hat{H}_{\mathrm{JC}} = \hbar\omega_R \hat{M} + \frac{\hbar\Delta}{2}\hat{\sigma}_z - \iota\hbar \frac{\Omega_0}{2} \left( \hat{\sigma}^\dagger \hat{a} - \hat{\sigma}\hat{a}^\dagger \right)} \tag{8-43}$$

[30] A simple argument for the validity of the rotating wave approximation when the coupling is not too strong, and the system is close to resonance, may be the following. Standard time-independent perturbation theory is employed to evaluate how much the energy $E_n$ of an eigenstate $|\psi_n\rangle$ of the Hamiltonian $\hat{H}_0$ is altered, by the addition of a small perturbation in the form of $\hat{H}_I$ with its different terms. First and second-order energy corrections are given by (see e.g. D.J. Griffiths, Introduction to quantum mechanics, Prentice Hall (1995); chapter 6):

$$E^{(1)} = \langle\psi_n| \hat{H}_I |\psi_n\rangle \qquad\qquad E^{(2)} = \sum_{m\neq n} \frac{|\langle\psi_m| \hat{H}_I |\psi_n\rangle|^2}{E_n - E_m}$$

where the set of the $|\psi_m\rangle$ is the complete basis of the eigenstates of the Hamiltonian. Given the form of $\hat{H}_I$ the first-order correction $E^{(1)}$ is zero. The second-order correction is made of contributions inversely proportional to the energy difference between the levels coupled by the matrix elements $\langle\psi_m| \hat{H}_I |\psi_n\rangle$. Thus, contributions from the resonant interactions are much more relevant than the anti-resonant ones.

The eigenstates of this full Hamiltonian are called the **dressed states**. They are important because the evolution of an arbitrary state can be conveniently computed by decomposing it on the basis of the dressed states. We address in the following the problem of calculating the eigenstates and their eigenvalues (i.e. their energies).
We have already observed that the interaction Hamiltonian $\hat{H}_I$ only connects pairs of states with the same number $(n+1)$ of total excitations such as $|g, n+1\rangle$ and $|e, n\rangle$. Formally, this is linked also to the fact that the Hamiltonian $\hat{H}_I$ commutes with the operator $\hat{M}$. This observation allows to write the overall Hamiltonian $\hat{H}_{JC} = \hat{H}_0 + \hat{H}_I$ as a sum of independent Hamiltonians, each acting on a different doublet of states (leaving aside the state $|g, 0\rangle$ which is the only one with a total number of excitations equal to zero):

$$\hat{H}_{JC} = -\frac{\hbar\Delta}{2} |g, 0\rangle \langle g, 0| + \sum_n \hat{H}_n \tag{8-44}$$

$$\hat{H}_n = \hbar\omega_R(n+1)\hat{\mathbb{I}}_n + \hat{V}_n \tag{8-45}$$

In Eq. (8-45) we have indicated with $\hat{\mathbb{I}}_n$ the identity operator acting on the subspace with $n+1$ excitations:

$$\mathbb{I}_n = |e, n\rangle \langle e, n| + |g, n+1\rangle \langle g, n+1| \tag{8-46}$$

and with $\hat{V}_n$ the operator:

$$\begin{aligned}\hat{V}_n &= \hbar\frac{\Delta}{2}\hat{\sigma}_z - \iota\hbar\frac{\Omega_n}{2}\left(|e, n\rangle \langle g, n+1| - |g, n+1\rangle \langle e, n|\right) = \\ &= \frac{\hbar}{2}\left(+\Delta |e, n\rangle \langle e, n| - \Delta |g, n+1\rangle \langle g, n+1| - \iota\Omega_n |e, n\rangle \langle g, n+1| + \iota\Omega_n |g, n+1\rangle \langle e, n|\right)\end{aligned} \tag{8-47}$$

where $\Omega_n = \sqrt{n+1}\,\Omega_0$ is defined as the *n-photon Rabi frequency.*
Given the structure of the full Hamiltonian, the set of its eigenstates is the union of the eigenstates of the individual $\hat{H}_n$ (plus the state $|g, 0\rangle$). Then, it is not difficult to note that the eigenstates of $\hat{H}_n$ are the eigenstates of $\hat{V}_n$. Indeed, if $|\psi\rangle$ is an eigenvector of $\hat{V}_n$, namely $\hat{V}_n |\psi\rangle = \lambda |\psi\rangle$, we have also $\hat{H}_n |\psi\rangle = \hbar\omega_R(n+1)\hat{\mathbb{I}}_n |\psi\rangle + \hat{V}_n |\psi\rangle = (\hbar\omega_R(n+1) + \lambda) |\psi\rangle$. Namely, $|\psi\rangle$ is also an eigenvector of $\hat{H}_n$ with eigenvalue $E = \hbar\omega_R(n+1) + \lambda$.
The eigenstates of $\hat{V}_n$, which are the same as those of $\hat{H}_n$, are calculated in Box 8-3 and read:

$$\boxed{\begin{aligned}|+, n\rangle &= \cos\frac{\theta_n}{2} |e, n\rangle + \iota \sin\frac{\theta_n}{2} |g, n+1\rangle \\ |-, n\rangle &= \sin\frac{\theta_n}{2} |e, n\rangle - \iota \cos\frac{\theta_n}{2} |g, n+1\rangle\end{aligned}} \tag{8-48}$$

As it is customary, they have been written as a function of the parameter $\theta_n$ defined from:

$$\cos\theta_n = \frac{\Delta}{\sqrt{\Delta^2 + \Omega_n^2}} \qquad \sin\theta_n = \frac{\Omega_n}{\sqrt{\Delta^2 + \Omega_n^2}} \tag{8-49}$$

These states correspond for $\hat{V}_n$ respectively to the eigenvalues:

$$\lambda_+ = +\frac{\hbar}{2}\sqrt{\Delta^2 + \Omega_n^2} \qquad \lambda_- = -\frac{\hbar}{2}\sqrt{\Delta^2 + \Omega_n^2} \tag{8-50}$$

The eigenvalues of $\hat{H}_n$ are derived straightforwardly from the former:

$$\boxed{E_n^{\pm} = \hbar\omega_R(n+1) \pm \frac{\hbar}{2}\sqrt{\Delta^2 + \Omega_n^2}} \tag{8-51}$$

These are indeed *the energies of the dressed states.*

Box 8-3 - Eigenvalues and eigenvectors of $\hat{V}_n$

We adopt a matrix notation where the generic state corresponds to a column vector as $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \equiv c_1 \left|e, n\right\rangle + c_2 \left|g, n+1\right\rangle$. The operator $\hat{V}_n$ becomes a matrix: $V_n = \frac{\hbar}{2} \begin{bmatrix} +\Delta & -\iota\Omega_n \\ +\iota\Omega_n & -\Delta \end{bmatrix}$.

For simplicity of calculation we will work here on $\frac{2}{\hbar}V_n$, namely we drop the multiplicative factor $\frac{\hbar}{2}$ on the left. The eigenvectors will be the same as those of $V_n$. The eigenvalues shall be later multiplied back by a factor $\frac{\hbar}{2}$. In detail, to find the **eigenvalues** of $\frac{2}{\hbar}V_n$ we impose:

$$\begin{vmatrix} +\Delta - \lambda & -\iota\Omega_n \\ +\iota\Omega_n & -\Delta - \lambda \end{vmatrix} = 0 \qquad \rightarrow \qquad -(\Delta - \lambda)(\Delta + \lambda) - \Omega_n^2 = -\Delta^2 + \lambda^2 - \Omega_n^2 = 0$$

which gives as a result $\lambda_+ = \sqrt{\Delta^2 + \Omega_n^2}$ and $\lambda_- = -\sqrt{\Delta^2 + \Omega_n^2}$. The eigenvalues of $V_n$ are the latter multiplied by $\frac{\hbar}{2}$.

To find the **eigenvectors** we impose:

$$\begin{bmatrix} +\Delta & -\iota\Omega_n \\ +\iota\Omega_n & -\Delta \end{bmatrix} \cdot \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \lambda_\pm \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \pm\sqrt{\Delta^2 + \Omega_n^2} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

This is a linear system of two equations in the unknown $c_1$ and $c_2$, which is intrinsically underdetermined. Considering only one equation is thus sufficient. The first one reads:

$$\Delta\, c_1 - \iota\Omega_n\, c_2 = \pm\sqrt{\Delta^2 + \Omega_n^2}\, c_1$$

At this point we can conveniently define $\cos\theta_n = \dfrac{\Delta}{\sqrt{\Delta^2 + \Omega_n^2}}$ and $\sin\theta_n = \dfrac{\Omega_n}{\sqrt{\Delta^2 + \Omega_n^2}}$; the former equation becomes:

$$\cos\theta_n c_1 - \iota \sin\theta_n c_2 = \pm c_1 \qquad \rightarrow \qquad -\iota\sin\theta_n c_2 = (\pm 1 - \cos\theta_n) c_1$$

The solution corresponding to the *positive eigenvalue* is $c_1 = -\iota\frac{\sin\theta_n}{1-\cos\theta_n}c_2$, which can be elaborated as:

$$\begin{aligned} c_1 &= -\iota\frac{\sin\theta_n}{1-\cos\theta_n}c_2 = -\iota\frac{\sqrt{1-\cos^2\theta_n}}{1-\cos\theta_n}c_2 = -\iota\frac{\sqrt{(1+\cos\theta_n)(1-\cos\theta_n)}}{1-\cos\theta_n}c_2 = \\ &= -\iota\sqrt{\frac{1+\cos\theta_n}{1-\cos\theta_n}}c_2 = -\iota\sqrt{\frac{1+\cos\theta_n}{2}}\sqrt{\frac{2}{1-\cos\theta_n}}c_2 = \frac{\cos(\theta_n/2)}{+\iota\sin(\theta_n/2)}c_2 \end{aligned}$$

The solution corresponding to the *negative eigenvalue*, on the other hand, is $c_1 = +\iota\frac{\sin\theta_n}{1+\cos\theta_n}c_2$, which becomes:

$$\begin{aligned} c_1 &= \iota\frac{\sin\theta_n}{1+\cos\theta_n}c_2 = \iota\frac{\sqrt{1-\cos^2\theta_n}}{1+\cos\theta_n}c_2 = \iota\frac{\sqrt{(1+\cos\theta_n)(1-\cos\theta_n)}}{1+\cos\theta_n}c_2 = \\ &= \iota\sqrt{\frac{1-\cos\theta_n}{1+\cos\theta_n}}c_2 = \iota\sqrt{\frac{1-\cos\theta_n}{2}}\sqrt{\frac{2}{1+\cos\theta_n}}c_2 = \frac{\sin(\theta_n/2)}{-\iota\cos(\theta_n/2)}c_2 \end{aligned}$$

Therefore, indicating with $\left|+\right\rangle$ and $\left|-\right\rangle$ respectively the two eigenvectors relative to the eigenvalues $\lambda_+$ and $\lambda_-$, we can write:

$$\left|+\right\rangle \equiv \begin{bmatrix} \cos(\theta_n/2) \\ +\iota\sin(\theta_n/2) \end{bmatrix} \qquad \left|-\right\rangle \equiv \begin{bmatrix} \sin(\theta_n/2) \\ -\iota\cos(\theta_n/2) \end{bmatrix}$$

In case of **large detuning** $\Delta \gg \Omega_n$ the eigenstates and their energies tend to be the same as those of the uncoupled case. Indeed $\cos\theta_n \to 1$ and $\sin\theta_n \to 0$, namely $\theta_n \to 0$, and one has:

$$|+,n\rangle \to |e,n\rangle \qquad\qquad |-,n\rangle \to -\iota\,|g,n+1\rangle \tag{8-52}$$

$$E_n^{\pm} = \hbar\omega(n+1) \pm \frac{\hbar}{2}\Delta \tag{8-53}$$

Note that the phase term $-\iota$ applied to the state $|g,n+1\rangle$ is not relevant. In practice, with large detuning the atom-field interaction plays a negligible role in determining the dynamics of the system. The opposite limiting case is the case of **resonance** $\Delta = 0$. Here, $\cos\theta_n/2 \to 0$ and $\sin\theta_n/2 \to 1$, which means $\theta \to \pi/2$ and $\theta/2 \to \pi/4$. The eigenstates and related energies are:

$$|+,n\rangle \to \frac{|e,n\rangle + \iota\,|g,n+1\rangle}{\sqrt{2}} \qquad\qquad |-,n\rangle \to \frac{|e,n\rangle - \iota\,|g,n+1\rangle}{\sqrt{2}} \tag{8-54}$$

$$E_n^{\pm} = \hbar\omega(n+1) \pm \frac{\hbar}{2}\Omega_n \tag{8-55}$$

Note that without the interaction (i.e. in the uncoupled case) in the same resonance condition $\Delta = 0$ the two levels would have been degenerate with energy $E_n^{\pm} = E_{g,n+1} = E_{e,n} = \hbar\omega(n+1)$. The interaction removes the degeneracy and the two dressed states become separated by $\hbar\Omega_n$.

## 8.4 Rabi oscillations induced by $n$ photons

One peculiar phenomenon predicted by the Janyes-Cummings model are Rabi oscillations at different frequencies depending on the initial state of the system.
We consider the resonant case $\Delta = 0$ and the system initially prepared in the state $|e,n\rangle$, namely with the atom in excited states and the radiation field with $n$ photons. The dressed states in this case are given by Eq. (8-54), thus the state at time $t = 0$ can be written as:

$$|\psi(0)\rangle = |e,n\rangle = \frac{1}{\sqrt{2}}\left(|+,n\rangle + |-,n\rangle\right) \tag{8-56}$$

The evolution in time of this system is obtained by multiplying each dressed state (which is the eigenstate of the interacting system) by a phase term $e^{-\iota Et/\hbar}$ where $E$ is its energy.

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(e^{-\iota\frac{E_n^+}{\hbar}t}\,|+,n\rangle + e^{-\iota\frac{E_n^-}{\hbar}t}\,|-,n\rangle\right) \tag{8-57}$$

We read the energies of the dressed states in Eq. (8-55) and we substitute:

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}e^{-\iota\hbar\omega(n+1)t}\left(e^{-\iota\frac{\Omega_n}{2}t}\,|+,n\rangle + e^{+\iota\frac{\Omega_n}{2}t}\,|-,n\rangle\right) \tag{8-58}$$

At this point we get back to the basis of the states $|e,n\rangle$ and $|g,n+1\rangle$:

$$\begin{aligned}
|\psi(t)\rangle &= \frac{1}{\sqrt{2}}e^{-\iota\hbar\omega(n+1)t}\left(e^{-\iota\frac{\Omega_n}{2}t}\frac{|e,n\rangle + \iota\,|g,n+1\rangle}{\sqrt{2}} + e^{+\iota\frac{\Omega_n}{2}t}\frac{|e,n\rangle - \iota\,|g,n+1\rangle}{\sqrt{2}}\right) = \\
&= e^{-\iota\hbar\omega(n+1)t}\left(\frac{e^{+\iota\frac{\Omega_n}{2}t} + e^{-\iota\frac{\Omega_n}{2}t}}{2}\,|e,n\rangle + \frac{e^{+\iota\frac{\Omega_n}{2}t} + e^{-\iota\frac{\Omega_n}{2}t}}{2\iota}\,|g,n+1\rangle\right) = \\
&= e^{-\iota\hbar\omega(n+1)t}\left[\cos\left(\frac{\Omega_n}{2}t\right)|e,n\rangle + \sin\left(\frac{\Omega_n}{2}t\right)|g,n+1\rangle\right]
\end{aligned} \tag{8-59}$$

Therefore, at an arbitrary time $t$, the probabilities to measure the atom in the excited or in the ground state are respectively:

$$p_e(t) = |\langle e|\psi(t)\rangle|^2 = \cos^2\left(\frac{\Omega_n}{2}t\right) = \frac{1+\cos(\Omega_n t)}{2} \tag{8-60}$$

$$p_g(t) = |\langle g|\psi(t)\rangle|^2 = \sin^2\left(\frac{\Omega_n}{2}t\right) = \frac{1-\cos(\Omega_n t)}{2} \tag{8-61}$$

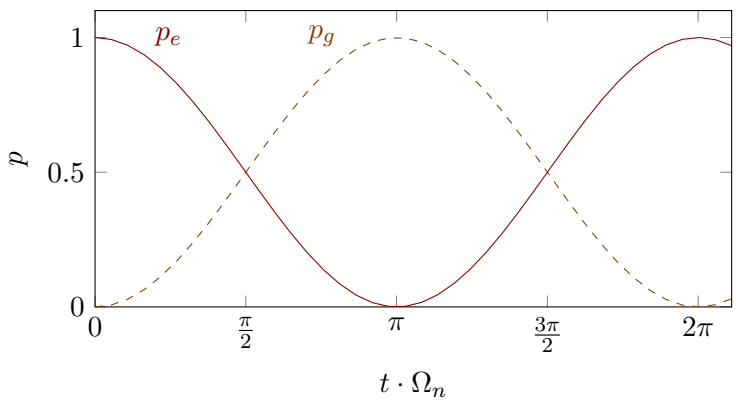


Figure 11: Probability for the atom to be measured in the excited state $p_e$, rather than in the ground state $p_g$, in case the state of the system in $t = 0$ is $|e, n\rangle$. $\Omega_n = \Omega_0\sqrt{n+1}$ is the $n$-photon Rabi frequency.

This is an oscillation between the two states $|e, n\rangle$ and $|g, n+1\rangle$, occurring at an angular frequency $\Omega_n \propto \sqrt{n+1}$ (Fig. 11). It is the quantum version of the Rabi oscillations.
A particularly interesting case is the oscillation occurring when the system is prepared initially on the state $|e, 0\rangle$, which happens between the states $|e, 0\rangle$ and $|g, 1\rangle$, at a frequency $\Omega_0$ corresponding to the vacuum Rabi frequency. This is a sort of oscillatory spontaneous emission. The atom initially is excited while the field is in the vacuum state. At first it emits a photon and the system goes in the state $|g, 1\rangle$. However, after some time we find again the system in $|e, 0\rangle$ as the photon is cyclically emitted and re-absorbed.
It is finally worth mentioning that a system initially in the state $|g, 0\rangle$ will remain indefinitely in that state and will not undergo any oscillation. The state $|g, 0\rangle$ indeed is also a dressed state of the full Hamiltonian.

## 8.5 Collapses and revivals of Rabi oscillations in a coherent field

The Jaynes-Cummings model enables to study even complex conditions, where the atom or the field are prepared in non-trivial quantum states. Let us consider an atom initially prepared in the excited state $|e\rangle$, interacting with a field initially prepared in the coherent state $|\alpha\rangle = \sum_n c_n |n\rangle$, with:

$$c_n = e^{-\frac{|\alpha|^2}{2}} \frac{\alpha^n}{\sqrt{n!}}$$

The initial state ($t = 0$) of the whole system can be written as:

$$|\psi(0)\rangle = \sum_{n=0}^{+\infty} c_n |e, n\rangle \tag{8-62}$$

Each of these states $|e, n\rangle$ evolves in time according to the $n$-photon Rabi oscillations studied in Section 8.4. Due to the linearity of the Schroedinger equation, to find $|\psi(t)\rangle$ it is sufficient to substitute in Eq. (8-62) the states their individual time-evolution of the states $|e, n\rangle$ according to Eq. (8-59):

$$|\psi(t)\rangle = \sum_{n=0}^{+\infty} e^{-\imath\hbar\omega(n+1)t} c_n \left[\cos\left(\frac{\Omega_n}{2}t\right)|e, n\rangle + \sin\left(\frac{\Omega_n}{2}t\right)|g, n+1\rangle\right] \tag{8-63}$$

The probability to measure the atom in the excited state, at an arbitrary time $t$, is actually the sum

of the probabilities of the system being in the possible states $|e, n\rangle$, namely:

$$p_e(t) = \sum_n |\langle e, n|\psi(t)\rangle|^2 = \sum_n |c_n|^2 \cos^2\left(\frac{\Omega_n}{2}t\right) \tag{8-64}$$

$$p_e(t) = \sum_n p_\alpha(n) \frac{1 + \cos(\Omega_0\sqrt{n+1}\,t)}{2} \tag{8-65}$$

where $p_\alpha(n) = |c_n|^2 = e^{-|\alpha|^2}\frac{|\alpha|^{2n}}{n!}$ coincides with the probability of measuring $n$ photons in the coherent state $|\alpha\rangle$. This is indeed a sum of terms oscillating at different frequencies $\Omega_0\sqrt{n+1}$, wieighted by the $p_\alpha(n)$ coefficients. We recall that, due to the Poissonian statistics, for a coherent state the expectation value of the photon number $\langle\hat{n}\rangle$ and its standard deviation $\Delta n$ are:

$$\langle\hat{n}\rangle = |\alpha|^2 = \bar{n} \qquad\qquad \Delta n = \sqrt{\langle\Delta n^2\rangle} = \sqrt{\bar{n}}$$

Thus, the frequencies that have the largest weight in the sum of Eq. (8-65) are those right around $n \simeq \bar{n}$, within an interval of $\pm\Delta n$. Other frequencies are present with smaller and smaller weight, the more $n$ becomes different from $\bar{n}$.
This combination of incommensurate oscillations give rise to peculiar effects. Figure 12 plots $p_e(t)$ as given by Eq. (8-65), in the case $\bar{n} = |\alpha|^2 = 5$, which is an example of a coherent state with a small mean photon number, and in case of a coherent state a bit more intense with $\bar{n} = |\alpha|^2 = 30$.
A first feature that we note in the graphs is that, at short times, $p_e$ oscillates at a frequency close to $\Omega_{\bar{n}} = \Omega_0\sqrt{\bar{n}+1}$; however, oscillations are rapidly quenched and $p_e(t)$ becomes almost stationary at 0.5. Indeed, at time $t = 0$ all terms in Eq. (8-65) start oscillating in phase; Rabi oscillations thus begin between the ground and the excited state of the atom. Since the most significant terms have frequencies close to $\Omega_0\sqrt{\bar{n}+1}$, this is the apparent oscillation frequency. However, the different frequency components after a while become out of phase, and this is the reason for the observed **collapse** of the Rabi oscillations.
We can estimate the collapse time $t_C$ by calculating the time for which there is a full $2\pi$ interval of possible phases in the range of the relevant frequencies between $\Omega_{\bar{n}+\sqrt{\bar{n}}}$ and $\Omega_{\bar{n}-\sqrt{\bar{n}}}$. For the oscillation at frequency $\Omega_n$ the accumulated phase after a time $t$ is:

$$\phi = \Omega_0\sqrt{n+1}t \tag{8-66}$$

By differentiating this expression (with respect to $\phi$ and $n$) we obtain an approximate formula for the relative dephasing between two oscillations with indices that differ by $\Delta n$:

$$\Delta\phi = \Omega_0\frac{1}{2\sqrt{n+1}}t\Delta n \tag{8-67}$$

We thus retrieve $t_C$ by substituting $\Delta\phi = 2\pi$, $\Delta n = 2\sqrt{\bar{n}}$ and $n = \bar{n}$:

$$t_C = \frac{2\pi}{\Omega_0}\frac{2\sqrt{\bar{n}+1}}{2\sqrt{\bar{n}}} \tag{8-68}$$

$$\boxed{t_C \simeq \frac{2\pi}{\Omega_0}} \tag{8-69}$$

We see that this collapse time does not depend on the intensity of the coherent state, but only on the conditions of the field-charge interaction, which determine $\Omega_0$. Note however that, the larger is $\bar{n} = |\alpha|^2$, the shorter is the periodicity of the observed oscillations at frequency $\Omega_{\bar{n}}$. Thus, within this same time $t_C$ we will observe more oscillations (around $\sqrt{\bar{n}}$) if the coherent state has a large $\bar{n}$.
A second feature that we observe in the graphs is that, some time after the collapse, there is a **revival** of the Rabi oscillations. This occurs because the oscillations at different frequency become again approximately in phase after a while; in detail, this happens when the difference in phase between

successive terms in Eq. (8-65) is close to $2\pi$. In this view, we can estimate the revival time $t_R$ by imposing $\Delta\phi = 2\pi$ and $\Delta n = 1$ in Eq. (8-67):

$$2\pi = \Omega_0 \frac{1}{2\sqrt{n+1}} t_R \qquad \rightarrow \qquad t_R = \frac{4\pi}{\Omega_0}\sqrt{n+1} \tag{8-70}$$

The above equation actually shows us that the re-phasing cannot happen perfectly for all oscillatory terms, because the condition of $t_R$ in Eq. (8-70) depends on $n$.
However, as already discussed, the frequency components that play a significant role in determining the dynamics are those with $n$ comprised in the interval between $\bar{n} - \sqrt{\bar{n}}$ and $\bar{n} + \sqrt{\bar{n}}$. In addition, especially if $\bar{n}$ is not too small, it is indeed not to important if we substitute in Eq. (8-70) $n = \bar{n}$, $n = \bar{n} \pm 1$, $n = \bar{n} \pm 2$ and so on. Namely, the group of adjacent frequencies around $\Omega_{\bar{n}}$ are all approximately back in phase with good approximation at a time $t_R$ defined by:

$$\boxed{t_R = \frac{4\pi}{\Omega_0}\sqrt{\bar{n}}} \tag{8-71}$$

The fact that rephasing is not perfect for all frequencies makes the revival peak smaller than the initial Rabi oscillations.
By observing Fig. (12)b, we note that the revival peak is followed by a new collapse after an additional time of approximately $t_C$, and then a second revival is observed at a time $t \simeq 2t_R$. Indeed this is a general feature. Rabi oscillations in this condition exhibit a series of revivals separated by idling periods. The revivals occurs approximately at times multiple of $t_R$. In each revival, however, the condition of rephasing is reached with a worse approximation, hence the peaks are wider and less pronounced as the time goes.
We further note that the collapses and revivals of Rabi oscillations do not depend on the initial phase terms of the $c_n$ coefficients, which characterize the Fock-state expansion of the initial state of the field. As a matter of fact, similar phenomena can be observed also for other forms of excitation of the field, consisting in the superposition of several Fock states.
Collapses and revivals of Rabi oscillations cannot be explained by classical models and are directly linked to the field quantization, which leads to the discrete set of Rabi frequencies $\Omega_n$. However, the model discussed above allows to explain also the **classical limit**, which occurs when $\Omega_0 \rightarrow 0$ and $\bar{n} \rightarrow +\infty$. To make $\Omega_0$ small we need a weak interaction between the field and the atom, and as well a large quantization volume $V$ that decrease the quantization constant $\mathcal{E}$ in the electric field expression. In this condition, $t_C \propto \frac{1}{\Omega_0} \rightarrow +\infty$. A large $\bar{n}$, corresponding to a coherent field assimilated to a classical wave, allows to observe a large number $\sqrt{\bar{n}}$ of oscillation before going out of phase. This tends to agree with the Rabi oscillation predicted by the semi-classical model, which do not show collapses.

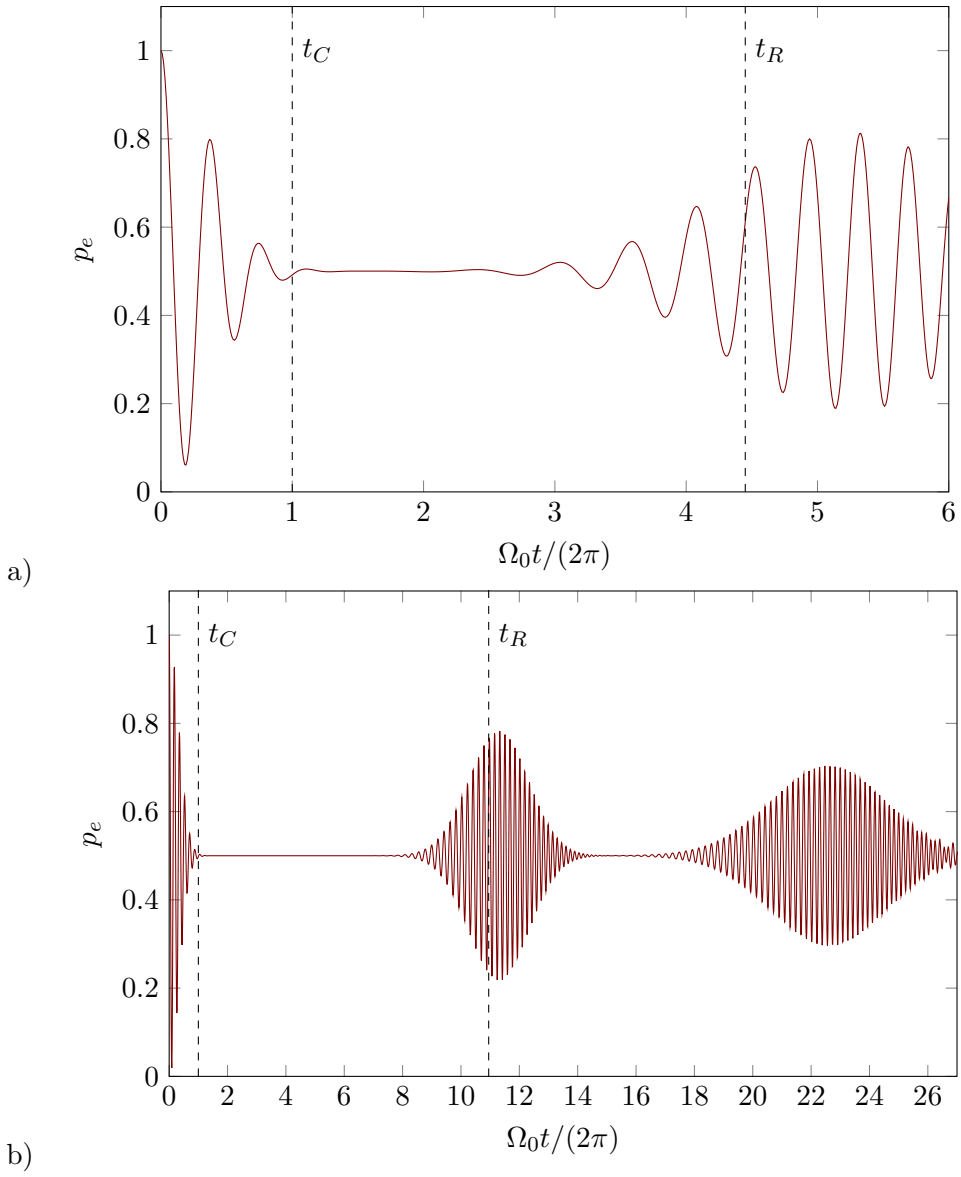


Figure 12: Collapses and revivals of the Rabi oscillations. The two graphs plots the probability $p_e(t)$, given by Eq. (8-65), for the two-level atom to be detected in the excited state, in case at $t = 0$ it is prepared in the excited state and the field is prepared in a coherent state. Graph in panel (a) is calculated for a coherent state with $\bar{n} = 5$, graph in panel (b) for the case $\bar{n} = 30$. Dashed vertical lines indicates the approximate collapse time $t_C$ and revival time $t_R$ as estimated by Eqs. (8-69) and (8-71).

# Notes on Quantum Optics - Appendix

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*Notes for the course 'Quantum Optics and Information' - A.Y. 2023/2024. These notes are provided for personal use only, with no claim of completeness or absence of errors.*

# A Fundamentals of Quantum Mechanics

We recall in this Appendix the postulates of Quantum Mechanics and a few fundamental theoretical results. The discussion will be succinct and has the purpose of fixing the formalism used in these Notes.

## A.1 System state and observable quantities

**Postulate I**

The state of a physical system is represented by a normalized **vector**, defined in a suitable Hilbert space.

Such vectors may be indeed of finite dimension or of infinite dimension, depending on the system that we want to describe. Using the Dirac notation, they are indicated as **kets**:

$$|\psi\rangle$$

In case of finite dimension, they can be associated to **complex vectors**, e.g. column vectors in $\mathbb{C}^n$ such as $\underline{c} = \begin{bmatrix} \vdots \end{bmatrix}$. The association is done in the following way. We take a complete orthonormal basis of the Hilbert space, consisting in the vectors $\{|e_n\rangle\}$. Since they are orthonormal, it holds $\langle e_m|e_n\rangle = \delta_{mn}$. We can decompose any vector $|\psi\rangle$ on this basis as:

$$|\psi\rangle = \sum_n c_n \, |e_n\rangle \tag{A-1}$$

The complex vector $\underline{c}$ associated to $|\psi\rangle$ is the column vector containing the coefficients $\underline{c} = \begin{bmatrix} c_1 \\ \vdots \\ c_N \end{bmatrix}$.
On the other hand, in case of infinite dimension, a ket is associated to a function of one or more continuous coordinates, e.g. $\psi(x)$, called the **wavefunction**.
In addition to the main Hilbert space, where the kets live, it is useful to define the dual space of the **bras**:

$$\langle\psi|$$

To every ket corresponds one bra. In case of finite dimensions, it is the Hermitian conjugate vector (i.e. the transposed complex-conjugate vector) $\underline{c}^\dagger = [\ldots]$; thus, the space of the bras is the space of the row vectors in $\mathbb{C}^n$. In case of infinite dimension, a bra is a linear functional with the form $\langle\psi| \rightarrow \int_{-\infty}^{+\infty} \psi^*(x,t) \ldots dx$. In fact, more rigorously, also in the case of finite dimension we may think of the bra not just as a row vector, but as a linear map: the row-column multiplication having as a left term that row vector.
The scalar product between two vectors can be introduced by exploiting the formalism of kets and bras. The scalar product of the vectors $|\phi\rangle$ and $|\psi\rangle$ (in this order) is the application of the bra associated to $|\phi\rangle$ to the ket $|\psi\rangle$. In Dirac notation we write the so-called **braket** to indicate the scalar product:

$$\langle\phi|\psi\rangle$$

In the case of finite dimension, if $|\psi\rangle \equiv \underline{\psi}$ and $|\phi\rangle \equiv \underline{\phi}$, we will have $\langle\phi|\psi\rangle = \underline{\phi}^\dagger \cdot \underline{\psi}$. In case of infinite dimension, with $|\psi\rangle \equiv \psi(x)$ and $|\phi\rangle \equiv \phi(x)$, we will have $\langle\phi|\psi\rangle = \int_{-\infty}^{+\infty} \phi^*(x)\psi(x)\, dx$ .
Note that the exchanging the terms in the scalar product transforms the result in its complex conjugate:

$$\langle\phi|\psi\rangle = (\langle\psi|\phi\rangle)^* \tag{A-2}$$

The scalar product induces a **norm** on the Hilbert space. Quantum states must have unit norm:

$$\|\psi\| = \sqrt{\langle\psi|\psi\rangle} = 1 \tag{A-3}$$

In the finite-dimension case, this implies $\sum_n |c_n|^2 = 1$. In the infinite-dimension case, it corresponds to $\int_{-\infty}^{+\infty} |\psi(x)|^2 dx = 1$.

**Postulate II**

Each observable physical quantity $Q$ is associated to one **linear Hermitian operator** $\hat{Q}$ acting on the Hilbert space of the states.

By definition, an operator is Hermitian if it satisfies $\hat{Q}^\dagger = \hat{Q}$. In the finite-dimensional case, $\hat{Q}$ is associated to a Hermitian **square matrix**. In fact, if we choose an arbitrary basis $\{|e_n\rangle\}$ of the Hilbert space, any linear operator can be written as:

$$\hat{Q} = \sum_{m,n} q_{m,n} |e_m\rangle \langle e_n| \tag{A-4}$$

This expression is justified as $\hat{Q}$ has to produce a generic linear operation on each vector of the basis:

$$\hat{Q} |e_n\rangle = \sum_m q_{m,n} |e_m\rangle \tag{A-5}$$

and must act linearly on any superposition of these vectors. If we now consider a generic state as in Eq. (A-1) and the operation $|\psi'\rangle = \hat{Q} |\psi\rangle$ we have:

$$|\psi'\rangle = \hat{Q} |\psi\rangle = \sum_k \hat{Q} |e_k\rangle = \sum_{k,m,n} c_k q_{m,n} |e_m\rangle \underbrace{\langle e_n|e_k\rangle}_{=\delta_{nk}} = \sum_{m,n} c_n q_{m,n} |e_m\rangle = \sum_m c'_m |e_m\rangle \tag{A-6}$$

with $c'_m = \sum_n q_{m,n} c_n$. Considering the complex vector $\underline{c} = [c_n]$ associated to $|\psi\rangle$ and the complex vector $\underline{c}' = [c'_n]$ associated to $|\psi'\rangle$, Eq. (A-6) shows that $|\psi'\rangle = \hat{Q} |\psi\rangle$ corresponds to:

$$\underline{c}' = Q\underline{c} \tag{A-7}$$

with the square matrix $Q = [q_{m,n}]$.
In general, sequential application of two operators to a given vector is not commutative. One defines the **commutator** of the operators $\hat{A}$ and $\hat{B}$ as:

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} \tag{A-8}$$

The commutator has the following useful properties:

$$[\hat{A}, \hat{A}] = 0 \qquad [\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$$
$$[\alpha\hat{A}, \hat{B}] = \alpha[\hat{A}, \hat{B}] \qquad [\hat{A} + \hat{B}, \hat{C}] = [\hat{A}, \hat{C}] + [\hat{B}, \hat{C}]$$

where $\alpha$ is a complex number and $\hat{C}$ a third operator.
Sometimes it proves useful also the **anticommutator**:

$$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A} \tag{A-9}$$

**Postulate III**

A measurement of the observable $Q$ necessarily gives one of the **eigenvalues** of $\hat{Q}$.

We remind that, by definition, $\lambda$ is an eigenvalue of $\hat{Q}$ associated to the eigenvector $|e_\lambda\rangle$ if they satisfy the relation:

$$\hat{Q}\,|e_\lambda\rangle = \lambda\,|e_\lambda\rangle \tag{A-10}$$

**Proposition A.1** *Hermitian operators can only have* real *eigenvalues.*

This is proven easily. Let $|e_\lambda\rangle$ be an eigenvector, then:

$$\langle e_\lambda|\,\hat{Q}\,|e_\lambda\rangle = \langle e_\lambda|\,\lambda\,|e_\lambda\rangle = \lambda\langle e_\lambda|e_\lambda\rangle = \lambda \tag{A-11}$$

If we now exploit the fact that $\hat{Q} = \hat{Q}^\dagger$:

$$\lambda = \langle e_\lambda|\,\hat{Q}\,|e_\lambda\rangle = \langle e_\lambda|\,\hat{Q}^\dagger\,|e_\lambda\rangle = (\langle e_\lambda|\,\hat{Q}\,|e_\lambda\rangle)^* = \lambda^* \tag{A-12}$$

But $\lambda = \lambda^*$ if and only if $\lambda$ is real. ■

Even though the state of the system is described by complex numbers, the measurement results must indeed be real numbers.

**Postulate IV**

If the system is described by the state $|\psi\rangle$, the measurement of $Q$ will give the eigenvalue $\lambda$ with **probability**:

$$p_\lambda = |\langle\psi|e_\lambda\rangle|^2 \tag{A-13}$$

where $|e_\lambda\rangle$ is the eigenvector associated to $\lambda$.

The postulates of Quantum Mechanics do not allow for any way to predict the measurement results in a deterministic way, in the general case. The measurement outcome is a genuinely random occurence, following the probability distribution given by Eq. (A-13). As a matter of fact, if we express the ket $|\psi\rangle$ as a linear combination of the eigenvectors of $\hat{Q}$:

$$|\psi\rangle = \sum_\lambda c_\lambda\,|e_\lambda\rangle \tag{A-14}$$

the probability to obtain $\lambda$ as a measurement result is exactly the modulus squared of the related coefficient:

$$p_\lambda = |c_\lambda|^2 \tag{A-15}$$

The coefficients assume the meaning of a probability once they are squared in modulus; they are called more precisely **probability amplitudes**. Note that the normalization condition implies, consistently, that $\sum_n p_n = \sum_n |c_n|^2 = 1$.

**Proposition A.2** *The* ***expectation value*** *for a measurement of the observable $Q$ in a system described by the state $|\psi\rangle$ is given by:*

$$\langle Q\rangle = \langle\psi|\,\hat{Q}|\psi\rangle \tag{A-16}$$

The expectation value is the sum of the possible measurement results, weighted by their probability:

$$\langle Q\rangle = \sum_{\lambda} \lambda p_{\lambda} \tag{A-17}$$

Indeed, this is what we obtain if we develop (A-16):

$$\begin{aligned}\langle\psi|\,\hat{Q}|\psi\rangle &= \left(\sum_n \langle e_n|\,c{*}_n\right)\hat{Q}\left(\sum_m c_m\,|e_m\rangle\right) = \left(\sum_n \langle e_n|\,c_n^*\right)\left(\sum_m c_m \underbrace{\hat{Q}\,|e_m\rangle}_{\lambda_m|e_m\rangle}\right) = \\ &= \sum_{m,n} c_n^* c_m \lambda_m \underbrace{\langle e_n|e_m\rangle} = \delta_{mn} = \sum_n c_n^* c_n \lambda_n = \sum_n |c_n|^2 \lambda_n = \sum_n p_{\lambda_n}\lambda_n\end{aligned} \tag{A-18}$$

■

**Proposition A.3** *A global phase applied to the state does not alter any measurement probability, for any possible observable $Q$.*

Let us consider a state $|\psi'\rangle = e^{\iota\beta}\,|\psi\rangle$. For any possible eigenvector $|e_\lambda\rangle$ one writes:

$$p'_\lambda = |\langle\psi'|e_\lambda\rangle|^2 = \langle e_\lambda|\psi'\rangle\langle\psi'|e_\lambda\rangle = \langle e_\lambda|\,e^{\iota\beta}\,|\psi\rangle\,\langle\psi|\,e^{-\iota\beta}\,|e_\lambda\rangle = \underbrace{e^{\iota\beta}e^{-\iota\beta}}_{=1}|\langle\psi|e_\lambda\rangle|^2 = p_\lambda \tag{A-19}$$

■

Global phases are not measurable and do not have physical meaning. Care should be taken in distinguishing global phases, applied equally to all coefficients $c_n$ of a quantum state, from phases applied to individual coefficients $c_n$: these other phase terms have instead profound influence on measurement probability and have full physical significance.

**Proposition A.4** *Consider an operator:*

$$\hat{R} = f(\hat{Q}) = \sum_{n=0}^{+\infty} b_n\,\hat{Q}^n \tag{A-20}$$

*namely that is a function of a single other operator $\hat{Q}$, expressible as a power series. In addition, let $|e_\lambda\rangle$ be an eigenvector of $\hat{Q}$ associated to the eigenvalue $\lambda$. Then, $|e_\lambda\rangle$ is also an eigenvector of $\hat{R}$ associated to the eigenvalue $f(\lambda)$:*

$$\hat{Q}\,|e_\lambda\rangle = \lambda\,|e_\lambda\rangle \qquad \Rightarrow \qquad \hat{R}\,|e_\lambda\rangle = f(\lambda)\,|e_\lambda\rangle \tag{A-21}$$

*with $f(\lambda) = \sum_{n=0}^{+\infty} b_n\,\hat{\lambda}^n$.*

Indeed:

$$\hat{R}\,|e_\lambda\rangle = \sum_{n=0}^{+\infty} b_n\,\hat{Q}^n\,|e_\lambda\rangle = \sum_{n=0}^{+\infty} b_n\,\lambda^n\,|e_\lambda\rangle = f(\lambda)\,|e_\lambda\rangle \tag{A-22}$$

■

This allows to associate licitly a physical observable $R$ that is function of a single physical observable $Q$, such as $R = f(Q) = \sum_n b_n Q^n$, to the operator $\hat{R}$ defined as the analogous power series of $\hat{Q}$. This does not hold straightforwardly for functions of more than one observables, as operators do not necessarily commute (see also Section A.3).

**Postulate V**

After a measurement of $Q$ that has given the eigenvalue $\lambda_k$, the system state is described precisely by the eigenvector $|e_k\rangle$ associated to that eigenvalue. The system state has **collapsed** on $|e_k\rangle$.

If, before a measurement, a system is already in the state $|e_k\rangle$, because of Postulate IV the measurement will give with *certainty* the result $\lambda_k$. In addition, after the measurement, the system will remain on the same initial state $|e_k\rangle$. In all other cases, the measurement will provide a random result among the eigenvalues of $Q$, according to Eq. (A-13), and upon the measurement the state of the system will suddenly change in accordance to the random measurement result.
This is an utterly different concept of measurement with respect to classical physics, where results are deterministic (if the state of the system is known) and the state of the system, in principle, may not be affected by the measurement process.

## A.2 Time evolution

**Postulate VI**

The time evolution of the state vector is governed by the Schroedinger equation:

$$\iota\hbar\frac{d}{dt}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle \tag{A-23}$$

where $\hat{H}$ is the Hamiltonian operator, namely it is an observable associated to the total energy of the system.

Postulates V and VI demonstrate the profound dichotomy, existent in Quantum Mechanics, between the change in the quantum state given by a measurement process, and the evolution of the state produced by the action of the Hamiltonian. In the first case there is a sudden and abrupt change in the quantum state; in the second one there is a continuous progression.

**Proposition A.5** *If $\hat{H}$ does not depend on time, the solution to the Schroedinger equation is given by:*

$$|\psi(t)\rangle = e^{-\iota\hat{H}t/\hbar}|\psi(0)\rangle = \hat{U}|\psi(0)\rangle \tag{A-24}$$

*Namely, the state evolution is given by the application of the **time-evolution** operator $\hat{U} = e^{-\iota\hat{H}t/\hbar}$ on the initial state at $|\psi(0)\rangle$.*

The time evolution operator can be expressed as:[22]

$$\hat{U} = e^{-\iota\hat{H}t/\hbar} = \sum_{n=0}^{+\infty}\left(\frac{\hat{H}}{\iota\hbar}\right)^n\frac{t^n}{n!} \tag{A-25}$$

Therefore, Eq. (A-24) can be developed as:

$$\begin{aligned}\frac{d}{dt}\left(\hat{U}|\psi(0)\rangle\right) &= \frac{d\hat{U}}{dt}|\psi(0)\rangle = \frac{d}{dt}\left(\sum_{n=0}^{+\infty}\left(\frac{\hat{H}}{\iota\hbar}\right)^n\frac{t^n}{n!}\right)|\psi(0)\rangle = \left(\sum_{n=1}^{+\infty}\left(\frac{\hat{H}}{\iota\hbar}\right)^n\frac{nt^{n-1}}{n!}\right)|\psi(0)\rangle = \\ &= \frac{\hat{H}}{\iota\hbar}\left(\sum_{n=1}^{+\infty}\left(\frac{\hat{H}}{\iota\hbar}\right)^{n-1}\frac{t^{n-1}}{(n-1)!}\right)|\psi(0)\rangle = \frac{\hat{H}}{\iota\hbar}e^{-\iota\hat{H}t/\hbar}|\psi(0)\rangle = \frac{\hat{H}}{\iota\hbar}\hat{U}|\psi(0)\rangle\end{aligned} \tag{A-26}$$

which solves Eq. (A-23). ■

[22] Note that the exponential of an operator is defined by the Taylor expansion $e^{\hat{A}} = \sum_{n=0}^{+\infty}\frac{\hat{A}^n}{n!}$.

The operator $\hat{U}$ is unitary and intrinsically conserves the norm of the state. One can also observe that:

$$[\hat{H}, \hat{U}] = 0 \tag{A-27}$$

as $\hat{U}$ is written as a linear combination of the powers of $\hat{H}$. Finally, the Hermitian conjugate of $\hat{U}$ is simply $\hat{U}^\dagger = e^{+\iota \hat{H} t/\hbar}$.
If the the initial state is an eigenstate of the energy, i.e. $|\psi(0)\rangle = |e\rangle$ with $\hat{H}|e\rangle = E|e\rangle$, then Proposition A.4 gives:

$$|\psi(t)\rangle = e^{-\iota \hat{H} t/\hbar}|e\rangle = e^{-\iota \frac{E}{\hbar} t}|\psi(0)\rangle \tag{A-28}$$

which means that the state of the system only acquires a global phase term with passing time. Its energy does not (quite obviously) change in time. Because of Proposition A.3, no changes are measurable on any other observable. Energy eigenstates are called **stationary states**.
In a more general situation, the state vector is a linear combination of the energy eigenstates $|e_n\rangle$, which form a complete basis of the Hilbert space. Each of them will evolve by acquiring a phase term related to its eigenvalue $E_n$ (being $\hat{H}|e_n\rangle = E_n|e_n\rangle$).

$$|\psi(0)\rangle = \sum_n c_n |e_n\rangle \qquad \rightarrow \qquad |\psi(t)\rangle = \sum_n e^{-\iota E_n t/\hbar} c_n |e_n\rangle = \sum_n c'_n(t)|e_n\rangle \tag{A-29}$$

The probability to measure a certain value $E_n$ of the total energy is $p(E_n) = |c'_n|^2 = |e^{-\iota E_n t/\hbar} c_n|^2 = |c_n|^2$, so it is constant and it does not change in time. In this sense energy is conserved in the quantum mechanical system during the unitary evolution given by the Schroedinger equation.
In general, given an observable $Q$ associated to its Hermitian operator $\hat{Q}$, its mean value evolves as:

$$\langle Q(t)\rangle = \langle\psi(t)|\hat{Q}|\psi(t)\rangle = \langle\psi(0)|\hat{U}^\dagger \hat{Q}\hat{U}|\psi(0)\rangle \tag{A-30}$$

We may actually define also a **time-dependent operator**:

$$\hat{Q}(t) = \hat{U}^\dagger \hat{Q}\hat{U} \tag{A-31}$$

and write the mean value as:

$$\langle Q(t)\rangle = \langle\psi(0)|\hat{Q}(t)|\psi(0)\rangle \tag{A-32}$$

In this expression the time-dependence has been attributed to the operator instead of the state vector. This way of representing the time-evolution (i.e. by attributing it to the operators instead of the state vectors) is called the **Heisenberg picture**. The other representation in which the time evolution is attributed to state vectors, with operators constant in time, is called the **Schroedinger picture**. There exists also a more general representation called the 'interaction picture', of which the Heisenberg and Schroedinger ones are special cases (see Box A-1).

**Proposition A.6** *In the Heisenberg picture, the operators evolve in time according to:*

$$\frac{d}{dt}\hat{Q}(t) = \frac{\iota}{\hbar}\left[\hat{H}, \hat{Q}\right] \tag{A-33}$$

Taking the derivative of Eq. (A-31) with respect to time:

$$\frac{d}{dt}\hat{Q}(t) = \frac{d\hat{U}^\dagger}{dt}\hat{Q}\hat{U} + \hat{U}^\dagger\hat{Q}\frac{d\hat{U}}{dt} = \frac{\iota}{\hbar}\hat{H}\hat{U}^\dagger\hat{Q}\hat{U} + \hat{U}^\dagger\hat{Q}\left(-\frac{\iota}{\hbar}\hat{H}\right)\hat{U} \tag{A-34}$$

We can factor out $\frac{\iota}{\hbar}$ and, as $\hat{H}$ commutes with $\hat{U}$, we can reorder the expression as:

$$\frac{d}{dt}\hat{Q}(t) = \frac{\iota}{\hbar}\left(\hat{H}\hat{U}^\dagger\hat{Q}\hat{U} - \hat{U}^\dagger\hat{Q}\hat{U}\hat{H}\right) = \frac{\iota}{\hbar}\left(\hat{H}\hat{Q}(t) - \hat{Q}(t)\hat{H}\right) \tag{A-35}$$

And we have proven Eq. (A-33). ■

Box A-1 - The interaction picture

Let us consider a Hamiltonian that is the sum of two parts:

$$\hat{H} = \hat{H}_0 + \hat{H}_1$$

In a typical application, the term $\hat{H}_0$ may represent the Hamiltonian of several separated physical systems, while $\hat{H}_1$ may describe their interaction. However, what follows holds mathematically for any kind of Hamiltonian that is sum of two parts. Let us further define the time-evolution operator that includes only $\hat{H}_0$:

$$\hat{U}_0 = e^{-\iota \frac{\hat{H}_0}{\hbar} t}$$

For any Hermitian operator $\hat{Q}$ associated to an observable quantity, we can write the expectation value as:

$$\langle Q \rangle = \langle \psi(t) | \hat{Q} | \psi(t) \rangle = \langle \psi(t) | \hat{U}_0 \hat{U}_0^\dagger \hat{Q} \hat{U}_0 \hat{U}_0^\dagger | \psi(t) \rangle$$

We can define a new ket $|\psi'\rangle$ and a new operator $\hat{Q}'$ (in general, both time dependent):

$$\boxed{|\psi'\rangle = \hat{U}_0^\dagger |\psi(t)\rangle \qquad \hat{Q}' = \hat{U}_0^\dagger \hat{Q} \hat{U}_0}$$

The expectation value becomes:

$$\boxed{\langle Q \rangle = \langle \psi' | \hat{Q}' | \psi' \rangle}$$

This new representation of the states and of the operators is called the **interaction picture**. Note that, by using the Schroedinger equation, we can derive the evolution of $|\psi'\rangle$ as follows.

$$\frac{d}{dt} |\psi'\rangle = \frac{d}{dt} \left( \hat{U}_0^\dagger |\psi(t)\rangle \right) = \underbrace{\frac{d}{dt} \hat{U}_0^\dagger}_{=+\frac{\iota}{\hbar} \hat{H}_0 \hat{U}_0} |\psi(t)\rangle + \hat{U}_0^\dagger \underbrace{\frac{d}{dt} |\psi(t)\rangle}_{=-\frac{\iota}{\hbar}(\hat{H}_0 + \hat{H}_1)|\psi(t)\rangle} =$$

$$= \frac{\iota}{\hbar} \hat{H}_0 \hat{U}_0 |\psi(t)\rangle - \frac{\iota}{\hbar} \hat{U}_0^\dagger \hat{H}_0 |\psi(t)\rangle - \frac{\iota}{\hbar} \hat{U}_0^\dagger \hat{H}_1 |\psi(t)\rangle =$$

$$= \frac{\iota}{\hbar} \hat{H}_0 |\psi'\rangle - \frac{\iota}{\hbar} \hat{U}_0^\dagger \hat{H}_0 \hat{U}_0 |\psi'\rangle - \frac{\iota}{\hbar} \hat{U}_0^\dagger \hat{H}_1 \hat{U}_0 |\psi'\rangle$$

At this point we consider that $\hat{U}_0$ and $\hat{H}_0$ commute, so $\hat{U}_0^\dagger \hat{H}_0 \hat{U}_0 = \hat{U}_0^\dagger \hat{U}_0 \hat{H}_0 = \hat{H}_0$, and we conclude:

$$\boxed{\iota \hbar \frac{d}{dt} |\psi'\rangle = \hat{H}' |\psi'\rangle}$$

where $\hat{H}' = \hat{U}_0^\dagger \hat{H}_1 \hat{U}_0$. This is a sort of modified Schroedinger equation for $|\psi'\rangle$, where the Hamiltonian $\hat{H}$ is substituted with $\hat{H}'$.

The *Heisenberg picture* is the limiting condition in which we consider $\hat{H}_0 \equiv \hat{H}$ and $\hat{H}_1 = 0$. From the Schroedinger equation $|\psi(t)\rangle = \hat{U}_0 |\psi(0)\rangle$ so the state vector $|\psi'\rangle = \hat{U}_0^\dagger |\psi(t)\rangle = \hat{U}_0^\dagger \hat{U}_0 |\psi(0)\rangle = |\psi_0\rangle$ becomes constant, and the time dependence is transferred entirely on the operator $\hat{Q}'$.

The *Schroedinger picture* corresponds to $\hat{H}_0 = 0$ and $\hat{H}_1 \equiv \hat{H}$. Unless the Hamiltonian itself depends on time, the time dependence is attributed entirely on the state vector.

The interaction picture indeed opens the way to intermediate cases, in which we transfer to the operator the time dependence given by the $\hat{H}_0$, which may be the one we are less interested in, for the application. In this way the time evolution of $|\psi'\rangle$, carrying the effect of the interaction part $\hat{H}_1$, may be studied conveniently.

## A.3 Measurements and uncertainties

The non-commutativity features of the quantum mechanical operators bring important consequences for that regards the simultaneous measurements of different quantities. As early understood by Heisenberg two so-called incompatible quantities cannot be measured together with arbitrary precision, because of fundamental and intrinsic limitations. What is widely known as the 'Heisenberg uncertainty principle', valid for position and momentum, can be generalized to an arbitrary pair of quantities and it is actually a theorem, namely a consequence of the Postulates discussed before.

**Proposition A.7** *Let $\hat{A}$ and $\hat{B}$ the operators associated to two different observable quantities. The product of the uncertainties $\sigma_A^2$ and $\sigma_B^2$ (intended as the variances) in the measurements of the two quantities satisfies the inequality:*

$$\sigma_A^2 \sigma_B^2 \geq \left( \frac{1}{2\iota} \langle [\hat{A}, \hat{B}] \rangle \right)^2 \tag{A-36}$$

Let us consider the generic ket $|\phi\rangle$, which may be not normalized to 1:

$$|\phi\rangle = (\hat{A} + \iota\lambda\hat{B}) |\psi\rangle \tag{A-37}$$

with $|\psi\rangle$ an arbitrary state vector and $\lambda \in \mathbb{R}$.
While the norm of $|\phi\rangle$ might be different from 1, it must surely be $\|\phi\|^2 = \langle\phi|\phi\rangle \geq 0$. Therefore, considering that $\hat{A}^\dagger = \hat{A}$ and $\hat{B}^\dagger = \hat{B}$ as they are Hermitian:

$$\begin{aligned}\langle\phi|\phi\rangle = \langle\psi| (\hat{A} - \iota\lambda\hat{B})(\hat{A} + \iota\lambda\hat{B}) |\psi\rangle = \langle\psi| \hat{A}^2 |\psi\rangle + \iota\lambda \langle\psi| \hat{A}\hat{B} - \hat{B}\hat{A} |\psi\rangle + \lambda^2 \langle\psi| \hat{B}^2 |\psi\rangle = \\ = \langle\hat{A}^2\rangle + \iota\lambda\langle[\hat{A}, \hat{B}]\rangle + \lambda^2\langle\hat{B}^2\rangle\end{aligned} \tag{A-38}$$

The equation in $\lambda$ should always give real solutions, so its determinant should be non-positive:

$$\left(\iota\langle[\hat{A}, \hat{B}]\rangle\right)^2 - 4\langle\hat{A}^2\rangle\langle\hat{B}^2\rangle \leq 0 \tag{A-39}$$

$$\langle\hat{A}^2\rangle\langle\hat{B}^2\rangle \geq \left(\frac{1}{2\iota}\langle[\hat{A}, \hat{B}]\rangle\right)^2 \tag{A-40}$$

Let us consider now the pair of operators:

$$\hat{A}' = \hat{A} - \langle A\rangle \tag{A-41}$$

$$\hat{B}' = \hat{B} - \langle B\rangle \tag{A-42}$$

Equation (A-40) holds for any pair of operators, so it can be written also for operators $\hat{A}'$ and $\hat{B}'$:

$$\langle\hat{A}'^2\rangle\langle\hat{B}'^2\rangle \geq \left(\frac{1}{2\iota}\langle[\hat{A}', \hat{B}']\rangle\right)^2 \tag{A-43}$$

At this point we observe that:

$$\sigma_A^2 \equiv \langle\hat{A}'^2\rangle \qquad \sigma_B^2 \equiv \langle\hat{B}'^2\rangle \qquad [\hat{A}, \hat{B}] = [\hat{A}', \hat{B}']$$

and by substitution of these relations in Eq. (A-43) we have proven Eq. (A-36).■
A small formal note may be done on the significance of the imaginary unit inside the square, on the right side of Eq. (A-36). Operators $\hat{A}$ and $\hat{B}$ are Hermitian so we can write:

$$([\hat{A}, \hat{B}])^\dagger = [\hat{B}^\dagger, \hat{A}^\dagger] = [\hat{B}, \hat{A}] = -[\hat{A}, \hat{B}] \tag{A-44}$$

If we define $\hat{C} = [\hat{A}, \hat{B}]$ the former relation says $\hat{C}^\dagger = -\hat{C}$, which means that $\hat{C}$ is *anti-hermitian*. If we now define $\hat{C} = \iota\hat{D}$, the operator $\hat{D}$ is Hermitian as we get $\hat{D}^\dagger = (-\iota\hat{C})^\dagger = \iota\hat{C}^\dagger = -\iota\hat{C} = \hat{D}$. For the expectation value of the commutator, it holds:

$$\langle[\hat{A}, \hat{B}]\rangle = \langle\hat{C}\rangle = \langle\iota\hat{D}\rangle = \iota\langle\hat{D}\rangle = \iota\delta \tag{A-45}$$

with $\delta = \langle \hat{D} \rangle \in \mathbb{R}$ as $\hat{D}$ is Hermitian. Hence, $\langle [\hat{A}, \hat{B}] \rangle$ is a purely imaginary number, and it is meaningful to divide it by $\iota$ before squaring it.
In case the two operators commute, the lower limit of the product of the uncertainties provided by Eq. (A-36) reaches zero. In fact, for two commuting operators it is possible to prove the following:

**Proposition A.8** *Two commuting operators $\hat{A}$ and $\hat{B}$ share (at least) one complete set of eigenvectors.*

We see this in the case $\hat{A}$ has only non-degenerate eigenvalues. This means that we assume that two eigenvectors $|e_1\rangle \neq |e_2\rangle$ of $\hat{A}$ associated to the same eigenvalue are necessarily parallel:

$$\begin{cases} \hat{A}\,|e_1\rangle = \lambda\,|e_1\rangle \\ \hat{A}\,|e_2\rangle = \lambda\,|e_2\rangle \end{cases} \qquad \Leftrightarrow \qquad |e_1\rangle = \gamma\,|e_2\rangle \quad \text{with} \quad \gamma \in \mathbb{C} \tag{A-46}$$

In this case, let us take any eigenvector $|e\rangle$ of $\hat{A}$, associated to the eigenvalue $\lambda$; then, we can write:

$$0 = [\hat{A}, \hat{B}]\,|e\rangle = \hat{A}\hat{B}\,|e\rangle - \hat{B}\hat{A}\,|e\rangle = \hat{A}\hat{B}\,|e\rangle - \lambda\hat{B}\,|e\rangle = \hat{A}\,|e'\rangle - \lambda\,|e'\rangle \tag{A-47}$$

where we have defined $|e'\rangle = \hat{B}\,|e\rangle$. The former relation means that $|e'\rangle$ is again an eigenvector of $\hat{A}$ associated to the same eigenvalue $\lambda$:

$$\hat{A}\,|e'\rangle = \lambda\,|e'\rangle \tag{A-48}$$

because of Eq. (A-46), this implies:

$$|e'\rangle = \hat{B}\,|e\rangle = \gamma\,|e\rangle \tag{A-49}$$

namely, $|e\rangle$ is also an eigenvalue of $\hat{B}$ with eigenvalue $\gamma$. As this holds for any eigenvector $|e\rangle$, this means that the complete basis of the eigenvectors $|e_n\rangle$ of $\hat{A}$ associated to the eigenvalues $\lambda_n$, is also a complete basis of eigenvectors of $\hat{B}$ associated to the eigenvalues $\gamma_n$.
The proof may be extended to the case of degenerate eigenvalues. A complete basis of shared eigenvectors can be still found. However, in this case, eigenvectors associated to the same degenerate eigenvalue $\lambda$ for the operator $\hat{A}$ may be associated to different eigenvalues for the operator $\hat{B}$. ■
This Proposition means that the lower bound of $\sigma_A^2\sigma_B^2 = 0$, if $\hat{A}$ and $\hat{B}$ commute, can be reached for both $\sigma_A^2 = \sigma_B^2 = 0$. Indeed, if the system state coincides with one of the common eigenvectors, the measurement of $\hat{A}$ gives with certainty as a result the associated eigenvalue, and does not change the state of the system; the same can be said about a measurement of $\hat{B}$. The two measurements could be done in succession in either order, or simultaneously, providing with certainty the respective eigenvalues. The two quantities $A$ and $B$ are said to be **compatible observables**.